cft-anyons Lab Book

From category data to provably rigorous CFTs

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Contents

1 Frontmatter, North Star, and Claim Status

This is a reproducible research lab book. Its guiding goal — the *north star* — is a single, concrete construction problem:

> Given a (unitary) fusion or modular tensor category $\mathcal{C}$, together with whatever additional data is needed (OPE coefficients / conformal data), construct a family of microscopic (lattice) models and their symmetry generators whose continuum limit *provably* yields a mathematically rigorous QFT/CFT that (i) realises the input category data and (ii) carries a full projective unitary representation of the symmetries.

The working belief is that this is achievable at least for *rational* CFTs, and perhaps for all QFTs. This is a programme, not a settled theory: every step toward the goal is a question, proposal, or hypothesis until a local source, a local derivation, or a reproducible run supports it.

1.1 The Two Success Criteria

A construction is only a candidate solution if it meets both, non-negotiably:

1. **Provability.** The continuum limit is established rigorously (an operator-algebraic / wavelet renormalisation limit with theorems, not a heuristic scaling argument).
2. **Fidelity.** The realised continuum content faithfully reproduces the input category data $\mathcal{C}$, and the symmetry representation is *full* and projective-unitary.

1.2 Claim Status

No claim is treated as established until it is marked *Source-local* or *Checked* and linked to a local source, derivation, test, or run artifact. The status labels are fixed in `CONVENTIONS.md`: *Question*, *Proposal*, *Source-local*, *Checked*, and *Rejected*. These labels are part of the scientific record; they separate programme-level conjecture from results.

1.3 Evidence Chain

Every substantive assertion should be traceable through:

1. a local source manifest under `references/` (the kept papers are the ground truth),
2. a convention entry in `CONVENTIONS.md`,
3. a derivation, script, test, or run bundle when computation is involved,
4. an index row in `INDEX.md`, and
5. a report shard in this lab book.

Missing links are allowed during exploration only when they are explicit.

2 Programme Map

This shard is a map of questions, not a list of results. The north star (§1 / CA-00) factors into a pipeline of stages; each stage is an open problem with its own conventions, sources, and checkable invariants.

2.1 The Pipeline

$$\underbrace{\mathcal{C}\ (+\,\text{OPE/conformal data})}_{\text{input}} \longrightarrow \underbrace{\text{microscopic models}}_{\text{lattice Hilbert space + Hamiltonian}} \longrightarrow \underbrace{\text{lattice symmetry generators}}_{\text{e.g. Koo–Saleur } L_n}$$

$$\longrightarrow \underbrace{\text{provable continuum limit}}_{\text{OAR / wavelets}} \longrightarrow \underbrace{\text{rigorous (C)FT}}_{\text{conformal net / VOA / Wightman}}$$

2.2 Stage 1 — Category Input

- Which categories are in initial scope: multiplicity-free unitary fusion categories (Fibonacci, Ising/TL), then modular tensor categories?
- What auxiliary data beyond $\mathcal{C}$ is genuinely required (OPE coefficients, central charge, conformal weights), and for which target CFTs?
- Which gauge and indexing conventions are fixed before any F/R-symbol is written (see `CONVENTIONS.md`)?

2.3 Stage 2 — Microscopic Models

- Which family realises $\mathcal{C}$ on the lattice: anyon chains, string-nets, or another construction — and how is the family indexed by scale?
- What is the lattice Hilbert space (fusion-tree basis) and the Hamiltonian, and which invariants pin them (fusion-rule consistency, pentagon, projector idempotence)?

2.4 Stage 3 — Lattice Symmetry Generators

- Which lattice operators approximate the continuum symmetry generators (Koo–Saleur lattice Virasoro L_n, $\bar{L}_n$, and kin)?
- What convergence statement is the target, and what is checkable at finite size (commutator algebra, central charge extraction)?

2.5 Stage 4 — Provable Continuum Limit

- Which rigorous renormalisation framework carries the limit (operator- algebraic renormalisation, wavelet / multi-scale entanglement)?
- What must be proved: existence of the limit, the limiting Hilbert space, and convergence of the symmetry generators to a projective unitary representation?

2.6 Stage 5 — Rigorous (C)FT Target

- What is the target object: a conformal net, a vertex operator algebra, a Wightman theory, or an Osterwalder–Schrader theory?
- What is the fidelity statement: that the target's superselection / representation category recovers $\mathcal{C}$?

2.7 First Decisions

The first technical session should not start with formulas. It should register the sources for one worked input category (Fibonacci or Ising/TL), fix the minimum conventions (F/R-symbol gauge, vacuum index, fusion-tree ordering), and identify the smallest invariant that can be checked end-to-end on one stage of the pipeline.

3 Many-Particle Hilbert Spaces: the AND/OR/Vacuum Grammar

3.1 Sources and Dependencies

This is a pre-topic foundation for the programme (see CA-01, the programme map): how to build the Hilbert space of an indefinite number of particles of arbitrary types. The guiding idea is that two logical connectives map cleanly onto two Hilbert-space constructions, and the no-particle system is the complex line.

This shard treats the *finite* grammar and its semantics; the rigorous *indefinite*-particle (Hilbert-completed) case is CA-03. It harmonises with, and re-proves nothing from, the finite "AND/OR many-particle grammar" developed by the author in the sibling project (notation $\mathcal{V}$, $\mathcal{V}_n$, AND $= \otimes$, OR $= \oplus$, $\mathbf{1} = \mathbb{C}$); see that project's shard `AQM-70`. Statistics (Bose / Fermi / para) are deferred throughout.

3.2 The Three Primitives

Let particle *types* be (separable, complex) Hilbert spaces; a type's space is the state space of one particle of that type. Three primitives generate composite systems:

- **OR = direct sum $\oplus$.** "Either a particle of type 1 or of type 2" is $\mathcal{H}_1 \oplus \mathcal{H}_2$.
- **AND = tensor product $\otimes$.** "A particle of type 1 *and* a particle of type 2" is $\mathcal{H}_1 \otimes \mathcal{H}_2$.
- **vacuum = $\mathbb{C}$.** The no-particle system is the complex line.

Crucially, this grammar fixes *indefinite particle number*; exchange symmetry (statistics) is a *separate, later* step, realised as a quotient or subspace (CA-03, §"Symmetries come later"). Keeping the two apart is the whole point.

3.3 Two Monoidal Structures and Their Units

On the category Hilb of complex Hilbert spaces (with the obvious unitary coherence isomorphisms) there are two symmetric monoidal structures:

$$(\mathrm{Hilb},\ \oplus,\ \{0\}) \qquad \text{and} \qquad (\mathrm{Hilb},\ \otimes,\ \mathbb{C}),$$

with the two units *distinct*:

$$A \oplus \{0\} \ \cong\ A \qquad (\oplus\text{-unit is the } \textit{zero space } \{0\}),$$

$$A \otimes \mathbb{C} \ \cong\ A \qquad (\otimes\text{-unit is the } \textit{vacuum } \mathbb{C}).$$

The two are linked by a *distributivity* natural isomorphism, which on Hilbert spaces is *unitary*:

$$\delta :\ A \otimes (B \oplus C) \ \xrightarrow{\cong}\ (A \otimes B) \oplus (A \otimes C), \qquad A \otimes \{0\} \ \cong\ \{0\}.$$

Together $(\mathrm{Hilb}, \oplus, \{0\}, \otimes, \mathbb{C})$ is a symmetric *rig* (bimonoidal) category. Its coherence — that all diagrams built from associators, unitors, symmetries, and δ commute — is standard (Laplaza; Kelly) and is *not* re-proved here; downstream we use only the elementary, manifestly unitary associativity/unitor isomorphisms and the degree-0 identification $\mathbb{C} \otimes A \cong A$.

Vacuum = $\mathbb{C}$, not $\{0\}$. The vacuum is the $\otimes$-unit $\mathbb{C}$ (the empty tensor product, one particle's worth of "nothing"), a *one-dimensional* space carrying the unit vector Ω. It is *not* the $\oplus$-unit $\{0\}$ (the empty direct sum, the impossible system). Empty $\otimes$ is $\mathbb{C}$; empty $\oplus$ is $\{0\}$. Conflating them is the first pitfall this grammar guards against (cf. CONVENTIONS.md (a), which fixes the vacuum *object* as $\mathbb{C}$).

3.4 The Grammar and its Carrier

Fix a set $\{K_x\}_{x\in X}$ of single-particle type spaces (the *atoms*). Grammar expressions are

$$E ::= 0 \;\mid\; \mathbf{1} \;\mid\; K \;\mid\; \mathrm{AND}(E,F) \;\mid\; \mathrm{OR}(E,F),$$

where 0 is the zero atom, $\mathbf{1}$ is the vacuum atom ($\mathbf{1}=\mathbb{C}$), and K ranges over the atoms. The *carrier* $\mathcal{V}(E)$ is defined recursively:

$$\begin{gathered}\mathcal{V}(0)=\{0\}, \qquad\qquad \mathcal{V}(\mathbf{1})=\mathbb{C}, \qquad\qquad \mathcal{V}(K)=K,\\ \mathcal{V}(\mathrm{AND}(E,F))=\mathcal{V}(E)\otimes\mathcal{V}(F), \quad \mathcal{V}(\mathrm{OR}(E,F))=\mathcal{V}(E)\oplus\mathcal{V}(F).\end{gathered}$$

Thus AND is independent coexistence and OR is alternatives. Parenthesisation and order are part of the written expression; the rig coherence above guarantees that re-bracketing or re-ordering a *fixed* expression changes $\mathcal{V}(E)$ only by a canonical unitary isomorphism, so $\mathcal{V}$ is well defined up to canonical unitary isomorphism. (Morally $\mathcal{V}$ is the evaluation of the free symmetric rig category on $\{K_x\}$ into Hilb; we use this only as motivation, not as a load-bearing theorem.)

3.5 Particle-Number Grading

Declare each atom K to be *one* particle and $\mathbf{1}$ to be *zero* particles. The carrier then carries a particle-number grading $\mathcal{V}_n$:

$$\begin{gathered}\mathcal{V}_0(\mathbf{1})=\mathbb{C}, \quad \mathcal{V}_n(\mathbf{1})=\{0\}\ (n\neq 0), \qquad \mathcal{V}_1(K)=K, \quad \mathcal{V}_n(K)=\{0\}\ (n\neq 1), \quad \mathcal{V}_n(0)=\{0\},\\ \mathcal{V}_n(\mathrm{OR}(E,F))=\mathcal{V}_n(E)\oplus\mathcal{V}_n(F), \qquad \mathcal{V}_n(\mathrm{AND}(E,F))=\bigoplus_{i+j=n}\mathcal{V}_i(E)\otimes\mathcal{V}_j(F).\end{gathered}$$

For a *finite* expression the parse tree is finite, so only finitely many sectors are nonzero and $\mathcal{V}(E)=\bigoplus_{n\geq 0}\mathcal{V}_n(E)$ is a *finite* direct sum. The integer n is a grammar count of one-particle atoms; it is not a dynamics, a statistics rule, or (yet) an indefinite-particle space.

Worked atoms. A one-particle "menu" over a finite type set X is $P=\mathrm{OR}_{x\in X}K_x$, with $\mathcal{V}(P)=\bigoplus_{x\in X}K_x$ (the infinite, ℓ^2-completed menu is CA-03). The expression $\mathbf{1}\,\mathrm{OR}\,K$ has carrier $\mathbb{C}\oplus K$ — a "maybe" particle (either zero or one particle of type K), as in the sibling shard's $\mathbb{C}\oplus\mathbb{C}^d$. Tensoring N "maybe" atoms, $\mathrm{AND}_{j=1}^{N}(\mathbf{1}\,\mathrm{OR}\,K)$, has carrier $(\mathbb{C}\oplus K)^{\otimes N}$, whose grade-$n$ sector (by distributivity) is $\bigoplus_{|S|=n}\bigotimes_{x\in S}K$ over n-subsets of the N slots — the finite "occupation" picture.

3.6 Lamport Dependency Skeleton

$$\begin{aligned}
D1 &: \text{atoms } \{K_x\}_{x\in X} \text{ (single-particle Hilbert spaces); } \mathbf{1}=\mathbb{C} \text{ (vacuum)},\ 0=\{0\}.\\
D2 &: E ::= 0 \mid \mathbf{1} \mid K \mid \mathrm{AND}(E,F) \mid \mathrm{OR}(E,F).\\
D3 &: \mathcal{V}:\ \mathcal{V}(0)=\{0\},\ \mathcal{V}(\mathbf{1})=\mathbb{C},\ \mathcal{V}(\mathrm{AND})=\otimes,\ \mathcal{V}(\mathrm{OR})=\oplus.\\
D4 &: \mathcal{V}_0(\mathbf{1})=\mathbb{C},\ \mathcal{V}_1(K)=K;\ \mathcal{V}_n(\mathrm{OR})=\oplus,\ \mathcal{V}_n(\mathrm{AND})=\textstyle\bigoplus_{i+j=n}\mathcal{V}_i\otimes\mathcal{V}_j.\\
D5 &: (\mathrm{Hilb},\oplus,\{0\},\otimes,\mathbb{C}) \text{ a symmetric rig category, with}\\
&\quad\ \delta: A\otimes(B\oplus C)\xrightarrow{\cong}(A\otimes B)\oplus(A\otimes C) \text{ unitary and natural.}\\
T1 &: \text{(D5) coherence (Laplaza; Kelly)} \Rightarrow \mathcal{V} \text{ well defined up to canonical unitary iso.}\\
T2 &: \text{(D4) finite } E:\ \mathcal{V}(E)=\textstyle\bigoplus_{n\geq 0}\mathcal{V}_n(E) \text{ a finite sum (finite particle number).}
\end{aligned}$$

3.7 Scope and Deferral

Two extensions are deliberately out of this shard:

- **Indefinite particle number.** A genuine superposition over *unboundedly* many particle numbers has infinite support and does not live in the finite direct sum above; it requires the Hilbert (ℓ^2) completion. That is the content of CA-03, and is the new rigorous part of this pre-topic.
- **Statistics.** AND and OR choose no exchange symmetry. Bosonic / fermionic / para sectors arise later, by acting with the permutation group on tensor slots and passing to an invariant subspace or quotient (CA-03). This grammar only sets up the carrier those symmetries act on.

Pitfalls. (i) The $\oplus$-unit is $\{0\}$; the $\otimes$-unit (the vacuum) is $\mathbb{C}$ — distinct objects. (ii) Distributivity is an *isomorphism*, not an equality; all sector identities below hold up to canonical unitary iso. (iii) Everything here is *finite*; do not silently read the grading as an indefinite-particle space.

4 Indefinite Particle Number: the Full Fock Completion

4.1 Sources and Dependencies

This shard completes CA-02's finite AND/OR/vacuum grammar to a genuine *indefinite*-particle Hilbert space. It depends on CA-02 (the two monoidal structures, the units $\{0\}$ and $\mathbb{C}$, the carrier $\mathcal{V}$ and grading $\mathcal{V}_n$). The named Fock construction is anchored locally:

The sibling project's finite grammar (`AQM-70`) is recovered as the finite-truncation of the object built here.

4.2 Two Direct Sums — the Load-Bearing Distinction

Let $\{W_n\}_{n\geq 0}$ be a countable family of Hilbert spaces.

- The **algebraic direct sum** $\bigoplus_n^{\mathrm{alg}} W_n$ consists of finite-support sequences (w_n) ($w_n \in W_n$, $w_n = 0$ for all but finitely many n). It is a pre-Hilbert space, generally *incomplete*.
- The **Hilbert (ℓ^2) direct sum** $\widehat{\bigoplus}_n W_n$ consists of sequences with $\sum_n \|w_n\|^2_{W_n} < \infty$, with inner product $\langle (w_n), (v_n)\rangle = \sum_n \langle w_n, v_n\rangle_{W_n}$ (absolutely convergent by Cauchy–Schwarz). It is *complete*, hence a Hilbert space.

$\bigoplus_n^{\mathrm{alg}} W_n$ is a *dense* subspace of $\widehat{\bigoplus}_n W_n$, and the latter is its completion (standard functional analysis).

Why indefinite particle number forces $\widehat{\bigoplus}$. A state with nonzero amplitude in *unboundedly* many particle-number sectors (a genuine superposition over indefinitely many particles) has infinite support: it lies in $\widehat{\bigoplus}$ but not in $\bigoplus^{\mathrm{alg}}$. "Indefinite particle number" $\Leftrightarrow$ "not confined to finitely many sectors" $\Rightarrow$ the ℓ^2 completion is *required*. This is exactly what CA-02's finite grammar lacks.

4.3 Completed Tensor Powers

For Hilbert spaces H, K, the *completed* (Hilbert) tensor product $H \mathbin{\widehat{\otimes}} K$ is the completion of the algebraic tensor product under the inner product determined by $\langle a\otimes b,\, c\otimes d\rangle = \langle a, c\rangle_H \langle b, d\rangle_K$ (well defined and positive definite). For finite-dimensional factors $\widehat{\otimes} = \otimes$; the completion bites only for infinite-dimensional one-particle spaces. Define the *completed tensor powers*

$$H^{\widehat{\otimes} 0} := \mathbb{C} \ \ \text{(the vacuum sector)}, \qquad H^{\widehat{\otimes} n} := \underbrace{H \mathbin{\widehat{\otimes}} \cdots \mathbin{\widehat{\otimes}} H}_{n},$$

the bracketing immaterial up to the canonical unitary associativity isomorphism.

4.4 The Full Fock Space

Definition. For a one-particle Hilbert space H, the *full Fock space* is

$$\boxed{\mathfrak{F}(H) \ := \ \widehat{\bigoplus}_{n\geq 0} H^{\widehat{\otimes} n}}, \qquad H^{\widehat{\otimes} 0} = \mathbb{C},$$

i.e. $\psi = (\psi_n)_{n\geq 0}$ with $\psi_n \in H^{\widehat{\otimes} n}$ and $\|\psi\|^2 = \sum_n \|\psi_n\|^2 < \infty$. The unit vector $\Omega = (1, 0, 0, \dots)$ in the degree-0 sector $\mathbb{C}$ is the *vacuum*. This is the *distinguishable* Fock space (the author's term in the local source `1910.10619`); in operator algebra it is also called the *free* or *Boltzmann* Fock space (standard terminology, not anchored in a local source), namely the Hilbert completion of the tensor algebra $T(H) = \bigoplus_n H^{\otimes n}$. It is *not* the symmetric (bosonic) or antisymmetric (fermionic) Fock space; see "Exchange sectors come later" below.

Compiler status. In the compiler grammar this is not an additional primitive constructor. It is the Hilbert completion of the ordinary tensor-power expressions $\mathbf{1}, H, H^{\widehat{\otimes}2}, \ldots$: a countable OR over all finite AND powers. If a one-particle symmetry representation is present, CA-05 makes this same object a graded direct-sum representation.

Theorem (Fock is a Hilbert space). $\mathfrak{F}(H)$ is a Hilbert space; (i) the algebraic direct sum $\bigoplus_n^{\mathrm{alg}} H^{\widehat{\otimes}n}$ (the finite-particle vectors) is *dense*; (ii) distinct-n sectors are *mutually orthogonal*, $\langle\psi, \varphi\rangle = \sum_n \langle\psi_n, \varphi_n\rangle$, so $H^{\widehat{\otimes}m} \perp H^{\widehat{\otimes}n}$ for $m \neq n$; (iii) on the degree-0 sector the inner product is the standard one on $\mathbb{C}$ and Ω is a unit vector. *Proof.* An instance of the ℓ^2-direct-sum facts above with $W_n = H^{\widehat{\otimes}n}$: completeness, density of the algebraic sum, and summand orthogonality are exactly those statements. □

The $1/N!$ caveat (a convention trap). Schottenloher's inner product on $D = \bigoplus_N H_N$ weights sector N by $1/N!$; that factor is a *symmetric*-Fock normalisation correcting the $N!$ overcounting of symmetric tensors. The full Fock space above uses the *plain* ℓ^2 sum with *no* combinatorial weight. Do not copy the $1/N!$ into the distinguishable case.

4.5 The Grammar Generates the Indefinite-Particle Space

Let the one-particle menu over a countable type set X be $P = \mathrm{OR}_{x\in X} K_x$, with carrier $\mathcal{V}(P) = \widehat{\bigoplus}_{x\in X} K_x$ (the ℓ^2 direct sum; an ordinary finite direct sum when X is finite). The indefinite-particle space over this menu is $\mathfrak{F}(\mathcal{V}(P))$. Expanding each completed tensor power by distributivity (CA-02's δ, extended unitarily to the completions) gives the $(n, \text{type-word})$-graded decomposition

$$\mathcal{V}(P)^{\widehat{\otimes}n} \cong \widehat{\bigoplus}_{w\in X^n} K_w, \quad K_w := K_{x_1} \widehat{\otimes} \cdots \widehat{\otimes} K_{x_n}, \qquad \Longrightarrow \qquad \mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus}_{n\geq 0} \widehat{\bigoplus}_{w\in X^n} K_w.$$

Each of the n tensor slots independently chooses a type $x_i \in X$ (two slots may choose the same type; nothing is symmetrised). The grading is $\mathcal{V}_n = \widehat{\bigoplus}_{w\in X^n} K_w$ (n = particle number, $w \in X^n$ = ordered type-word) — the ℓ^2-completed, all-n analogue of CA-02's finite grade-n sector. The sibling project's finite expressions ($P_X^{\mathrm{AND}\, m}$, the truncation $T_{\leq M}$) are precisely the finite-support / finite-M restrictions of this object; that is the exact harmonisation: `AQM-70` is the $\bigoplus^{\mathrm{alg}}$/finite-M truncation of $\mathfrak{F}$.

4.6 Exchange Sectors Come Later

The full Fock space already carries symmetry representations when H does: internal symmetries act sectorwise by tensor powers (CA-05), and for each n the permutation group S_n acts unitarily on $H^{\widehat{\otimes}n}$ by permuting tensor slots. What is deferred is the *sector selection.* Bosonic and fermionic Fock spaces are the images of the per-sector orthogonal projectors

$$\Gamma_{\mathrm{s}}(H) = P_{\mathrm{sym}}\, \mathfrak{F}(H) \quad \text{(symmetric / Bose)}, \qquad \Gamma_{\mathrm{a}}(H) = P_{\mathrm{anti}}\, \mathfrak{F}(H) \quad \text{(antisymmetric / Fermi)},$$

with para-statistics the other Young-symmetry isotypic components. This is the "impose an equivalence relation under particle exchange" step. We define no projector and prove nothing here:

the construction of these symmetry quotients/subspaces, and the genuinely anyonic case (where standard Fock space is not the right object), are deferred to later shards.

4.7 Lamport Dependency Skeleton

$$\begin{aligned}
&D6: \bigoplus^{\mathrm{alg}}(\text{finite support}) \ \subset\ \widehat{\bigoplus}(\ell^2):\ \langle (w_n),(v_n)\rangle = \textstyle\sum_n \langle w_n, v_n\rangle.\\
&D7: \text{completed tensor power } H^{\widehat{\otimes} n},\ H^{\widehat{\otimes} 0} = \mathbb{C}\ (\text{vacuum}).\\
&D8: \mathfrak{F}(H) := \widehat{\bigoplus}_{n\geq 0} H^{\widehat{\otimes} n}\ (\text{full / distinguishable Fock});\ \Omega = (1,0,\dots).\\
&D9: P = \mathrm{OR}_{x\in X} K_x; \quad \mathfrak{F}(\mathcal{V}(P)) = \widehat{\bigoplus}_{n\geq 0} \mathcal{V}(P)^{\widehat{\otimes} n}.\\
&T3: (\text{D6,D7})\ \mathfrak{F}(H)\ \text{Hilbert};\ \bigoplus^{\mathrm{alg}}\ \text{dense};\ H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n}\ (m \neq n).\\
&T4: \text{indefinite particle no.} \Leftrightarrow \text{infinite support} \Rightarrow \widehat{\bigoplus}\ (\text{not } \bigoplus^{\mathrm{alg}}).\\
&T5: (\text{D8},\delta)\ \mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus}_{n\geq 0} \widehat{\bigoplus}_{w\in X^n} K_w\ ((n, \text{type-word})\ \text{grading}).\\
&T6: \Gamma_{\mathrm{s}} = P_{\mathrm{sym}}\mathfrak{F},\ \Gamma_{\mathrm{a}} = P_{\mathrm{anti}}\mathfrak{F}\ \text{deferred (statistics later)}.
\end{aligned}$$

4.8 Scope and Pitfalls

In scope: the indefinite-particle direction, i.e. countable OR (the Fock completion). **Out of scope:** countable AND (an *infinite tensor product*), which is analytically a different object (von Neumann's infinite tensor product, needing a stabilising reference sequence) — not developed here.

Pitfalls: (i) *algebraic* $\bigoplus^{\mathrm{alg}}$ (finite particle number, dense, incomplete) vs. *completed* $\widehat{\bigoplus}$ (indefinite, complete) — the distinction this shard exists to make. (ii) For infinite-dimensional one-particle H, the AND of CA-02 must mean the *completed* $\widehat{\otimes}$, not the algebraic $\otimes$. (iii) The $1/N!$ symmetric-Fock weight must not be imported into the distinguishable case. (iv) All sector identities hold on dense subspaces and extend by continuity; states are ℓ^2-summable and cross-sector inner products vanish.

5 A Hilbert-Space Compiler Contract

5.1 Status and Local Anchors

Status: Proposal / contract shard. This shard introduces no new mathematical classification theorem. It records a design target for the pre-work sequence around CA-02 and CA-03: a typed grammar of many-particle expressions should be compilable to Hilbert-space data and observable-algebra data with explicit evidence status.

Local anchors are CA-02 for the finite AND/OR/vacuum grammar, CA-03 for the Hilbert direct-sum Fock completion, and `CONVENTIONS.md` entries (f) and (g) for symmetry and sector semantics. The first executable check for this sequence is the finite averaging projector in `test/runtests.jl`, backed by `src/CftAnyons.jl`.

5.2 Mini North Star

Proposal: build a *Hilbert-space compiler*. The user supplies an expression in a typed grammar, together with the atom table and required policies; the compiler returns the Hilbert carrier, its observable algebra policy, its symmetry representation data, and any sector projection or quotient presentation that was requested.

The motivating analogy is "regular expressions for quantum mechanics": a small syntax has compositional semantics. The analogy is only a prompt. Here the syntax is typed and semantics-bearing; every constructor must say what it does to Hilbert carriers, particle gradings, observable algebras, symmetries, and evidence metadata.

5.3 Input Record

A compiled expression is not just a string. It has an input record:

$$\mathsf{Input} = (\mathsf{Atoms},\ E,\ \mathsf{Completion},\ \mathsf{ObsPolicy},\ \mathsf{Symmetry},\ \mathsf{SectorPolicy}).$$

Atoms. A table of atoms. At minimum, each atom names a Hilbert space K_x. Later entries may include a group representation, a particle degree, a source locator, or category data.

E. A grammar expression built from $0, \mathbf{1}, K_x, \mathrm{AND}, \mathrm{OR}$, and later sector constructors. Fock-style all-n notation is treated as a derived completion of tensor powers, not as a primitive grammar constructor.

$\mathsf{Completion}$. Whether $\oplus, \otimes$ are finite Hilbert operations, Hilbert completions $\widehat{\bigoplus}, \widehat{\otimes}$, or a finite cutoff. CA-03 fixes the first infinite completion of the derived all-tensor-power construction used here.

$\mathsf{ObsPolicy}$. The observable-algebra choice. There is no silent default: full $\mathcal{B}(H)$, block-preserving, tensor-local, symmetry-invariant, and compressed sector algebras are different outputs.

$\mathsf{Symmetry}$. Optional unitary or projective-unitary representation data, handled according to CONVENTIONS.md (f).

$\mathsf{SectorPolicy}$. Optional projection, subspace, or quotient data; see CONVENTIONS.md (g).

5.4 Output Record

The intended output is a typed record, not only a Hilbert-space dimension:

$$\mathsf{Compile}(E) = (H_E,\ \{H_{E,n}\}_{n\geq 0},\ \mathcal{A}_E,\ U_E,\ P_E,\ Q_E,\ \mathsf{Evidence}_E).$$

H_E. The Hilbert carrier. CA-02 and CA-03 define the first carrier semantics.

$\{H_{E,n}\}$. The particle-number grading when the atom table declares particle degrees.

$\mathcal{A}_E$. The selected observable algebra. This slot is mandatory once any observable claim is made.

U_E. The propagated symmetry representation, if symmetry data were supplied.

P_E. The selected sector projection, if a sector constructor was used.

Q_E. The quotient presentation, if one was requested and its closed relation subspace was named.

$\mathsf{Evidence}_E$. Source-local, checked, proposal, question, or rejected evidence tags for the pieces above.

5.5 Observable-Algebra Policies

The carrier alone does not determine the observable algebra relevant to a model. For example, $H \oplus K$ supports at least:

$$\mathcal{B}(H \oplus K), \qquad \mathcal{B}(H) \oplus \mathcal{B}(K), \qquad \text{off-diagonal creation/annihilation blocks between } H \text{ and } K.$$

All three are legitimate in different contexts; none is forced by the word "OR" alone. Similarly, $H \otimes K$ supports the full algebra $\mathcal{B}(H \otimes K)$ and the tensor-local subalgebra generated by $\mathcal{B}(H) \otimes I$ and $I \otimes \mathcal{B}(K)$.

Compiler rule: no report statement may infer an observable algebra merely from a carrier constructor. The statement must name the observable policy, or remain a carrier-only statement.

5.6 Lamport Dependency Skeleton

$$\begin{aligned}
D10 &: \mathsf{Input} = (\mathsf{Atoms}, E, \mathsf{Completion}, \mathsf{ObsPolicy}, \mathsf{Symmetry}, \mathsf{SectorPolicy}).\\
D11 &: \mathsf{Compile}(E) = (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Evidence}_E).\\
D12 &: \mathsf{ObsPolicy} \in \{\text{carrier-only}, \mathcal{B}(H), \text{block}, \text{tensor-local}, \text{invariant}, \text{compressed}\}.\\
T7 &: \text{A carrier claim needs } H_E\text{; an observable claim also needs } \mathcal{A}_E.\\
T8 &: \text{The compiler analogy is a proposal, not a theorem of canonical quantization.}
\end{aligned}$$

5.7 Boundary

This contract does not yet compile fusion-category labels, anyonic braid statistics, Hamiltonians, continuum limits, or CFT data. It only fixes the pre-work target: future shards should be able to say exactly which input record they compile and which output slots they have actually justified.

6 Symmetry-Decorated Many-Particle Grammar

6.1 Sources and Dependencies

This shard extends CA-02 and CA-03 by decorating atoms with symmetry data. It uses CONVENTIONS.md (f).

6.2 Decorated Atoms

Fix a topological group G. A symmetry-decorated atom is a pair

$$(K_x, U_x), \qquad U_x : G \to U(K_x),$$

where K_x is the atomic Hilbert space and U_x is a unitary representation. If the intended symmetry is projective, the pre-work convention is to choose a central extension

$$1 \to U(1) \to E \to G \to 1$$

and use an honest unitary representation $U_x : E \to U(K_x)$. The expression is then compiled over E, with a note that the induced physical symmetry is projective over G.

The vacuum atom is $(\mathbb{C}, \mathbf{1})$, where $\mathbf{1}(g)z = z$. The zero atom carries the unique representation on $\{0\}$.

6.3 Propagation Through OR and AND

If E, F have compiled carriers and representations (H_E, U_E) and (H_F, U_F) for the same group or chosen central extension, define

$$H_{\mathrm{OR}(E,F)} = H_E \oplus H_F, \qquad U_{\mathrm{OR}(E,F)}(g) = U_E(g) \oplus U_F(g),$$

and

$$H_{\mathrm{AND}(E,F)} = H_E \mathbin{\widehat{\otimes}} H_F, \qquad U_{\mathrm{AND}(E,F)}(g) = U_E(g) \mathbin{\widehat{\otimes}} U_F(g).$$

For finite-dimensional factors, $\widehat{\otimes} = \otimes$. In infinite-dimensional Hilbert spaces, the completed tensor product is the one already required by CA-03.

These formulas are the Hilbert-space version of the monoidal structure on $\mathrm{Rep}^{\dagger}(G)$: direct sums are block actions and tensor products are diagonal actions. If two atoms use different groups, different central extensions, or incompatible projective multipliers, the compiler must stop and ask for a common symmetry object before forming a single U_E.

6.4 The Derived Full-Fock Representation

For an ordinary unitary representation $U : G \to U(H)$, the full-Fock object is itself a graded representation:

$$\mathfrak{F}(U)(g) = \bigoplus_{n \geq 0} U(g)^{\widehat{\otimes} n} \quad \text{on} \quad \mathfrak{F}(H) = \widehat{\bigoplus}_{n \geq 0} H^{\widehat{\otimes} n},$$

with $U(g)^{\widehat{\otimes} 0} = \mathrm{id}_{\mathbb{C}}$. This preserves the particle-number grading. This is not a new grammar constructor: it is the direct sum of the tensor-power representations already produced by OR and AND, completed as in CA-03. In this pre-work shard, no Shale–Stinespring-type implementability theorem is claimed for general canonical field algebras. Implementing a one-particle unitary on a chosen CAR/CCR observable algebra is a separate field-algebra question.

6.5 Intertwiners and Symmetry-Preserving Compilation

A morphism between decorated atoms or compiled expressions is symmetry-preserving only when it is an intertwiner:

$$TU_E(g) = U_F(g)T \quad \text{for all } g.$$

This matters for compiler optimisations. Rebracketing and distributivity isomorphisms used by CA-02 are acceptable in the decorated grammar only when they also intertwine the propagated actions. The standard Hilbert-space associators, unitors, symmetries, and distributivity maps do so for diagonal direct-sum and tensor-product actions.

6.6 Lamport Dependency Skeleton

$D13$: decorated atom $(K_x, U_x),\ U_x : G \to U(K_x)$.
$D14$: projective symmetry over G is compiled via $U_x : E \to U(K_x)$ for a named central extension E.
$D15$: $U_{E\,\mathrm{OR}\,F} = U_E \oplus U_F, \quad U_{E\,\mathrm{AND}\,F} = U_E \mathbin{\widehat{\otimes}} U_F$.
$D16$: $\mathfrak{F}(U) = \widehat{\bigoplus}_{n\geq 0} U^{\widehat{\otimes} n}, \quad U^{\widehat{\otimes} 0} = 1_{\mathbb{C}}$.
$T9$: The propagated symmetry preserves the CA-02/CA-03 particle grading.
$T10$: A single compiled U_E requires a common group or central extension.

6.7 Boundary

This shard does not yet impose invariance, select charge sectors, form quotient spaces, or define symmetry-invariant observable algebras. Those are sector operations, not mere symmetry propagation, and are treated in CA-06.

7 Sectors, Projections, Subspaces, and Quotients

7.1 Sources and Dependencies

This shard depends on CA-04 and CA-05 and fixes the first sector semantics for the Hilbert-space compiler. It uses CONVENTIONS.md (g).

7.2 The Triangle

Let H be a Hilbert space. The compiler treats a sector first as an orthogonal projection

$$P = P^* = P^2 \in \mathcal{B}(H).$$

It may then expose either of the equivalent Hilbert-space presentations

$$H_P := \operatorname{Ran}(P) \subset H, \qquad H/\ker(P) = H/\operatorname{Ran}(1-P).$$

The map

$$[v] \in H/\ker(P) \longmapsto Pv \in \operatorname{Ran}(P)$$

is a well-defined unitary isomorphism when the quotient has the Hilbert quotient norm. Thus the safe slogan is not "symmetry equals quotient" but:

$$\text{projection} \iff \text{closed subspace} \iff \text{quotient by the named orthogonal complement.}$$

7.3 Finite-Group Invariants

Let G be finite and $U : G \to U(H)$ a unitary representation. Define

$$P_G = \frac{1}{|G|} \sum_{g \in G} U(g).$$

Then P_G is the orthogonal projection onto the invariant subspace

$$H^G = \{v \in H : U(g)v = v \ \forall g \in G\}.$$

Proof. Idempotence follows by reindexing:

$$P_G^2 = \frac{1}{|G|^2} \sum_{g,h} U(gh) = \frac{1}{|G|} \sum_{k} U(k) = P_G.$$

Self-adjointness follows from unitarity and $g \mapsto g^{-1}$:

$$P_G^* = \frac{1}{|G|} \sum_{g} U(g)^* = \frac{1}{|G|} \sum_{g} U(g^{-1}) = P_G.$$

For every k, $U(k)P_G = P_G$, so $\operatorname{Ran}(P_G) \subset H^G$. If $v \in H^G$, then $P_G v = v$, so $H^G \subset \operatorname{Ran}(P_G)$. $\square$

7.4 Coinvariant Quotient

For the same finite action, let

$$R_G = \operatorname{span}\{U(g)v - v : g \in G,\ v \in H\}.$$

Then $R_G \subseteq \ker(P_G)$ because $P_G U(g) = P_G$. Conversely, if $w \in \ker(P_G)$, then

$$w = \frac{1}{|G|} \sum_{g \in G} \bigl(w - U(g)w\bigr) \in R_G.$$

Therefore $R_G = \ker(P_G) = (H^G)^\perp$, and the coinvariant quotient

$$H_G := H/R_G$$

is canonically unitarily isomorphic to the invariant subspace H^G. For compact non-finite groups the analogous statement requires a Haar-integral and analytic hypotheses; this shard records only the finite checked case.

7.5 Observable-Algebra Distinctions

There are three different symmetry-related algebras that the compiler must not identify silently:

$$\mathcal{B}(H), \qquad \mathcal{B}(H)^G := \{A : U(g)AU(g)^* = A \ \forall g\}, \qquad P_G\mathcal{B}(H)P_G \cong \mathcal{B}(H^G).$$

The first is the full algebra before imposing symmetry. The second is the algebra of invariant observables on the original carrier. The third is the compressed algebra acting on the invariant sector. A model may choose one of these, but the choice is an observable policy in the sense of CA-04, not a consequence of the word "symmetry" alone.

7.6 Checked Example

The current Julia check uses $G = C_2$ acting on $\mathbb{C}^2$ by the swap matrix S. The averaging projection is

$$P = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

The test checks $P^* = P = P^2$, checks $I - P$ is the complementary orthogonal projection, verifies $P(I - P) = 0$, and verifies that $(1, 1)$ is fixed while $(1, -1)$ is killed. This is the first executable invariant for the sector compiler.

7.7 Lamport Dependency Skeleton

$D17$: sector projection $P = P^* = P^2$; $\quad H_P = \operatorname{Ran}(P)$.
$D18$: $H/\ker(P) \xrightarrow{[v]\mapsto Pv} \operatorname{Ran}(P)$.
$D19$: $P_G = |G|^{-1} \sum_g U(g) \quad (G \text{ finite})$.
$D20$: $R_G = \operatorname{span}\{U(g)v - v\}$.
$L1$: $P_G = P_G^* = P_G^2, \quad \operatorname{Ran}(P_G) = H^G$.
$L2$: $R_G = \ker(P_G) = (H^G)^\perp, \quad H/R_G \cong H^G$.
$T11$: Projection, subspace, and quotient are equivalent only after the relation subspace is named.
$T12$: $\mathcal{B}(H)^G$ and $P_G\mathcal{B}(H)P_G$ are different observable policies.

7.8 Boundary

The finite averaging theorem is not a statement about spontaneous symmetry breaking, gauge constraints, BRST cohomology, braid statistics, or superselection sectors in a continuum CFT. It is the first Hilbert-space sector rule needed by the compiler.

8 Exchange Symmetry and Statistics as Sector Selection

8.1 Sources and Dependencies

This shard discharges part of the deferral in CA-03: ordinary Bose/Fermi statistics are treated as choices of irreducible exchange-symmetry sectors on the full distinguishable Fock carrier. It depends on CA-05 for internal symmetry propagation and CA-06 for projection/subspace/quotient semantics.

8.2 Two Symmetry Layers

There are two distinct actions on many-particle carriers:

$$\text{internal symmetry } G, \qquad \text{exchange symmetry } S_n.$$

If $U : G \to U(H)$ is an internal one-particle representation, then on the n-particle distinguishable sector $H^{\widehat{\otimes} n}$ it acts diagonally:

$$U_n(g) = U(g)^{\widehat{\otimes} n}.$$

The permutation group S_n acts by permuting tensor slots:

$$\Pi_n(\sigma)(v_1 \otimes \cdots \otimes v_n) = v_{\sigma^{-1}(1)} \otimes \cdots \otimes v_{\sigma^{-1}(n)}.$$

The two actions commute:

$$\Pi_n(\sigma)U_n(g) = U_n(g)\Pi_n(\sigma).$$

Thus internal charge sectors and exchange-statistics sectors can be imposed in either order in the ordinary tensor-slot setting.

8.3 Irrep-Labeled Exchange Projectors

The representation $\Pi_n : S_n \to U(H^{\widehat{\otimes} n})$ is the primary datum. Sector projections are supplied by irreducible representation data of S_n, not by specialised grammar constructors. For a one-dimensional unitary character $\chi : S_n \to U(1)$, CA-06 applied to the twisted representation $\overline{\chi}\,\Pi_n$ gives

$$P_{\chi,n} = \frac{1}{n!} \sum_{\sigma \in S_n} \overline{\chi(\sigma)}\,\Pi_n(\sigma),$$

whose range is the χ-sector

$$\{w \in H^{\widehat{\otimes} n} : \Pi_n(\sigma)w = \chi(\sigma)w \ \forall \sigma \in S_n\}.$$

For the trivial character 1, this is the symmetric projector:

$$P_{\mathrm{s},n} = \frac{1}{n!} \sum_{\sigma \in S_n} \Pi_n(\sigma),$$

and for the sign character sgn, this is the antisymmetric projector:

$$P_{\mathrm{a},n} = \frac{1}{n!} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma)\Pi_n(\sigma).$$

Their ranges are the symmetric and antisymmetric tensor powers:

$$\mathrm{Ran}(P_{\mathrm{s},n}) = H^{\widehat{\otimes}_s n}, \qquad \mathrm{Ran}(P_{\mathrm{a},n}) = \bigwedge^n H.$$

The corresponding Fock sectors are

$$\Gamma_{\mathrm{s}}(H) = \widehat{\bigoplus}_{n\geq 0} \mathrm{Ran}(P_{\mathrm{s},n}), \qquad \Gamma_{\mathrm{a}}(H) = \widehat{\bigoplus}_{n\geq 0} \mathrm{Ran}(P_{\mathrm{a},n}).$$

Thus "Bose" and "Fermi" name the trivial- and sign-irrep choices for ordinary exchange symmetry. The compiler should record the selected irrep/character and the resulting projection, not a special Bose/Fermi syntax node.

8.4 Quotient Presentations

For a one-dimensional character χ, the sector can also be presented as a quotient of the full tensor algebra by the closed linear span of relations

$$\Pi_n(\sigma)v - \chi(\sigma)v.$$

The trivial character recovers symmetric relations; the sign character recovers antisymmetric signed relations. In the compiler, these quotient presentations are secondary: the primary sector object is the orthogonal projector, because it fixes the inherited Hilbert structure and the compressed observable algebra.

8.5 Para and Anyonic Boundaries

Higher-dimensional irreducible S_n-sectors require central idempotents or matrix-block projectors for the group algebra of S_n. This is exactly the same representation/irrep principle, but this shard records it as a proposal rather than a completed construction; a later shard must source the convention and projector normalization before using it.

Genuinely anyonic exchange is not an S_n-projection problem on ordinary tensor slots. It involves braid-group or categorical data and, in this project, must be handled together with the fusion-category conventions for associators, braiding, and F/R-symbols. No braid-statistics projector is defined here.

8.6 Lamport Dependency Skeleton

$$\begin{aligned}
D21 &: U_n(g) = U(g)^{\widehat{\otimes} n} \quad \text{(internal symmetry on } H^{\widehat{\otimes} n}\text{)}.\\
D22 &: \Pi_n : S_n \to U(H^{\widehat{\otimes} n}) \quad \text{permutes tensor slots.}\\
D23 &: P_{\chi,n} = n!^{-1} \textstyle\sum_\sigma \overline{\chi(\sigma)}\Pi_n(\sigma) \quad (\chi \text{ a one-dimensional } S_n \text{ character}).\\
L3 &: \Pi_n(\sigma)U_n(g) = U_n(g)\Pi_n(\sigma).\\
L4 &: P_{\chi,n} \text{ is an orthogonal projection by CA-06 finite averaging.}\\
T13 &: \Gamma_{\mathrm{s}}, \Gamma_{\mathrm{a}} \text{ are the trivial/sign irrep sectors inside full Fock.}\\
T14 &: \text{Anyonic exchange remains deferred to categorical braid data.}
\end{aligned}$$

8.7 Compiler Consequence

The compiler now has its first nontrivial sector family: given the exchange representation $\Pi_n : S_n \to U(H^{\widehat{\otimes} n})$, a selected irrep character χ supplies $P_{\chi,n}$, its range, its quotient relations, and the chosen compressed observable algebra. There is no primitive Bose, Fermi, or Fock grammar node in this account; Fock is the derived graded representation of CA-05, and Bose/Fermi are names for particular irrep choices of exchange symmetry.

9 Scoping the Hilbert-Space Compiler

9.1 Status, Sources, and Dependencies

Status: specification shard. This shard turns CA-04's contract into a first partial compiler specification. Its scope is the pre-work grammar of CA-02–CA-07: finite Hilbert carriers, derived all-tensor-power completions, unitary/projective symmetries via central extensions, and irrep/projection sectors.

9.2 Compile Judgement

The compiler is a *partial* map. A successful judgement has the form

$$\Gamma; \Pi \vdash E \Downarrow (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Ev}_E),$$

where Γ is the atom table and Π is the policy table. A failed judgement has the form

$$\Gamma; \Pi \vdash E \Uparrow \mathsf{reason}.$$

The failure result is part of the specification: incompatible symmetry data, missing observable policy, or an ill-typed sector constructor must stop compilation.

The atom table entry for x has at least a Hilbert space K_x, a degree $d_x \in \mathbb{N}$ (usually 1), evidence metadata, and optionally symmetry data $U_x : G \to U(K_x)$ or a named projective central extension as in CA-05.

9.3 Surface Language for the First Compiler

The first surface language is deliberately small:

$$E ::= 0 \mid \mathbf{1} \mid x \mid \mathrm{OR}(E, F) \mid \mathrm{AND}(E, F) \mid \mathrm{Sector}_{G,\chi}(E).$$

Here G acts on the compiled carrier and χ is a one-dimensional unitary character; invariants are the case $\chi = 1$. Fock, finite truncation, symmetric sectors, and antisymmetric sectors are derived from OR/AND, representation propagation, and the generic sector operation.

For a one-particle atom x, define tensor-power expressions recursively by

$$x^{[0]} := \mathbf{1}, \qquad x^{[m+1]} := \mathrm{AND}(x^{[m]}, x).$$

Then

$$\mathsf{T}_{\leq M}(x) := \mathrm{OR}_{m=0}^{M}\, x^{[m]}, \qquad \mathsf{T}_{\infty}(x) := \text{Hilbert completion of } \mathrm{OR}_{m\geq 0}\, x^{[m]}.$$

$\mathsf{T}_{\infty}(x)$ is the usual full/distinguishable Fock space $\widehat{\bigoplus}_{m\geq 0} K_x^{\widehat{\otimes} m}$, but as a derived graded representation.

9.4 Constructor Rules

The carrier, grading, and symmetry rules are recursive:

$$H_0 = \{0\}, \quad H_{\mathbf{1}} = \mathbb{C}, \quad H_x = K_x,$$

$$H_{\mathrm{OR}(E,F)} = H_E \oplus H_F, \qquad H_{\mathrm{AND}(E,F)} = H_E \mathbin{\widehat{\otimes}} H_F.$$

The grading rules are CA-02's:

$$H_{\mathrm{OR}(E,F),n} = H_{E,n} \oplus H_{F,n}, \qquad H_{\mathrm{AND}(E,F),n} = \bigoplus_{i+j=n} H_{E,i} \mathbin{\widehat{\otimes}} H_{F,j}.$$

If U_E, U_F are representations of the same group or named central extension, the symmetry rules are

$$U_{\mathrm{OR}}(g) = U_E(g) \oplus U_F(g), \qquad U_{\mathrm{AND}}(g) = U_E(g) \mathbin{\widehat{\otimes}} U_F(g).$$

If the symmetry objects are not the same, the carrier may still compile, but the symmetry output is rejected unless the requested output was explicitly carrier-only.

For the derived tensor-power closures,

$$H_{\mathsf{T}_\infty(x)} = \widehat{\bigoplus}_{m\geq 0} K_x^{\widehat{\otimes} m}, \qquad H_{\mathsf{T}_{\leq M}(x)} = \bigoplus_{m=0}^{M} K_x^{\otimes m}.$$

For finite examples the compiler may report dimensions:

$$\begin{aligned}
\dim \mathrm{OR} &= \dim H_E + \dim H_F, \\
\dim \mathrm{AND} &= (\dim H_E)(\dim H_F), \\
\dim \mathsf{T}_{\leq M}(x) &= \sum_{m=0}^{M} (\dim K_x)^m.
\end{aligned}$$

These follow from choosing orthonormal bases: disjoint union gives direct-sum bases, ordered pairs give tensor-product bases, and truncation is a finite sum.

Sector constructors return projection witnesses supplied by representation data. For finite G, a unitary character $\chi : G \to U(1)$, and representation $U_E : G \to U(H_E)$, set

$$P_\chi = |G|^{-1} \sum_{g\in G} \overline{\chi(g)}\, U_E(g), \qquad H_{\mathrm{Sector}_{G,\chi}(E)} = \mathrm{Ran}(P_\chi).$$

The quotient relation is generated by $U_E(g)v - \chi(g)v$. The invariant sector is $\chi = 1$. For exchange sectors on $K^{\widehat{\otimes} n}$, $G = S_n$ and $U_E = \Pi_n$; the trivial and sign characters give

$$P_{\mathrm{s},n} = n!^{-1} \sum_{\sigma\in S_n} \Pi_n(\sigma), \qquad P_{\mathrm{a},n} = n!^{-1} \sum_{\sigma\in S_n} \mathrm{sgn}(\sigma)\Pi_n(\sigma).$$

The observable algebra is never inferred from these formulas. It is the policy slot Π_{obs}: carrier-only, full, block, tensor-local, invariant, or compressed.

9.5 Example 1: A Maybe Qubit

Let $Q = \mathbb{C}^2$ with basis e_0, e_1. Compile

$$E_1 = \mathrm{OR}(\mathbf{1}, Q).$$

The rule gives

$$H_{E_1} = \mathbb{C} \oplus \mathbb{C}^2, \qquad H_{E_1,0} = \mathbb{C}, \quad H_{E_1,1} = \mathbb{C}^2, \quad \dim H_{E_1} = 3.$$

With full observable policy, $\mathcal{A}_{E_1} = \mathcal{B}(\mathbb{C} \oplus \mathbb{C}^2)$. With block policy,

$$\mathcal{A}_{E_1}^{\mathrm{block}} = \mathcal{B}(\mathbb{C}) \oplus \mathcal{B}(\mathbb{C}^2).$$

Thus the same carrier has two different compiled observable outputs.

9.6 Example 2: Two Distinguishable Qubits

Compile $E_2 = \mathrm{AND}(Q, Q)$. The carrier is

$$H_{E_2} = Q \otimes Q, \qquad \{e_0 \otimes e_0, e_0 \otimes e_1, e_1 \otimes e_0, e_1 \otimes e_1\} \text{ is a basis,}$$

so $\dim H_{E_2} = 4$, all in degree 2. The tensor-local policy is the subalgebra generated by

$$\mathcal{B}(Q) \otimes I, \qquad I \otimes \mathcal{B}(Q),$$

whereas the full policy is $\mathcal{B}(Q \otimes Q)$. The compiler records which one was requested; it does not identify them by syntax alone.

9.7 Example 3: Derived Tensor-Power Closure

Compile the macro-expanded expression

$$E_3 = \mathsf{T}_{\leq 2}(Q) = \mathrm{OR}(\mathbf{1}, \mathrm{OR}(Q, \mathrm{AND}(Q, Q))).$$

The output carrier is

$$H_{E_3} = \mathbb{C} \oplus Q \oplus (Q \otimes Q),$$

with sector dimensions

$$\dim H_0 = 1, \qquad \dim H_1 = 2, \qquad \dim H_2 = 4, \qquad \dim H_{E_3} = 1 + 2 + 4 = 7.$$

If $U : G \to U(Q)$ is supplied, symmetry propagation gives

$$U_{E_3}(g) = 1_{\mathbb{C}} \oplus U(g) \oplus (U(g) \otimes U(g)).$$

No Bose/Fermi relation has been imposed yet; this is still the distinguishable graded representation.

9.8 Example 4: Two-Qubit Exchange-Irrep Sectors

Let S be the swap on $Q \otimes Q$. Since

$$S(e_0e_0) = e_0e_0, \quad S(e_1e_1) = e_1e_1, \quad S(e_0e_1) = e_1e_0, \quad S(e_1e_0) = e_0e_1,$$

where e_ie_j abbreviates $e_i \otimes e_j$, the exchange representation of S_2 has two one-dimensional irrep characters, trivial and sign, supplying projectors with ranges:

$$P_{\mathrm{s}} = \frac{1}{2}(I + S), \qquad P_{\mathrm{a}} = \frac{1}{2}(I - S)$$

$$\mathrm{Ran}(P_{\mathrm{s}}) = \mathrm{span}\{e_0e_0,\ (e_0e_1 + e_1e_0)/\sqrt{2},\ e_1e_1\},$$
$$\mathrm{Ran}(P_{\mathrm{a}}) = \mathrm{span}\{(e_0e_1 - e_1e_0)/\sqrt{2}\}.$$

Thus the compiler reports dimensions 3 and 1 for the trivial- and sign-irrep sectors. The Julia test suite checks the projection, orthogonality, trace/dimension, and basis-vector claims for this example.

9.9 Example 5: A Finite Invariant Quotient

Let $C_2 = \{1, s\}$ act on $H = \mathbb{C}^2$ by the swap matrix. Then

$$P_{C_2} = \frac{1}{2}(I + S) = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}.$$

Its range is $\mathrm{span}\{(1,1)\}$, and its kernel is $\mathrm{span}\{(1,-1)\}$. The quotient presentation is therefore

$$\mathbb{C}^2/\,\mathrm{span}\{(1,-1)\} \cong \mathrm{span}\{(1,1)\}.$$

This is $\mathrm{Sector}_{C_2,1}(H)$, the trivial-irrep sector of the C_2 representation. The invariant observable algebra on the original carrier consists of the operators commuting with S, while the compressed algebra on the invariant line is one-dimensional. These are different observable policies even in this small example.

9.10 Example 6: A Required Compile Error

Let A, B be atoms with projective symmetries over the same physical group G, but represented in the input by unrelated central extensions $E_A \to G$ and $E_B \to G$. The carrier of $\mathrm{OR}(A, B)$ is still $K_A \oplus K_B$. However, a request for one compiled symmetry representation must fail:

$$\Gamma; \Pi \vdash \mathrm{OR}(A, B) \Uparrow \text{“no common symmetry group or central extension specified”}.$$

This is not a missing feature; it is the strict semantics required by CA-05 and CONVENTIONS.md (f).

9.11 Lamport Dependency Skeleton

$D24$: $\Gamma; \Pi \vdash E \Downarrow (H_E, \{H_{E,n}\}, \mathcal{A}_E, U_E, P_E, Q_E, \mathsf{Ev}_E)$ or $\Gamma; \Pi \vdash E \Uparrow \mathsf{reason}$.
$D25$: $E ::= 0 \mid \mathbf{1} \mid x \mid \mathrm{OR} \mid \mathrm{AND} \mid \mathrm{Sector}_{G,\chi}$.
$D26$: $\mathsf{T}_{\le M}(x) = \mathrm{OR}_{m=0}^{M} x^{[m]}, \quad H_{\mathsf{T}_{\le M}} = \bigoplus_{m=0}^{M} K_x^{\otimes m}$.
$D27$: $P_\chi = |G|^{-1} \sum_g \overline{\chi(g)} U_E(g), \quad H_{\mathrm{Sector}_{G,\chi}} = \mathrm{Ran}(P_\chi)$.
$T15$: Finite dimensions obey sum, product, and $\sum_{m=0}^{M} d^m$ by explicit basis constructions.
$T16$: Irrep/character choices supply projection/subspace/quotient witnesses.
$T17$: Compiler failure is a specified outcome when requested structure is ill typed or unsupported.

9.12 Acceptance Criteria and Boundary

Future implementation work must preserve these acceptance criteria: every constructor fills typed output slots; every nontrivial mathematical output has an evidence tag; every quotient names its relation subspace; every observable claim names its observable policy; and every executable example asserts an invariant, not merely successful execution.

This compiler does not yet parse fusion-category simples, anyonic braid data, Hamiltonians, lattice limits, conformal nets, or VOA data. Those enter only after their source conventions and invariants are fixed. CA-08 is the finite Hilbert-space kernel that later shards must extend without weakening the output record, the evidence discipline, or the failure semantics.

10 Fibonacci Kinematics as a Fusion-Rule Sector

10.1 Status, Sources, and Dependencies

Status: checked kinematic shard. This shard confirms the side conjecture: before braiding, Hamiltonians, or continuum limits, the Fibonacci fusion-path Hilbert spaces are a sector construction from one physical atom τ. The vacuum $\mathbf{1}$ is still present as the tensor unit and as a charge label; it is not a second physical particle atom.

Dependencies: CA-02 for AND/OR/vacuum, CA-06 for projection/subspace semantics, CA-08 for compiler output slots, and CONVENTIONS.md (c) for the left-associated kinematic path basis.

10.2 One Physical Atom, Two Charge Labels

The physical input word is n copies of one anyon type:

$$\tau, \tau, \ldots, \tau.$$

The charge labels used to describe intermediate fusion outcomes are

$$L = \{\mathbf{1}, \tau\}.$$

Thus the proposed slogan is correct only with this distinction:

$$\text{one physical atom } \tau \quad + \quad \text{charge labels } \{\mathbf{1}, \tau\} \quad + \quad \text{fusion-rule sector selection.}$$

The vacuum $\mathbf{1}$ is required because it is both the tensor unit and a possible total charge. It is not an additional non-vacuum particle species.

10.3 Ambient Path Space and Admissibility Projector

Use the left-associated path basis fixed in CONVENTIONS.md (c). For n physical τ-atoms and total charge $c \in \{\mathbf{1}, \tau\}$, define the ambient word space

$$W_{n,c} = \ell^2\{(x_0, \ldots, x_n) : x_i \in \{\mathbf{1}, \tau\},\ x_0 = \mathbf{1},\ x_n = c\}.$$

The basis vector $|x_0, \ldots, x_n\rangle$ records the cumulative charge after successively fusing the first i copies of τ.

The Fibonacci fusion rules needed here are

$$\mathbf{1} \otimes \tau = \tau, \qquad \tau \otimes \mathbf{1} = \tau, \qquad \tau \otimes \tau = \mathbf{1} \oplus \tau.$$

Let $N_{ab}^c \in \{0, 1\}$ be these multiplicity-free fusion coefficients. Define the diagonal operator

$$P_{n,c}|x_0, \ldots, x_n\rangle = \left(\prod_{i=1}^{n} N_{x_{i-1},\tau}^{x_i}\right) |x_0, \ldots, x_n\rangle.$$

Since every diagonal entry is 0 or 1, $P_{n,c} = P_{n,c}^* = P_{n,c}^2$. The kinematic Fibonacci Hilbert space is therefore the sector

$$H_{n,c}^{\mathrm{Fib}} := \mathrm{Ran}(P_{n,c}) \subseteq W_{n,c}.$$

This is exactly the CA-06 projection/subspace semantics, but the projector is not supplied by a group character. It is supplied by categorical fusion-rule admissibility.

10.4 Dimension Recurrence

Let

$$a_n = \dim H^{\mathrm{Fib}}_{n,\mathbf{1}}, \qquad b_n = \dim H^{\mathrm{Fib}}_{n,\tau}.$$

At $n = 0$, no physical τ-atom has been fused, so

$$a_0 = 1, \qquad b_0 = 0.$$

Adding one more τ gives the transition equations

$$a_{n+1} = b_n, \qquad b_{n+1} = a_n + b_n.$$

The first equation says total charge $\mathbf{1}$ can only come from $\tau \otimes \tau \to \mathbf{1}$. The second says total charge τ can come from $\mathbf{1} \otimes \tau \to \tau$ or from $\tau \otimes \tau \to \tau$. Hence

$$b_{n+1} = b_n + b_{n-1}, \qquad b_0 = 0, \quad b_1 = 1.$$

Thus the total-τ sector dimensions are Fibonacci numbers:

$$0, 1, 1, 2, 3, 5, 8, \ldots.$$

The Julia test suite checks the pair sequence

$$(a_n, b_n) = (1, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 5), (5, 8), (8, 13)$$

for $0 \le n \le 7$, including the recurrence.

10.5 Small Examples

$n = 1$. For total charge τ, the ambient basis is $|\mathbf{1}, \tau\rangle$, and it is admissible. Thus $H^{\mathrm{Fib}}_{1,\tau} \cong \mathbb{C}$.

$n = 2$. The admissible paths are

$$|\mathbf{1}, \tau, \mathbf{1}\rangle \in H^{\mathrm{Fib}}_{2,\mathbf{1}}, \qquad |\mathbf{1}, \tau, \tau\rangle \in H^{\mathrm{Fib}}_{2,\tau}.$$

This is the kinematic content of $\tau \otimes \tau = \mathbf{1} \oplus \tau$.

$n = 3$. The total-τ ambient basis has two admissible paths:

$$|\mathbf{1}, \tau, \mathbf{1}, \tau\rangle, \qquad |\mathbf{1}, \tau, \tau, \tau\rangle.$$

The forbidden word $|\mathbf{1}, \mathbf{1}, \tau, \tau\rangle$, for example, is killed because $N^{\mathbf{1}}_{\mathbf{1},\tau} = 0$. Therefore $\dim H^{\mathrm{Fib}}_{3,\tau} = 2$.

10.6 Compiler Consequence

This is the first example where the Hilbert-space compiler wants category data. At the carrier level, it can compile the ambient finite word space $W_{n,c}$ and the diagonal sector projection $P_{n,c}$. But producing $P_{n,c}$ from the expression "n copies of τ" requires the fusion coefficients N^c_{ab}. That is already tensor-category input, not ordinary Hilbert-space grammar.

Moreover, the source makes clear that different fusion orders give different fusion-tree bases and are related by F-moves. Therefore a full Fibonacci compiler must eventually output not only

$$H^{\mathrm{Fib}}_{n,c}$$

but also the coherent recoupling data that identifies different bracketings: the F-symbols satisfying the pentagon equations, and later the R-symbols for braiding. The current shard is only kinematic: it builds the admissible sector and counts it.

10.7 Lamport Dependency Skeleton

$$\begin{aligned}
D28 &: L = \{\mathbf{1}, \tau\}, \quad \mathbf{1} \otimes \tau = \tau, \quad \tau \otimes \tau = \mathbf{1} \oplus \tau. \\
D29 &: W_{n,c} = \ell^2\{(x_0, \dots, x_n) : x_0 = \mathbf{1},\ x_n = c,\ x_i \in L\}. \\
D30 &: P_{n,c}|x\rangle = \textstyle\prod_i N^{x_i}_{x_{i-1},\tau}|x\rangle. \\
D31 &: H^{\mathrm{Fib}}_{n,c} = \mathrm{Ran}(P_{n,c}). \\
L5 &: P_{n,c} = P^*_{n,c} = P^2_{n,c}. \\
T18 &: \dim H^{\mathrm{Fib}}_{n,\tau} \text{ obeys } b_{n+1} = b_n + b_{n-1},\ b_0 = 0,\ b_1 = 1. \\
T19 &: \text{Full Fibonacci compilation needs } F\text{-coherent tensor-category data.}
\end{aligned}$$

10.8 Boundary

This shard does not fix the Fibonacci F-matrix gauge, any R-symbols, braid representations, golden-chain Hamiltonians, or a continuum limit. It only realises the fusion-path Hilbert spaces kinematically as admissibility sectors.

11 Why Lattice Symmetry Generators Are a Problem

11.1 Status, Sources, and Dependencies

Status: proposal shard with sourced boundary. The claim that a lattice boost should look like a first moment of the Hamiltonian density is an intuition until a local source, derivation, or test is attached. This shard sets the boundary for CA-10–CA-17.

Dependencies: CA-05 for symmetry as unitary/projective-unitary representation, CA-08 for compiler output slots, and CONVENTIONS.md (h) for lattice candidate generator signs.

11.2 The Minimal Quantum-Mechanical Input

An unadorned quantum system gives a Hilbert space and, once dynamics is chosen, a self-adjoint Hamiltonian. By Stone's theorem this is the same kind of datum as a strongly continuous unitary representation of the additive group $\mathbb{R}$:

$$U(t) = \exp(-itH).$$

Thus the one symmetry forced by the bare dynamical presentation is time translation. Everything beyond this requires extra structure: spatial coordinates, locality, a metric, a lattice graph, or a continuum target.

11.3 What the Lattice Adds

A lattice Hamiltonian is not merely a self-adjoint operator. It usually comes as a decomposition

$$H = \sum_j h_j$$

where each h_j is local near a site, bond, cell, or other geometric support. The cited lattice-fermion source uses exactly this pattern: a Hamiltonian is a sum of self-adjoint finite-support terms, together with an observable algebra whose generators are intended as discretisations of continuum fields.

This decomposition is extra data. Two different decompositions of the same operator H can lead to different candidate densities and hence different candidate symmetry generators. Therefore the compiler target is not

$$\text{Hilbert space plus one operator } H,$$

but rather

$$(\text{Hilbert space, lattice geometry, } \{h_j\}, \text{ scaling map}).$$

11.4 The Continuum Target

If the intended continuum theory is Lorentz-invariant in $d + 1$ dimensions, the target symmetry is not just $\mathbb{R}$-time translation. It is a projective-unitary representation of the Poincare group, lifted as necessary to a unitary representation of a covering group. Its translation subgroup has self-adjoint generators

$$P_0 = H, \qquad P_1, \ldots, P_d,$$

where the P_a are momenta.

Motivating question. If the microscopic input only supplies a lattice Hamiltonian density $H = \sum_j h_j$, how do we guess the lattice analogues of

$$P_a, \qquad M_{ab} \text{ rotations}, \qquad M_{0a} \text{ boosts?}$$

11.5 Intuition to Be Checked

Intuition 1: a rotation is a position-dependent spatial translation. In the plane, the vector field for an infinitesimal rotation about the origin is

$$v(x, y) = (-y, x).$$

Thus if we can identify local momentum densities, rotations should be spatial first moments of those densities.

Intuition 2: a boost is a position-dependent time translation. Since time translation is generated by energy, the one-dimensional boost candidate is a first moment of the energy density,

$$K_x \propto \sum_j x_j h_j.$$

Intuition 3: the commutator of a boost with time translation should produce spatial translation, up to sign and normalization. On the lattice this suggests that

$$i[H, K_x]$$

is a candidate momentum generator. CA-12 derives the nearest-neighbour formula from this ansatz and records the sign convention explicitly.

11.6 Source Boundary

Weinberg QFT Vol. 1 is now registered with targeted OCR anchors, and the original Koo–Saleur source is now registered from arXiv with source-TeX and PDF text anchors. The present shard still cites Schottenloher for continuum symmetry definitions, OAR/wavelet sources for scaling-limit translation recovery, and the local lattice-fermion source for rigorous free-fermion Koo–Saleur convergence. CA-14 cites the original Koo–Saleur paper for the source-local lattice Virasoro prototype.

12 The Continuum Bulk Symmetry Target

12.1 Status, Sources, and Dependencies

Status: sourced definitions plus checked local derivation. The Poincare group and the projective-unitary implementation are sourced. The commutator table below is a local vector-field derivation from the sourced metric-preserving description, and its signs are pinned by a Julia check.

Dependencies: CA-05, CA-10, and CONVENTIONS.md (h), (k).

12.2 Group-Level Target

For a $D = d + 1$ dimensional Lorentzian target, Schottenloher describes the Poincare group as the semidirect product of the Lorentz group and translations:

$$P = SO_0(1, d) \ltimes \mathbb{R}^{d+1},$$

with the usual replacement by the relevant covering group when the quantum symmetry is implemented projectively. The translation subgroup is abelian and has commuting self-adjoint generators P_μ. In the source convention,

$$P_0 = H, \qquad P_a \text{ are spatial momenta } (a = 1, \dots, d).$$

For Euclidean two-dimensional CFT language, the Euclidean motion group is generated by rotations and translations. This gives the Wick-rotated analogue: rotations and translations are the geometric target before scale/conformal extensions are added.

12.3 Vector-Field Derivation

Let $\eta = \mathrm{diag}(1, -1, \dots, -1)$, and lower indices by $x_\mu = \eta_{\mu\nu} x^\nu$. Use vector fields

$$P_\mu = \partial_\mu, \qquad M_{\mu\nu} = x_\mu \partial_\nu - x_\nu \partial_\mu, \qquad M_{\mu\nu} = -M_{\nu\mu}.$$

The source gives the metric-preserving Lie algebra condition $\omega^T \eta + \eta\omega = 0$. The $M_{\mu\nu}$ form the corresponding coordinate vector fields, while P_μ are translations.

Compute the Lie bracket of vector fields. Since the P_μ have constant coefficients,

$$[P_\mu, P_\nu] = 0.$$

For the mixed bracket,

$$\begin{aligned}[M_{\mu\nu}, P_\rho] &= -\partial_\rho(x_\mu)\partial_\nu + \partial_\rho(x_\nu)\partial_\mu \\ &= \eta_{\nu\rho} P_\mu - \eta_{\mu\rho} P_\nu.\end{aligned}$$

For Lorentz-Lorentz brackets,

$$\begin{aligned}[M_{\mu\nu}, M_{\rho\sigma}] = \eta_{\nu\rho} M_{\mu\sigma} &- \eta_{\mu\rho} M_{\nu\sigma} \\ &+ \eta_{\mu\sigma} M_{\nu\rho} - \eta_{\nu\sigma} M_{\mu\rho}.\end{aligned}$$

The Julia check evaluates representative brackets using this convention. For example in $3 + 1$ dimensions,

$$[M_{01}, P_0] = -P_1, \qquad [M_{01}, P_1] = -P_0.$$

12.4 Boost-Time Relation

With $K_a := M_{0a}$, the vector-field convention gives

$$[K_a, P_0] = -P_a.$$

Equivalently, $[P_0, K_a] = P_a$. The passage from this vector-field statement to self-adjoint commutator notation is the separate sign convention in CONVENTIONS.md (k): Stone generators use $U_X(t) = \exp(-itA_X)$, the spacetime flow is implemented passively, and the resulting target is

$$i[H, K_a] = P_a, \qquad i[P_b, K_a] = \delta_{ab} H.$$

Spatial rotations carry the additional convention $J_{ab} = -A_{M_{ab}}$, which is why the Gaussian one-particle table has $i[K_a, K_b] = J_{ab}$ even though the active vector fields obey $[M_{0a}, M_{0b}] = -M_{ab}$. CA-12 keeps finite-lattice signs explicit under CONVENTIONS.md (h).

12.5 Boundary

This shard does not claim that a finite lattice has an exact Poincare representation. It only identifies the continuum algebraic target whose relations a sequence of lattice candidates should approximate after scaling.

13 A One-Dimensional Boost Produces a Momentum Current

13.1 Status, Sources, and Dependencies

Status: checked local derivation. The identity in this shard is not quoted from a source. It is derived from the local Hamiltonian-density ansatz and checked by a small coefficient invariant.

Dependencies: CA-10, CA-11, and CONVENTIONS.md (h).

13.2 Setup

Let a finite open one-dimensional lattice have sites or bonds labelled $j = 1, \ldots, N$, real positions x_j, and a self-adjoint local-density decomposition

$$H = \sum_{j=1}^{N} h_j.$$

Assume the density terms commute at distance greater than one:

$$[h_j, h_k] = 0 \qquad \text{if } |j - k| > 1.$$

This is the finite-range simplification appropriate for the first derivation; longer-range densities only add longer-range edge terms.

Define the position-weighted energy moment

$$K_x = \sum_{j=1}^{N} x_j h_j.$$

This is the lattice boost candidate in the sense of CA-10: a boost is treated as a position-dependent time translation, and time translation is generated by energy density.

13.3 Commutator Calculation

Compute

$$\begin{aligned} i[H, K_x] &= i\left[\sum_j h_j, \sum_k x_k h_k\right] \\ &= i\sum_{j,k} x_k [h_j, h_k]. \end{aligned}$$

The terms with $j = k$ vanish, and by the nearest-neighbour assumption only pairs $(j, j+1)$ survive. For each adjacent pair,

$$i x_{j+1}[h_j, h_{j+1}] + i x_j [h_{j+1}, h_j] = i(x_{j+1} - x_j)[h_j, h_{j+1}].$$

Hence

$$i[H, K_x] = \sum_{j=1}^{N-1} (x_{j+1} - x_j)\, i[h_j, h_{j+1}].$$

Each $i[h_j, h_{j+1}]$ is self-adjoint because h_j, h_{j+1} are self-adjoint.

13.4 Uniform Spacing

If $x_j = aj$, then $x_{j+1} - x_j = a$ and

$$i[H, K_x] = a \sum_{j=1}^{N-1} i[h_j, h_{j+1}].$$

For unit spacing this gives the candidate bulk momentum

$$P_{\text{bulk}}^{\text{lat}} := \sum_{j=1}^{N-1} i[h_j, h_{j+1}],$$

with endpoint terms deliberately visible. Periodic chains require an extra boundary convention because the position function j is not single-valued on a circle.

13.5 Interpretation

The continuum bracket in CA-11 says that boost plus time translation gives a spatial translation, up to sign convention. The lattice calculation says that the same operation applied to the first energy moment produces the commutator of neighbouring energy densities. Therefore $i[h_j, h_{j+1}]$ is not guessed from thin air: it is the local current forced by the first-moment ansatz.

13.6 Boundary

This is not yet a theorem that $P_{\text{bulk}}^{\text{lat}}$ converges to continuum momentum. The missing ingredients are a scaling map, a domain for the candidate operators, a state/sector sequence, and convergence of the commutator relations. CA-15 turns these into acceptance checks.

14 Position-Dependent Bulk Generators

14.1 Status, Sources, and Dependencies

Status: proposal shard with local derivation pattern. The sourced continuum content is geometric: vector fields, translations, and rotations. The higher-dimensional lattice formulas are compiler targets, not established convergence claims, and this shard does not define a stress-energy tensor or a lattice momentum density.

Dependencies: CA-10–CA-12 and CONVENTIONS.md (h).

14.2 Continuum Picture

Translations are flows of constant vector fields. In the Euclidean plane, the infinitesimal motion

$$X(z) = c + i\theta z$$

contains both translations (c) and rotations ($i\theta z$). Written in real coordinates, the rotation part is the position-dependent vector field

$$(x, y) \mapsto (-\theta y, \theta x).$$

This is the precise continuum content of the slogan:

$$\text{rotation} = \text{position-dependent translation.}$$

14.3 Scalar Ramps

Suppose lattice density terms are attached to positions $r \in \Lambda \subset \mathbb{R}^d$. For a scalar function $f : \Lambda \to \mathbb{R}$, define

$$K_f := \sum_{r \in \Lambda} f(r) h_r.$$

The same derivation as CA-12 gives, for commuting non-overlapping densities,

$$i[H, K_f] = \sum_{\{r,s\}} (f(s) - f(r))\, i[h_r, h_s],$$

where the sum is over unordered overlapping pairs after a choice of orientation. This is a discrete-gradient response. Linear ramps

$$f_a(r) = r^a$$

are therefore the first candidates for spatial momentum components.

14.4 Boost Candidates

Boosts can be attempted directly from energy density:

$$K_a^{\text{lat}} := \sum_{r \in \Lambda} r^a h_r.$$

Then $i[H, K_a^{\text{lat}}]$ gives the a-direction current candidate. This is the higher-dimensional version of the one-dimensional derivation, but it depends on the chosen density supports and orientations.

14.5 Rotation Candidates

Rotations require one more layer. In continuum notation,

$$J_{ab} = \int (x_a p_b(x) - x_b p_a(x))\, dx.$$

Here this formula is only a continuum mnemonic for why rotations should involve an angular-momentum density. The Schottenloher anchors above support the underlying source vector fields and rotations, not a QFT stress-energy tensor, not a momentum-density formula, and not a lattice dictionary for $p_a(x)$. Thus an actual lattice rotation candidate must be routed through a later momentum-density/current convention rather than built directly from h_r. Once such a convention is fixed, the provisional compiler shape is:

$$J_{ab}^{\mathrm{lat}} \approx \sum_r \big(r_a p_b^{\mathrm{lat}}(r) - r_b p_a^{\mathrm{lat}}(r)\big),$$

where p_a^{lat} must come with its own source, derivation, or checked continuity-equation evidence.

14.6 Compiler Moral

The lattice Hilbert-space compiler cannot infer these operators from the carrier alone. It needs a lattice geometry, a local Hamiltonian-density decomposition, and a target vector-field algebra. The output is not an exact symmetry by default; it is a candidate list with proof obligations.

14.7 Boundary

No stress-energy convention is fixed here. No cell-orientation convention is fixed here. No periodic wrapping convention is fixed here. No higher-dimensional convergence theorem is fixed here. Those choices must be made in a later shard before any numerical or categorical lattice model uses these candidates as checked symmetries.

15 Koo–Saleur as the Prototype

15.1 Status, Sources, and Dependencies

Status: source-local prototype plus later rigorous precedent. The original Koo–Saleur paper is registered here and supports the lattice Virasoro prototype as a conjectural construction with numerical and analytic checks for XXZ/RSOS/Temperley–Lieb settings. This shard keeps that separate from the local lattice-fermion/OAR treatment, which states and proves a free-fermion Koo–Saleur convergence result.

Dependencies: CA-10–CA-13.

15.2 Continuum Side

In a two-dimensional CFT on the circle, the Hamiltonian is expressed as an integral of a Hamiltonian density,

$$H = \int h(x)\, dx.$$

The source identifies $h(x)$ as a local quantum field built from the chiral and anti-chiral energy-momentum tensors $T, \bar{T}$. Its Fourier modes H_k are linear combinations of the Virasoro generators $L_k, \bar{L}_k$.

Thus in this special setting the desired spacetime symmetries are not abstract: they are generated by modes of the stress tensor.

15.3 Lattice Side

The Koo–Saleur move is to imitate the continuum construction using the lattice Hamiltonian density. From a lattice density

$$h^{(N)} : \Lambda_N \to \mathcal{A}_N$$

one forms Fourier modes $H_k^{(N)}$. In the original Temperley–Lieb presentation, the Hamiltonian-limit density is represented by ground-state-subtracted $e_j - e_\infty$ terms, while the second stress-tensor component is represented by nearest-neighbour commutators $[e_j, e_{j+1}]$. The lattice Virasoro candidates $l_n, \bar{l}_{-n}$ are therefore Fourier sums of

$$(e_j - e_\infty) \ \pm \ \text{constant} \cdot [e_j, e_{j+1}],$$

with the $c/24$ zero-mode shift. The later free-fermion/OAR Definition 4.1 uses the same structural pattern: density modes plus commutator corrections against the lattice Hamiltonian mode $H_0^{(N)}$ produce lattice analogues of L_k and $\bar{L}_k$.

The important structural point for our programme is:

$$\text{lattice symmetry generator} = \text{weighted Hamiltonian density} + \text{commutator correction.}$$

This is the same motif as CA-12, but now with Fourier weights and a Virasoro target rather than a first moment and a Poincare translation target.

15.4 Scaling Representation Matters

The original source already builds in ground-state subtraction, normal ordering, and a double-limit process; it also warns that finite-L lattice quantities do not close as the Virasoro algebra and that a scaling limit of a commutator is not generally the commutator of scaling limits. The later rigorous free-fermion/OAR source similarly makes convergence depend on the correct scaling-limit representation and on subtracting ground-state expectation values, corresponding to normal ordering. Therefore the compiler must not output only formal sums. It must also output:

state sequence,
renormalisation/scaling maps,
vacuum subtraction policy,
operator topology target.

15.5 What This Foreshadows

Koo–Saleur is a prototype for the lattice-symmetry compiler:

$$\{h_j\} \longmapsto \{H_k^{(N)}, L_k^{(N)}, \bar{L}_k^{(N)}\} \longrightarrow \{L_k, \bar{L}_k\}.$$

The first arrow is algebraic and finite-scale. The second is analytic and requires a scaling theorem.

15.6 Boundary

The original Koo–Saleur source supports the prototype as a conjectural and numerically checked construction for XXZ/RSOS/Temperley–Lieb lattice models; it does not prove a general rigorous convergence theorem for arbitrary microscopic models. The convergence theorem cited here is still the later free-fermion/OAR result, and arbitrary anyon chains remain outside both sourced claims.

16 Acceptance Tests for Lattice Symmetry Candidates

16.1 Status, Sources, and Dependencies

Status: specification shard. This shard does not prove convergence for new models. It states the checks required before a candidate generator is promoted from proposal to checked continuum symmetry data.

Dependencies: CA-10–CA-14.

16.2 Local Finite-Scale Checks

At fixed lattice size N, a candidate generator G_N must first pass finite operator checks:

$$G_N = G_N^*, \qquad G_N \text{ is built from named local density terms}, \qquad \operatorname{supp}(G_N) \text{ policy is explicit.}$$

For unbounded thermodynamic-limit candidates, the finite check is only the first layer; domains must later be named.

16.3 Algebraic Checks

Given a target Lie algebra with generators X_a, the lattice candidates $G_a^{(N)}$ should satisfy commutator closure in a controlled sense:

$$i[G_a^{(N)}, G_b^{(N)}] \approx f_{ab}{}^{c} G_c^{(N)} + \text{central or boundary terms.}$$

The approximation relation must be specified: exact finite-size equality, matrix-element convergence, strong resolvent convergence, strong operator convergence on a core, or convergence after compression to a low-energy sector.

16.4 Scaling Checks

The OAR source shows that continuous spatial translations can be reconstructed from discrete dyadic translations in a suitable scaling limit, with the usual momentum operators as generators. It also states that analogous convergence for Lorentz or conformal transformations requires further work. This is the main warning for this programme:

$$\text{translation recovery} \not\Rightarrow \text{Poincare recovery.}$$

Therefore every candidate boost or rotation needs its own scaling statement. For a sequence G_N, the report must say:

$$G_N \xrightarrow{?} G$$

and name the topology, state sequence, renormalisation maps, and dense domain or core on which the statement is intended.

16.5 Observable Covariance Checks

The target is not only an operator on states. It is a symmetry action on the observable system. The evidence payload should include

$$\alpha_N^G(A) = e^{itG_N} A e^{-itG_N}$$

or the infinitesimal commutator $i[G_N, A]$, together with the expected transformation of local algebras. For translations this means moving supports. For rotations it means rotating supports. For boosts it means mixing time and space in the continuum representation.

16.6 Acceptance Payload

A lattice generator candidate is accepted into the compiler only with:

1. density provenance: the exact h_j and supports,
2. finite algebra checks: self-adjointness and commutators,
3. renormalisation maps and state sequence,
4. operator topology or matrix-element convergence target,
5. observable covariance statement,
6. boundary/central/vacuum-subtraction policy.

16.7 Boundary

Time-zero net axioms and Lieb-Robinson bounds give important locality control, but local causality is not the same as full Poincare covariance. The compiler must keep these evidence levels separate.

17 A Compiler Interface for Lattice Symmetry Candidates

17.1 Status, Sources, and Dependencies

Status: compiler specification shard. This shard extends CA-08 at the level of output slots. It is not yet an implementation.

Dependencies: CA-08 and CA-10–CA-15.

17.2 Input Contract

The lattice symmetry compiler receives a compiled Hilbert/QM system plus:

$$\begin{array}{ll} \Lambda_N & : \text{finite lattice, graph, or cell complex,} \\ x & : \Lambda_N \to \mathbb{R}^d \text{ position map,} \\ \mathcal{A}_N & : \text{observable algebra with locality map,} \\ H_N = \sum_r h_r & : \text{self-adjoint local Hamiltonian density,} \\ T & : \text{target symmetry algebra or group,} \\ S_N & : \text{state/sector sequence and scaling maps, if known.} \end{array}$$

If the Hamiltonian is supplied only as a black-box matrix, the compiler must fail for boost/rotation generation: the density decomposition is part of the mathematical data.

17.3 Candidate Generation

For one-dimensional open chains, the first candidate layer is deterministic:

$$K_x = \sum_j x_j h_j, \qquad P_x = i[H, K_x].$$

For uniform nearest-neighbour density this reduces to the CA-12 current sum.

For higher-dimensional lattices, scalar ramps $f_a(r) = r^a$ generate candidate momentum components

$$P_a^{\mathrm{lat}} := i[H, K_{f_a}].$$

Rotation candidates are then built only after the momentum-density/current layer has its own convention and evidence:

$$J_{ab}^{\mathrm{lat}} \approx \sum_r (r_a p_b^{\mathrm{lat}}(r) - r_b p_a^{\mathrm{lat}}(r)).$$

Until that layer is fixed, this is a deferred proposal slot rather than a formula emitted by the compiler. Koo–Saleur-style modes use Fourier weights instead of polynomial weights and include the commutator corrections specified by the cited Definition 4.1.

17.4 Output Contract

The compiler output for each candidate generator must include:

name	: for example $P_a, K_a, J_{ab}, L_k, \bar{L}_k$,
formula	: finite expression in h_r and commutators,
domain policy	: finite matrix, core, or unbounded deferral,
locality support	: site/bond/cell support accounting,
status	: Proposal, Source-local, or Checked,
evidence	: source lines, derivation, test, or run artifact,
continuum target	: target generator and expected bracket.

17.5 Failure Modes

The compiler must fail loudly when:

1. H has no density decomposition,
2. Λ_N has no position map but a boost/rotation is requested,
3. periodic boundary positions are requested without a branch convention,
4. the target algebra requires a source/convention not yet fixed,
5. a convergence claim is requested without scaling maps or states.

17.6 Compiler Principle

The finite expression and the continuum theorem are different compilation products. The first can often be generated algebraically from $(\Lambda_N, x, \{h_r\})$. The second is a proof obligation involving scaling maps, states, and operator topology. The compiler should output both slots, but it must never fill the proof slot with intuition.

18 Small Examples and the Lattice-Symmetry Research Queue

18.1 Status, Sources, and Dependencies

Status: worked examples and queue. The examples are deliberately small. They illustrate compiler behaviour, not new convergence theorems.

Dependencies: CA-10–CA-16 and CONVENTIONS.md (h), (j).

18.2 Example 1: Black-Box Quantum System

Input: a Hilbert space and a self-adjoint Hamiltonian H.

Output: time translation $U(t) = e^{-itH}$.

Failed request: infer boosts, rotations, or spatial translations. There is no lattice geometry, no position map, and no Hamiltonian density. The compiler must stop.

18.3 Example 2: Open Uniform Chain

Input:

$$H = \sum_{j=1}^{N} h_j, \qquad x_j = j, \qquad [h_j, h_k] = 0 \text{ for } |j-k| > 1.$$

Candidate boost:

$$K = \sum_j j h_j.$$

Candidate momentum:

$$P = i[H, K] = \sum_{j=1}^{N-1} i[h_j, h_{j+1}].$$

Status: checked finite derivation; no continuum convergence yet.

18.4 Example 3: Nonuniform Chain

Input: positions $x_1 < \cdots < x_N$. The same derivation gives

$$P_x = i[H, K_x] = \sum_{j=1}^{N-1} (x_{j+1} - x_j) i[h_j, h_{j+1}].$$

The coefficients are discrete spacings. The Julia test pins this invariant for uniform and nonuniform examples.

18.5 Example 4: Periodic Chain

Input: a circle with periodic density terms.

Problem: $x_j = j$ is not a single-valued function on the circle. A periodic boost-like first moment requires a branch cut, a sawtooth, Fourier modes, or a different construction. The same warning applies to finite periodic momentum-coordinate tests: a finite grid supports periodic symbols and Fourier eigenvalue checks, not an exact differential operator $X = i\partial_k$ with a canonical commutator. Koo–Saleur Fourier modes are the better sourced periodic prototype.

18.6 Example 5: Square Lattice

Input: positions $r = (m, n)$. Scalar ramps $f_x(r) = m$, $f_y(r) = n$ give two candidate momentum components via

$$P_a^{\mathrm{lat}} = i[H, \sum_r r^a h_r].$$

Rotation about the origin should then be built as a position-dependent translation from the two momentum-density components. Status: proposal until an orientation and density-localisation convention is fixed.

18.7 Example 6: Koo–Saleur Modes

Input: periodic one-dimensional critical Hamiltonian density. Fourier modes of the density and commutator corrections give candidates for $L_k, \bar{L}_k$. In the cited free-fermion/OAR setting, smeared approximants converge to smeared Virasoro generators. Status outside that setting: source-local precedent, not a general theorem.

18.8 Research Queue

1. Use registered Weinberg anchors before promoting exact Lorentz-generator formulas.
2. Use the registered original Koo–Saleur anchors before promoting model-specific KS claims.
3. Source or derive a general local energy-current theorem.
4. Fix periodic branch/sawtooth/Fourier first-moment conventions.
5. Define higher-dimensional oriented density supports.
6. Add numerical examples checking commutator closure in small chains.
7. Connect anyon-chain Hamiltonian densities to the same candidate layer.

18.9 Landing Point

The next compiler upgrade is therefore not only a Hilbert-space compiler and not yet only a tensor-category compiler. It is a lattice-QM compiler layer:

$$(\text{carrier}, \mathcal{A}_N, \Lambda_N, \{h_r\}, x, S_N) \longmapsto (\text{candidate bulk symmetries, evidence obligations}).$$

Only after those obligations are discharged can the continuum symmetry representation be claimed.

19 Why Galilean Symmetry Is a Different Lattice Target

19.1 Status, Sources, and Dependencies

Status: sourced boundary plus proposal shard. The Galilei group as a classical nonrelativistic symmetry group is sourced locally. The general projective-representation and central-extension mechanism is sourced locally. The specific mass central extension of the Galilei group is not yet sourced in this repository; CA-20 treats its coefficient as a checked canonical-commutator derivation, not as a literature claim about a named group.

Dependencies: CA-05 for projective symmetries via central extensions, CA-10–CA-17 for the Lorentzian lattice-symmetry programme, and CONVENTIONS.md (i).

19.2 The New Target

The previous block targeted Lorentz-invariant continuum systems. Its first lattice intuition was:

$$\text{Lorentz boost} \simeq \text{position-dependent time translation}, \qquad K_a^{\mathrm{Lor}} \sim \sum_x x_a h_x.$$

That is appropriate only when the continuum target mixes space and time in the Lorentzian way and the boost is controlled by the energy density.

The nonrelativistic target is different. Schottenloher lists the Galilei group as the symmetry group for free particles in classical nonrelativistic mechanics. For the lab-book compiler this means that a Galilean target is not obtained by renaming the Poincare block. It has absolute time, spatial translations, rotations, and boosts whose quantum implementation may be projective.

19.3 Immediate Consequence for Lattices

The Lorentzian boost ansatz used only the Hamiltonian density h_x. A Galilean boost needs more input. At time $t = 0$, the familiar nonrelativistic one-particle expression is controlled by the mass-weighted position, not by the energy-weighted position:

$$G_a(0) \sim mX_a.$$

For a many-body lattice this suggests a mass-density or particle-number-density first moment:

$$G_a^{\mathrm{lat}}(0) \sim \sum_{x\in\Lambda} x_a\,\mu_x, \qquad \mu_x = m\,n_x \quad \text{when the particles have common mass } m.$$

This is a proposal-level lattice formula until the model supplies the density μ_x, the conservation law, and the continuity/current relation.

19.4 Compiler Implication

The compiler rule is therefore stricter than in CA-16:

$$\begin{aligned} &\text{Lorentzian candidate layer:} && (\Lambda, x, \{h_x\}) \text{ may generate } K_a \sim \textstyle\sum_x x_a h_x,\\ &\text{Galilean candidate layer:} && (\Lambda, x, \{h_x\}) \text{ is not enough.}\end{aligned}$$

For Galilean boosts the input must also name:

$$\{\mu_x\} \text{ or } \{n_x\}, \qquad M = \sum_x \mu_x, \qquad [H, M] = 0,$$

plus, for a momentum candidate, a current witness for the discrete continuity equation. If those data are absent, the compiler must fail loudly rather than silently using the energy first moment.

19.5 Why Projective Data Matter Earlier

In the Hilbert-space grammar, projective symmetries were handled by passing to a central extension. Galilean symmetry is the first lattice target where that choice becomes operational: the mass parameter is expected to sit in the projective/central part of the quantum representation. Locally we can support only this precise statement:

projective quantum symmetries are represented by unitary representations of central extensions.

The stronger statement identifying the Galilei extension by its standard literature name remains a source-acquisition task.

19.6 Boundary for CA-18–CA-22

The follow-up block does four things:

CA-19	derives the unextended Galilei vector-field algebra,
CA-20	derives the mass central coefficient on a canonical core,
CA-21	derives the lattice first-moment continuity formula,
CA-22	turns the data into compiler rules and small examples.

No shard in this block claims that a finite lattice has exact Galilean symmetry unless the corresponding finite-dimensional representation and commutator checks are actually supplied.

20 The Galilei Algebra as Newtonian Vector Fields

20.1 Status, Sources, and Dependencies

Status: checked local derivation. The source anchor is that the Galilei group is the nonrelativistic free-particle symmetry group. The bracket table below is derived here from vector fields and checked by the Julia test suite.

Dependencies: CA-18 and CONVENTIONS.md (i).

20.2 Generators

Use Newtonian spacetime coordinates

$$(t, x_1, \ldots, x_d).$$

The convention fixed in CONVENTIONS.md (i) is:

$$\begin{array}{rcll} H &=& \partial_t, & \text{time translation,} \\ P_a &=& \partial_a, & \text{spatial translation,} \\ G_a &=& t\,\partial_a, & \text{Galilean boost,} \\ J_{ab} &=& x_a\partial_b - x_b\partial_a, \quad a < b, & \text{spatial rotation.} \end{array}$$

Here $a, b, c \in \{1, \ldots, d\}$. The boost vector field says that a boost changes position by a velocity times time while leaving absolute time fixed.

20.3 Translation and Boost Brackets

The translation fields have constant coefficients, hence

$$[H, P_a] = 0, \qquad [P_a, P_b] = 0.$$

Since $G_a = t\partial_a$,

$$\begin{aligned} [H, G_a] &= [\partial_t, t\partial_a] \\ &= \partial_t(t)\partial_a \\ &= P_a. \end{aligned}$$

The reverse bracket is therefore $[G_a, H] = -P_a$. Spatial translations do not differentiate the coefficient t, so

$$[P_a, G_b] = 0, \qquad [G_a, G_b] = 0.$$

This is the unextended vector-field algebra. It is not yet the projective quantum algebra; CA-20 adds the central term that is invisible to vector fields on spacetime.

20.4 Rotation on Vectors

For $J_{ab} = x_a\partial_b - x_b\partial_a$, compute against a translation:

$$\begin{aligned} [J_{ab}, P_c] &= -\partial_c(x_a)\partial_b + \partial_c(x_b)\partial_a \\ &= \delta_{bc}P_a - \delta_{ac}P_b. \end{aligned}$$

The same calculation applies to boosts because J_{ab} does not act on t:

$$[J_{ab}, G_c] = \delta_{bc}G_a - \delta_{ac}G_b.$$

Thus both P_c and G_c transform as spatial vectors.

20.5 Rotation-Rotation Bracket

For two rotations, expanding the vector-field bracket gives

$$\begin{aligned}[J_{ab}, J_{cd}] = \delta_{bc} J_{ad} - \delta_{ac} J_{bd} \\ + \delta_{ad} J_{bc} - \delta_{bd} J_{ac},\end{aligned}$$

with the convention $J_{ba} = -J_{ab}$ and $J_{aa} = 0$. In $d = 3$, one representative checked bracket is

$$[J_{12}, J_{23}] = J_{13}.$$

20.6 Checked Representatives

The test suite pins the signs with the following representative outcomes:

$$\begin{array}{rcl}[H, G_1] & = & P_1, \\ [G_1, H] & = & -P_1, \\ [J_{12}, P_1] & = & -P_2, \\ [J_{12}, P_2] & = & P_1, \\ [J_{12}, G_1] & = & -G_2, \\ [J_{12}, J_{23}] & = & J_{13}.\end{array}$$

It also checks that $[P_1, G_2] = 0$ and $[G_1, G_2] = 0$ in the unextended vector-field algebra.

20.7 Boundary

The signs here are signs for vector fields. A quantum model uses self-adjoint generators and unitary flows, so translating this table into operator commutators requires the Stone convention and the chosen $i[A, B]$ observable action. The report therefore cites this shard for the geometric algebra and CA-20 for the central quantum coefficient.

21 Projective Galilean Symmetry and the Mass Central Term

21.1 Status, Sources, and Dependencies

Status: sourced projective framework plus checked local derivation. Schottenloher supplies the representation-theoretic framework: projective symmetries need not lift to the original group, but lift through a central extension. The coefficient $m\delta_{ab}$ below is a local canonical commutator derivation checked in Julia. A local source specifically identifying the Galilei mass central extension by its standard name has not yet been registered.

Dependencies: CA-05, CA-18, CA-19, and CONVENTIONS.md (f), (i).

21.2 Why This Belongs in the Grammar

CA-05 made a conservative grammar choice: a projective symmetry is represented by an honest unitary representation of a named central extension

$$1 \longrightarrow U(1) \longrightarrow E \longrightarrow G \longrightarrow 1.$$

For Galilean symmetry this is not a decorative detail. The central parameter is physically interpreted as mass in the canonical nonrelativistic model, and it controls how boosts and momenta compose at the quantum level.

21.3 Canonical One-Particle Calculation

Work formally on a common invariant core where the position and momentum operators obey

$$[X_a, P_b] = i\delta_{ab} I.$$

For a particle of mass m, the time-zero boost generator is represented by

$$G_a(0) = mX_a.$$

Then

$$[G_a(0), P_b] = [mX_a, P_b]$$
$$= m[X_a, P_b]$$
$$= i\, m\, \delta_{ab} I.$$

The Julia check records exactly the real coefficient $m\delta_{ab}$.

21.4 Compatibility with Time Evolution

For the free Hamiltonian

$$H = \frac{1}{2m} \sum_b P_b^2,$$

the same canonical commutator gives

$$[H, mX_a] = \frac{1}{2} \sum_b [P_b^2, X_a]$$
$$= \frac{1}{2} \sum_b \big(P_b [P_b, X_a] + [P_b, X_a] P_b \big)$$
$$= \frac{1}{2}\big(-iP_a - iP_a \big) = -iP_a.$$

Thus

$$i[H, G_a(0)] = P_a.$$

This matches the CA-19 vector-field relation $[H, G_a] = P_a$ after applying the self-adjoint-generator convention used for observable evolution.

21.5 Central Extension Versus Vector Fields

The unextended vector-field algebra had $[G_a, P_b] = 0$. The canonical quantum calculation has instead

$$[G_a, P_b] = i\, m\, \delta_{ab} I.$$

The additional term is central: it is a scalar multiple of the identity on a fixed-mass sector. This is the algebraic reason the compiler must attach a central-extension witness to Galilean symmetry data rather than store only the spacetime vector fields.

21.6 Source Gap

The local source collection includes Bargmann's theorem about lifting projective representations under a Lie-algebra cohomology hypothesis, and it includes the general central-extension theorem for projective representations. It does not yet include a registered source for the standard mass central extension of the Galilei group. Until that source is acquired, this shard uses the cautious name *mass central term* and cites the checked canonical calculation as its local evidence.

21.7 Compiler Payload

A Galilean projective symmetry record must therefore include:

base group/algebra	Galilei target from CA-19,
central generator	M or I on a fixed-mass sector,
central charge	m or a mass operator,
raw commutator witness	$[G_a, P_b] = i\delta_{ab} M$,
sign convention	conversion to $i[A, B]$ stated explicitly.

An OR of sectors with different masses can still be a Hilbert direct sum, but the central element must then be represented as a block-diagonal operator rather than a single scalar charge.

22 Lattice Galilean Boosts from Mass-Density First Moments

22.1 Status, Sources, and Dependencies

Status: proposal plus checked local derivation. Local sources support Galilean invariance as a physical assumption in nonrelativistic fermion models and give examples where conservation laws produce continuity equations. The finite open-chain first-moment formula below is a local derivation checked by the Julia test suite.

Dependencies: CA-18–CA-20 and CONVENTIONS.md (h), (i).

22.2 Input Data

Let $\Lambda = \{1, \ldots, N\}$ be a finite open chain with positions x_j. A Galilean lattice candidate needs, in addition to the Hamiltonian, a conserved mass or particle-number density:

$$M = \sum_j \mu_j, \qquad [H, M] = 0.$$

For identical particles of mass m, a common special case is

$$\mu_j = m n_j,$$

where n_j is a number density.

The time-zero Galilean boost candidate is the first mass moment

$$G^{\mathrm{lat}}(0) = \sum_j x_j \mu_j.$$

This is the Galilean analogue of mX, not the Lorentzian energy moment $\sum_j x_j h_j$.

22.3 Discrete Continuity Assumption

Assume a nearest-neighbour open-chain continuity equation with bond currents $J_{j+1/2}$:

$$i[H, \mu_j] = J_{j-1/2} - J_{j+1/2},$$

where endpoint currents are either absent or treated as boundary terms. This assumption is model data. It does not follow merely from writing $H = \sum_j h_j$.

22.4 First-Moment Derivation

Multiply the continuity equation by x_j and sum:

$$\begin{aligned} i\left[H, \sum_j x_j \mu_j\right] &= \sum_j x_j (J_{j-1/2} - J_{j+1/2}) \\ &= \sum_{j=1}^{N-1} (x_{j+1} - x_j) J_{j+1/2} \quad + \quad \text{boundary terms.} \end{aligned}$$

For a unit-spaced open chain away from endpoints,

$$i[H, G^{\mathrm{lat}}(0)] \simeq \sum_{j=1}^{N-1} J_{j+1/2}.$$

This is the lattice momentum candidate associated with the Galilean boost. For nonuniform positions the coefficients are the spacings

$$\Delta x_j = x_{j+1} - x_j,$$

and the test suite records a coefficient check for this shard.

22.5 Relation to Center of Mass

The local paired-fermion source uses Galilean invariance to explain a response in terms of center-of-mass acceleration while relative motion is unaffected. That is exactly the intuition this shard turns into compiler data:

$$\text{Galilean boost} \quad \rightsquigarrow \quad \text{center-of-mass / mass-first-moment generator.}$$

The source does not prove the lattice formula above. It supports the physical role of center-of-mass motion under Galilean invariance; the lattice identity is the checked local derivation.

22.6 What Is and Is Not Proven

The finite identity proves only that, given a local continuity equation, the time derivative of the mass first moment is a current sum. It does not prove:

1. that the current sum converges to continuum momentum,
2. that the finite lattice has exact Galilei symmetry,
3. that the mass current equals momentum density in every model,
4. that boundary terms vanish without a stated boundary condition.

Those are acceptance-test obligations for later shards or model-specific examples.

23 Compiler Rules and Examples for Galilean Targets

23.1 Status, Sources, and Dependencies

Status: compiler specification shard. This shard does not implement the compiler. It specifies the extra fields a Galilean target requires, using the checked algebra and first-moment derivations from CA-19–CA-21.

Dependencies: CA-08, CA-16, and CA-18–CA-21.

23.2 Target Declaration

A Galilean compiler target is a structured request:

$$\mathrm{Target}(\mathrm{Gal}(d),\ \mathrm{projective},\ \mathrm{mass}).$$

It asks for:

H	time translation,
P_a	spatial translations / momenta,
J_{ab}	rotations,
G_a	Galilean boosts,
M	central mass generator.

The finite candidate layer is allowed to output formulas and witnesses. It is not allowed to assert continuum Galilean covariance without scaling-limit evidence.

23.3 Required Input Slots

Compared with CA-16, the Galilean target adds the following required slots:

μ_x or n_x	local mass or particle-number density,
$M = \sum_x \mu_x$	conserved total mass/number,
$[H, M] = 0$	finite commutator witness,
$J_{x,e}$	current witness for a discrete continuity equation,
m or M	central charge/operator for the projective representation,
boundary policy	open, periodic with branch, or thermodynamic-limit handling.

If any of the first four are absent, boosts and momenta for a Galilean target are underdetermined.

23.4 Generated Candidates

For an open chain, the deterministic candidate formulas are

$$G^{\mathrm{lat}}(0) = \sum_j x_j \mu_j,$$

and, when a continuity witness is supplied,

$$P^{\mathrm{lat}} := i[H, G^{\mathrm{lat}}(0)] = \sum_{j=1}^{N-1} (x_{j+1} - x_j) J_{j+1/2} \quad + \quad \text{boundary terms.}$$

Rotations in higher dimension require momentum or current-density data:

$$J_{ab}^{\mathrm{lat}} \approx \sum_x (x_a p_b(x) - x_b p_a(x)).$$

Thus rotations are second-stage candidates: first compile current/momentum density, then form the spatial first moment.

23.5 Example 1: One Free Particle

Input:

$$\mathcal{H} = L^2(\mathbb{R}^d), \qquad H = \frac{P^2}{2m}, \qquad [X_a, P_b] = i\delta_{ab} I.$$

Output at $t = 0$:

$$G_a = mX_a, \qquad [G_a, P_b] = im\delta_{ab} I, \qquad i[H, G_a] = P_a.$$

Status: checked formal derivation on a common core, with unbounded-operator domain work deferred.

23.6 Example 2: Number-Conserving Open Chain

Input:

$$H, \quad n_j, \quad [H, \sum_j n_j] = 0, \quad i[H, n_j] = J_{j-1/2} - J_{j+1/2}.$$

For identical mass m,

$$\mu_j = mn_j, \qquad G^{\mathrm{lat}}(0) = m \sum_j x_j n_j.$$

The compiler may output the current-sum momentum candidate with coefficient $m(x_{j+1} - x_j)$. It must still mark continuum Galilean covariance as a proof obligation.

23.7 Example 3: Hamiltonian Density Only

Input:

$$H = \sum_j h_j$$

with no n_j, no μ_j, and no current witness. For a Lorentzian target, CA-12 permits the energy-momentum proposal. For a Galilean target, the compiler must return:

```
CompileError: missing mass/number density and continuity witness.
```

Using $\sum_j x_j h_j$ here would be a convention bug.

23.8 Example 4: Spin Chain with No Particle Number

A spin chain may have a perfectly good Hamiltonian density and exact lattice translations. Unless a conserved density is explicitly declared as mass or particle number, it does not compile as a Galilean particle system. The correct output is either a failed Galilean target or a different target declaration.

23.9 Example 5: Direct Sum of Mass Sectors

For

$$\mathcal{H} = \mathcal{H}_{m_1} \oplus \mathcal{H}_{m_2},$$

the central generator is not one scalar if $m_1 \neq m_2$. It is

$$M = m_1 I_{\mathcal{H}_{m_1}} \oplus m_2 I_{\mathcal{H}_{m_2}}.$$

The compiler can carry the OR carrier, but an irreducible fixed-mass sector selection is a separate projection/sector step in the sense of CA-06.

23.10 Next Obligations

The next Galilean work needs:

1. a registered source for the standard mass central extension,
2. model examples with explicit lattice number-current continuity,
3. finite commutator tests for those models,
4. scaling maps proving that current sums converge to continuum momenta,
5. a decision on periodic boundary branch conventions.

Only after these are present should the compiler promote Galilean lattice candidates from formulas to continuum symmetry evidence.

24 Gaussian Boson Lorentz Generator Roadmap

24.1 Status, Sources, and Dependencies

Status: roadmap shard with checked seed. This shard scopes a multi-shard programme. CA-24 supplies the first checked symbol calculation. The stronger continuum conclusion is not yet proved for a general coefficient class.

Dependencies: CA-10–CA-17 for lattice Lorentz motivation and acceptance tests, and CONVENTIONS.md (j). This shard starts the Gaussian-boson block.

24.2 Problem Statement

We now restrict the lattice-symmetry problem to a solvable class: translation-invariant Gaussian bosonic lattice systems in spatial dimensions

$$d = 1, 2, 3, \qquad D = d + 1.$$

The physical target is Lorentz/Poincare covariance in the scaling limit. The expected endpoint is the scalar free field:

$$\omega(k)^2 = m^2 + |k|^2$$

after the correct normalization of lattice spacing, velocity, field strength, and mass.

The inputs are not merely a matrix Hamiltonian. The compiler receives:

$\Lambda_{\epsilon,L}$	periodic hypercubic lattice or infinite lattice,
q_x, p_x	canonical bosonic field variables,
V_r	translation-invariant quasi-local quadratic coefficients,
ϵ	lattice spacing,
S_ϵ	state/scaling-map data, when a continuum claim is requested.

The first coefficient tier is the scalar canonical Hamiltonian

$$H = \frac{1}{2}\sum_x p_x^2 + \frac{1}{2}\sum_{x,r} q_x V_r q_{x+r}.$$

Full bosonic BdG/pairing systems are a later tier, not silently included in the first convention.

24.3 North Star

For each admissible coefficient family $V_r^{(\epsilon)}$, compute the candidate generators as functions of the coefficients:

$$H, \quad P_a, \quad J_{ab}, \quad K_a,$$

then compute the residuals in the Poincare algebra. The residuals become conditions on the symbol

$$\omega_\epsilon(k)^2 = \sum_r V_r^{(\epsilon)} e^{i\epsilon k\cdot r}.$$

The end goal is a theorem-shaped statement:

Gaussian lattice coefficients satisfying locality, positivity,
single low-energy minimum, and isotropic Klein-Gordon Taylor data
$\Longrightarrow$ scaling limit is the free scalar/Klein-Gordon Poincare system.

CA-15 proof boundary. This is a theorem-shaped target, not a theorem already obtained from the symbol checks. A continuum claim still needs the CA-15 payload: scaling maps, state sequence, topology, operator domains, local observable covariance, and boundary/renormalization policy. The value of the Gaussian block is that the algebraic residuals can be made explicit before attempting that proof.

24.4 Shard Roadmap

The current Gaussian block is:

CA-24. Symbol calculus for the scalar coefficient tier, including the boost-time residual and the checked nearest-neighbour Klein-Gordon seed.

CA-25. Diagonalization into one-particle symbols, with the sourced free-scalar Fock construction separated from the local coefficient-symbol derivation and BdG pairing deferred.

CA-26. Smooth-patch one-particle generator algebra for H, P, J, K, exact brackets, dispersion-dependent residuals, and only formal $\mathrm{d}\Gamma$ bookkeeping on the algebraic finite-particle core.

CA-27. Lorentz residual conditions on coefficients: Taylor constraints, anisotropic failures, and doubler counterexamples.

CA-28. Numerical invariant suite for scalar symbols and finite periodic stiffness matrices, covering Klein-Gordon, anisotropic, and doubler examples.

CA-29. Real-space energy-density convention for the scalar Gaussian Hamiltonian: cell volume, bond sharing, boundary policy, vacuum subtraction, and improvement terms before any position-weighted use.

CA-30. One-dimensional stress-energy candidate dictionary: T_{00} from CA-29, checked finite-open-chain T_{01} from A4, and proposal-level T_{10} and T_{11} witnesses left for later work.

CA-31. One-dimensional current-symbol bridge: the translation-invariant integrated T_{01} current symbol equals the nearest-neighbour Klein-Gordon boost-time symbol, while finite open-chain $\mathrm{d}\Gamma(k)$ equivalence remains open.

CA-32. Higher-dimensional cell-current proposal: oriented $d = 2, 3$ support data, energy-flux equations, momentum-density witnesses, stress-current witnesses, and rotation/boundary diagnostics as compiler inputs rather than convergence claims.

The remaining queued follow-up block is proposal work, not completed stress-energy material:

Open-chain momentum equivalence. Upgrade the CA-31 symbol bridge to finite open-chain wave-packet and $\mathrm{d}\Gamma(k)$ comparison data, including boundary residuals.

Stress-energy numerical suite. Add numerical checks only after candidate formulas and source-backed expected identities exist.

Source acquisition. Use the newly registered discrete Gaussian/free-field Virasoro source only within its square-grid dGFF change-of-measure scope. Separate source anchors are still required for harmonic-chain energy-current formulas and any continuum stress-energy/Poincare-generator claim.

24.5 Side Quests

Source acquisition. The registered local OAR sources cover the free scalar lattice scaling limit and spacetime translations. They explicitly warn that Lorentz/conformal convergence requires further work. The discrete Gaussian free-field Virasoro source is now local; it supports lattice-level change-of-measure Virasoro claims for the square-grid dGFF, not a continuum stress-energy or Poincare-generator formula.

Generator ambiguity. A local Hamiltonian density is not unique:

$$h_x \mapsto h_x + \nabla \cdot b_x$$

changes first moments by boundary or current terms. The Gaussian block must track which density convention produced $K_a = \sum_x x_a h_x$.

Periodic positions. Numerical tests use periodic lattices and finite Brillouin grids. First moments, position coordinates, and finite-grid momentum-coordinate generators need a branch, sawtooth, Fourier interpolation, or another explicit periodic coordinate convention. Until that is fixed, early finite tests are symbol/eigenvalue checks on the periodic grid, or open-chain first-moment checks.

BdG pairing. A general quasi-local Hamiltonian in creation and annihilation operators may include pairing terms. The scalar q, p tier captures Klein-Gordon directly; BdG requires symplectic positivity and a stable bosonic Bogoliubov convention before generator formulas are promoted.

24.6 Numerical Verification Plan

The Julia layer should stay at the invariant level:

1. symbol from coefficients equals the analytic dispersion,
2. $\frac{1}{2}\nabla\omega_\epsilon(k)^2$ equals the smooth-patch boost-time residual,
3. $\frac{\sin(\epsilon k_a)}{\epsilon} \to k_a$ for the KG lattice,
4. anisotropic Hessians fail the Lorentz residual checks,
5. finite Fourier matrices diagonalize periodic stiffness matrices.

The first three checks are implemented with CA-24.

24.7 Acceptance Boundary

The roadmap rows above remain scoped by the CA-15 proof boundary stated next to the theorem-shaped goal. Passing symbol or finite-matrix tests is not yet a proof of Lorentz invariance.

25 Gaussian Boson Symbol Calculus and the Boost-Time Residual

25.1 Status, Sources, and Dependencies

Status: sourced model class plus checked flat local derivation. The standard lattice scalar-field Hamiltonian and dispersion are sourced. The boost-time residual calculation is a local one-particle derivation on flat $L^2(dk)$, checked by Julia on positive-dispersion patches. It is not yet a sourced OAR or mass-shell Poincare representation theorem, and massless zero-mode examples are coefficient-level tests only.

Dependencies: CA-11, CA-12, CA-23, and CONVENTIONS.md (h), (j), (k).

25.2 Coefficient Tier

Fix $d \in \{1, 2, 3\}$. On a periodic hypercubic lattice with spacing ϵ, use canonical fields

$$[q_x, q_y] = 0, \qquad [p_x, p_y] = 0, \qquad [q_x, p_y] = i\delta_{xy}.$$

The first scalar Gaussian tier is

$$H = \frac{1}{2}\sum_x p_x^2 + \frac{1}{2}\sum_{x,r} q_x V_r q_{x+r}, \qquad V_r = V_{-r} \in \mathbb{R}.$$

Quasi-locality means that V_r has enough decay, or finite range in the first examples, so that the Fourier symbol and its derivatives used below exist.

Define

$$\omega_\epsilon(k)^2 := \sum_r V_r e^{i\epsilon k\cdot r}.$$

For a stable scalar model, this symbol is non-negative, with the zero-mode policy stated separately in the massless case. The helper parameter m is a nonnegative mass label; the generator statements below use only the stricter case $\omega_\epsilon(k) > 0$ on the chosen patch.

25.3 Nearest-Neighbour Klein-Gordon Example

The sourced lattice scalar-field example has, in the notation of this shard,

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2}.$$

Therefore

$$\begin{aligned}\omega_{\epsilon,m}(k)^2 &= m^2 + \frac{2d}{\epsilon^2} - \frac{1}{\epsilon^2}\sum_{a=1}^{d}(e^{i\epsilon k_a} + e^{-i\epsilon k_a}) \\ &= m^2 + \frac{2}{\epsilon^2}\sum_{a=1}^{d}(1 - \cos(\epsilon k_a)).\end{aligned}$$

For fixed physical k and $\epsilon \to 0$,

$$\omega_{\epsilon,m}(k)^2 = m^2 + |k|^2 + O(\epsilon^2|k|^4).$$

25.4 Momentum-Space Position

After diagonalization, this shard uses the flat local momentum space $L^2(\mathcal{U}, dk)$. The one-particle Hamiltonian is multiplication by $\omega(k)$, with $\omega(k) > 0$ on $\mathcal{U}$, and

$$X_a = i\partial_{k_a}$$

on a common smooth core. The time-zero boost candidate corresponding to the energy first moment is the flat-measure symmetrized operator

$$K_a = \frac{1}{2}(X_a\omega + \omega X_a).$$

The symmetrization is the one-particle version of taking a self-adjoint first moment of the energy density. If the one-particle scalar product is changed, the adjoint and hence the boost formula must be rederived; no unitary equivalence with the OAR weighted space or Schottenloher mass-shell realization is asserted here.

25.5 Boost-Time Residual

For a test wavefunction ψ,

$$\begin{aligned}[K_a, \omega]\psi &= \frac{1}{2}(X_a\omega^2 - \omega X_a\omega)\psi \\ &= i\,\omega\,(\partial_a\omega)\psi.\end{aligned}$$

Thus

$$i[\omega, K_a] = \omega\,\partial_a\omega = \frac{1}{2}\partial_a\omega(k)^2.$$

This is the algebraic output of the boost-time commutator. With the vector-field-to-Stone sign map of CONVENTIONS.md (k), the continuum Poincare target is $i[H, K_a] = P_a$. Taking $P_a = k_a$, one needs

$$\frac{1}{2}\partial_a\omega(k)^2 = k_a.$$

Integrating over k gives

$$\omega(k)^2 = m^2 + |k|^2$$

on the connected low-energy patch. With speed c, the right side is c^2k_a, giving $\omega(k)^2 = m^2 + c^2|k|^2$.

25.6 Lattice Residual

For the nearest-neighbour lattice Klein-Gordon symbol,

$$\frac{1}{2}\partial_a\omega_{\epsilon,m}(k)^2 = \frac{\sin(\epsilon k_a)}{\epsilon}.$$

Therefore the boost-time commutator has the correct continuum limit:

$$\frac{\sin(\epsilon k_a)}{\epsilon} = k_a + O(\epsilon^2 k_a^3).$$

The Julia tests check this in $d = 1, 2, 3$, both from the closed formula and by evaluating the coefficient symbol $\sum_r V_r e^{i\epsilon k\cdot r}$.

25.7 Rotation Residual

For the one-particle rotation candidate

$$J_{ab} = X_a P_b - X_b P_a, \qquad P_a = k_a,$$

one obtains

$$i[\omega, J_{ab}] = k_b \partial_a \omega - k_a \partial_b \omega.$$

Thus rotations commute with the Hamiltonian exactly when the dispersion is radial on the momentum region under consideration. For a hypercubic lattice this is not exact over the full Brillouin zone, but the Klein-Gordon expansion is isotropic to quadratic order at $k = 0$.

25.8 Boundary

This shard is a one-particle symbol calculation. The many-body generators are second quantizations plus vacuum-energy and domain policies. Real-space locality of $K_a = \sum_x x_a h_x$, boundary terms, and convergence of the second-quantized operators are deferred to later shards.

26 Gaussian Boson Diagonalization and the One-Particle Symbol

26.1 Status, Sources, and Dependencies

Status: sourced template plus flat local coefficient derivation. The standard lattice free-scalar Hamiltonian, Fock representation, continuum one-particle space, and second-quantized Hamiltonian are sourced. The extension from nearest-neighbour coefficients to an arbitrary real translation-invariant scalar symbol is a local algebraic calculation on the flat $L^2(dk)$ symbol space fixed in CONVENTIONS.md (j).

Dependencies: CA-23–CA-24 and CONVENTIONS.md (a), (f), (j).

26.2 Scalar Coefficient Tier

Fix $d \in \{1, 2, 3\}$. The first Gaussian tier is

$$H = \frac{1}{2}\sum_x p_x^2 + \frac{1}{2}\sum_{x,r} q_x V_r q_{x+r}, \qquad V_r = V_{-r} \in \mathbb{R}.$$

On an infinite translation-invariant lattice, or on a periodic lattice before taking volume limits, the Fourier transform sends the potential kernel to

$$\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}.$$

The reality condition $V_r = V_{-r}$ makes this symbol real:

$$\omega_\epsilon(k)^2 = V_0 + \sum_{r>0} 2V_r \cos(\epsilon k \cdot r).$$

The stability condition for this tier is

$$\omega_\epsilon(k)^2 \geq 0$$

on the Brillouin zone. The `mass` parameter is a nonnegative label; in the nearest-neighbour Klein-Gordon family, $m > 0$ gives the positive-dispersion case used by the current generator statements. For a general coefficient symbol the required condition remains $\omega_\epsilon(k) > 0$ on the chosen patch. When $m = 0$, zero modes are coefficient data only until a zero-mode convention excludes or otherwise controls them.

26.3 Mode Decoupling

The Hamilton equations are a local derivation from the canonical commutators:

$$\dot{q}_x = p_x, \qquad \dot{p}_x = -\sum_r V_r q_{x+r}.$$

Fourier transforming gives

$$\dot{\widehat{q}}(k) = \widehat{p}(k), \qquad \dot{\widehat{p}}(k) = -\omega_\epsilon(k)^2 \widehat{q}(k),$$

and hence

$$\ddot{\widehat{q}}(k) + \omega_\epsilon(k)^2 \widehat{q}(k) = 0.$$

Thus each momentum mode is an oscillator with frequency $\omega_\epsilon(k)$. This is the algebraic reason the compiler can reduce the scalar Gaussian tier to multiplication by a dispersion symbol.

26.4 Complex Structure and Fock Space

For $\omega > 0$ on the chosen momentum patch, the oscillator complex structure is

$$(q, p) \longmapsto (-\omega p, \omega^{-1} q).$$

The sourced free-scalar construction uses this pattern with $\gamma_m(k) = \sqrt{|k|^2 + m^2}$ in the continuum and with $\gamma_m^{(N)}(k)$ on the lattice. It also realizes the Weyl algebra in a bosonic Fock representation by creation and annihilation operators. The source scalar product is the OAR one on real Cauchy data, with $\gamma_m^{-1/2}\widehat{q} + i\gamma_m^{1/2}\widehat{p}$; the text then realizes the representation on a mass-independent flat L^2 Fock space. This shard uses only the resulting flat symbol bookkeeping and does not prove that the boost formula of CA-24/CA-26 intertwines the OAR or mass-shell representation.

The compiler-level output for the positive-dispersion scalar tier is therefore:

$\mathfrak{h}_\epsilon$	flat one-particle momentum space on the chosen patch or grid,
h_ϵ	multiplication by $\omega_\epsilon(k)$,
$\mathcal{F}_+(\mathfrak{h}_\epsilon)$	symmetric Fock completion,
H_ϵ	$d\Gamma(h_\epsilon) + E_{\epsilon,0}$ if a vacuum-energy policy is named.

The additive constant $E_{\epsilon,0}$ is not part of the commutator algebra for the generators considered here, but it is part of a Hamiltonian-density normal-ordering policy and must be recorded before using ground-state energies as numerical claims.

26.5 What Is Not Yet Included

This shard does not define the full bosonic BdG tier. A general quadratic Hamiltonian in creation and annihilation operators may include pairing terms, and then the correct diagonalization is a symplectic/Bogoliubov problem with positivity, implementability, and domain conditions. Those conventions are not fixed here.

The current compiler rule is deliberately narrower:

$$\text{scalar } (q, p) \text{ tier} \quad \Rightarrow \quad \omega_\epsilon(k)^2 \text{ symbol} \quad \Rightarrow \quad h_\epsilon = \omega_\epsilon(k).$$

BdG inputs must fail closed or route to a later convention until that tier is written.

26.6 Compiler Consequence

For the positive-dispersion scalar Gaussian block, later candidate generators can be constructed first on the one-particle smooth momentum core and then second-quantized. The coefficient data enter only through $\omega_\epsilon(k)^2$ and its derivatives, while the continuum proof still requires the CA-15 payload: scaling maps, state convergence, domains, and observable covariance.

27 One-Particle Generator Algebra for Gaussian Bosons

27.1 Status, Sources, and Dependencies

Status: flat local derivation with limited Julia checks. The Poincare target, Stone convention, and free-boson Fock realization are sourced. The commutator identities in this shard are local differential-operator derivations on a flat $L^2(dk)$ smooth momentum-space core. The Julia suite currently checks coefficient/symbol consequences: Klein-Gordon symbols, boost-time symbols, sampled rotation-Hamiltonian residuals, sampled boost-boost residual coefficient data, low-energy Hessian examples, finite periodic massive spectra, the massless doubler coefficient-rejection witness, and the Gaussian tolerance policy. It does not check the full generator bracket table, finite coordinate commutators, or any second-quantized identity. The smooth-core derivation is not yet a sourced OAR or mass-shell Poincare representation theorem.

Dependencies: CA-11, CA-24–CA-25, and CONVENTIONS.md (h), (j), (k).

27.2 Momentum Core

There are two separate settings.

Smooth patch. Work on a connected smooth momentum patch $\mathcal{U}$ where $\omega(k) > 0$, with the flat measure dk. Let

$$\mathcal{D}_0 = C_c^\infty(\mathcal{U}) \subset L^2(\mathcal{U}, dk)$$

and define

$$X_a = i\partial_{k_a}, \qquad h = \omega(k), \qquad p_a = k_a.$$

The one-particle Hamiltonian and momenta are multiplication operators:

$$H^{(1)} = h, \qquad P_a^{(1)} = p_a.$$

The rotation and time-zero boost candidates are

$$J_{ab}^{(1)} = X_a P_b - X_b P_a, \qquad K_a^{(1)} = \frac{1}{2}(X_a h + h X_a).$$

These are written in the self-adjoint sign convention of CONVENTIONS.md (k); in particular J_{ab} is the Gaussian rotation sign, not the raw lowered-index CA-11 vector field M_{ab}. All commutators below are identities on $\mathcal{D}_0$; promoting them to self-adjoint generator statements is a separate domain theorem. Promoting this flat formula to the OAR one-particle Hilbert space or Schottenloher's mass-shell/free-boson Poincare representation additionally requires a sourced or locally derived unitary intertwiner.

Finite periodic grids. On a finite periodic Gaussian lattice, the dual grid is used for multiplication symbols and Fourier eigenvalue checks. The current finite suite does not place the differential operator $X_a = i\partial_{k_a}$ on that grid and does not claim an exact finite $[X_a, p_b] = i\delta_{ab}$ commutator. Periodic first-moment or momentum-coordinate generator tests require an explicit branch, sawtooth, Fourier-interpolation, or comparable periodic-coordinate convention before they can be stated. Thus the exact differential commutators below are smooth-patch algebra; the finite periodic checks remain symbol/eigenvalue checks.

27.3 Exact Translation and Rotation Brackets

On the smooth core, because h and p_a are multiplication operators,

$$i[h, p_a] = 0.$$

The canonical differential identity is the smooth-patch identity

$$[X_a, p_b] = i\delta_{ab}.$$

Therefore

$$i[P_c, J_{ab}] = \delta_{ac}P_b - \delta_{bc}P_a.$$

This is independent of the dispersion. It is kinematical on the smooth patch: it comes from the chosen momentum coordinate and the differential representation of rotations, in the rotation sign convention of CONVENTIONS.md (k).

27.4 Boost-Momentum Bracket

For the boost candidate,

$$\begin{aligned}[P_b, K_a] &= \frac{1}{2}\big([P_b, X_a]h + h[P_b, X_a]\big) \\ &= -i\delta_{ab}h.\end{aligned}$$

Hence

$$i[P_b, K_a] = \delta_{ab}H.$$

This is the boost-momentum line of the self-adjoint target table in CONVENTIONS.md (k). On the smooth patch, this part of the Poincare algebra is exact for every positive scalar dispersion because it only uses K_a as the symmetrized first moment of energy.

27.5 Boost-Time Residual

CA-24 derived

$$i[H, K_a] = \frac{1}{2}\partial_a\omega(k)^2.$$

The continuum target with speed $c = 1$ wants $P_a = k_a$. The residual is

$$R_a(k) := \frac{1}{2}\partial_a\omega(k)^2 - k_a.$$

For speed c, replace k_a by c^2k_a. This is the first algebraic filter on Hamiltonian coefficients. Julia now exposes it both as the boost-time symbol $B_a = \frac{1}{2}\partial_a\omega^2$ and as the residual vector $R_a = B_a - c^2k_a$, still only at coefficient-symbol level.

27.6 Rotation-Hamiltonian Residual

The Hamiltonian commutes with rotations iff the dispersion is radial on the patch:

$$i[h, J_{ab}] = k_b\partial_a\omega - k_a\partial_b\omega.$$

Equivalently,

$$k_b\partial_a\omega^2 - k_a\partial_b\omega^2 = 0$$

where $\omega > 0$. Hypercubic lattice symbols usually fail this globally on the Brillouin zone but may satisfy it to leading order at a low-energy minimum. The Julia helper records the sampled antisymmetric matrix

$$2(k_bB_a - k_aB_b), \qquad B_a = \frac{1}{2}\partial_a\omega^2,$$

so the factor of two follows the boost-time convention instead of being re-derived independently.

27.7 Boost-Boost Residual

Write

$$B_a(k) := \frac{1}{2}\partial_a \omega(k)^2, \qquad R_a(k) := B_a(k) - k_a.$$

Equivalently $B_a = h\partial_a h$. Since

$$K_a = i\left(h\partial_a + \frac{1}{2}\partial_a h\right),$$

the commutator of the first-order differential parts gives

$$i[K_a, K_b] = J_{ab} + R_b X_a - R_a X_b.$$

The leading $+J_{ab}$ is the sign dictated by CONVENTIONS.md (k): CA-11 has $[M_{0a}, M_{0b}] = -M_{ab}$, while the self-adjoint rotation here is $J_{ab} = -A_{M_{ab}}$. Here the possible divergence term vanishes because R is a gradient residual: $\partial_a R_b = \partial_b R_a$. Thus the boost-boost bracket closes to the rotation bracket whenever the boost-time residual R_a vanishes on the patch. The Julia helper records only the sampled coefficient tensor for the first-order residual $R_b X_a - R_a X_b$: its (a, b, c) entry is the coefficient of X_c for the ordered boost pair (a, b). It is not a finite-grid coordinate commutator and does not check the boost-boost operator bracket. The remaining caveat is analytic, not algebraic: all of these are unbounded operators and still require a common self-adjointness/core theorem before this becomes a continuum representation statement.

27.8 Second Quantization

Let

$$\mathcal{F}_{\mathrm{alg}}(\mathcal{D}_0) := \bigoplus_{n=0}^{\mathrm{fin}} \mathcal{D}_0^{\otimes_s n} \subset \mathcal{F}_+\big(L^2(\mathcal{U}, dk)\big)$$

be the algebraic finite-particle core. The formal many-body candidates are

$$H = d\Gamma(h), \quad P_a = d\Gamma(p_a), \quad J_{ab} = d\Gamma(J_{ab}^{(1)}), \quad K_a = d\Gamma(K_a^{(1)}),$$

with a separate vacuum-energy convention if absolute Hamiltonian values are used. The formal core identity is

$$[d\Gamma(A), d\Gamma(B)]\psi = d\Gamma([A, B])\psi, \qquad \psi \in \mathcal{F}_{\mathrm{alg}}(\mathcal{D}_0),$$

whenever the one-particle commutator identity is already established on $\mathcal{D}_0$ and the displayed products preserve that algebraic domain. This is a local $d\Gamma$ bookkeeping statement on the algebraic finite-particle core, not a Julia-checked invariant and not a self-adjoint many-body representation theorem. The current tests do not lift residuals to Gaussian Fock space; domain/self-adjointness for the unbounded $d\Gamma$ generators and the sourced OAR/mass-shell bridge remain open proof obligations.

28 Lorentz Residual Conditions on Gaussian Coefficients

28.1 Status, Sources, and Dependencies

Status: checked local coefficient calculus. The sourced massive free scalar model supplies the target dispersion. The general coefficient residuals are local Taylor calculations from $\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}$. The examples in this shard are checked in Julia; the massless doubler is only a coefficient-level rejection witness, not a positive-dispersion generator example.

Dependencies: CA-24–CA-26 and CONVENTIONS.md (j).

28.2 Gradient Residual

For a scalar coefficient family,

$$\omega_\epsilon(k)^2 = \sum_r V_r e^{i\epsilon k\cdot r}.$$

The boost-time residual from CA-26 is

$$R_a(k) = \frac{1}{2}\partial_a \omega_\epsilon(k)^2 - c^2 k_a.$$

Explicitly,

$$\frac{1}{2}\partial_a \omega_\epsilon(k)^2 = \frac{i\epsilon}{2}\sum_r r_a V_r e^{i\epsilon k\cdot r}.$$

When $V_r = V_{-r}$, this is real and can be written as a sine sum:

$$\frac{1}{2}\partial_a \omega_\epsilon(k)^2 = -\epsilon \sum_{r>0} r_a V_r \sin(\epsilon k\cdot r),$$

where $r > 0$ denotes any fixed choice of one representative from each $\{r, -r\}$ pair.

28.3 Low-Energy Hessian Condition

At a symmetric minimum $k = 0$, the first derivative vanishes. The Hessian is

$$\partial_a\partial_b \omega_\epsilon(0)^2 = -\epsilon^2 \sum_r r_a r_b V_r.$$

Therefore the quadratic Lorentz-speed condition is

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = c^2 \delta_{ab}.$$

This is the first finite-dimensional coefficient test. In compiler form:

$$\mathsf{HessResidual}_{ab} := -\frac{\epsilon^2}{2}\sum_r r_a r_b V_r - c^2\delta_{ab}.$$

It must vanish at the chosen low-energy point before the model can be routed to a Klein-Gordon continuum target.

It is not a positivity or patch certificate. The Hessian only sees the Taylor coefficient at the chosen point; a finite-grid symbol can still be negative elsewhere or have additional low-energy minima.

28.4 Nearest-Neighbour Klein-Gordon

For

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2},$$

one obtains

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = \delta_{ab}.$$

The gradient residual is the stronger finite-ϵ expression

$$R_a(k) = \frac{\sin(\epsilon k_a)}{\epsilon} - k_a,$$

which vanishes as $O(\epsilon^2 k_a^3)$ for fixed physical momentum.

28.5 Anisotropic Counterexample

Let the nearest-neighbour coefficients carry different speeds c_a:

$$\omega_\epsilon(k)^2 = m^2 + \frac{2}{\epsilon^2}\sum_a c_a^2(1 - \cos(\epsilon k_a)).$$

Then

$$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r = c_a^2 \delta_{ab}.$$

Unless all c_a agree with the target speed, the boost-time residual cannot match a single Lorentz cone. The Julia test suite checks this as a deliberate failure case, not as an approximate success.

28.6 Hessian-Passing Instability Witness

In $d = 1$, at $\epsilon = 1$, consider the scalar coefficients

$$V_0 = 0, \qquad V_{\pm 1} = 1, \qquad V_{\pm 2} = -\frac{1}{2}.$$

The symbol is

$$\omega(k)^2 = 2\cos k - \cos(2k).$$

Its low-energy Hessian data pass the speed-one condition:

$$-\frac{1}{2}\sum_r r^2 V_r = 1.$$

However, the $L = 4$ periodic sample has values

$$\omega(0)^2 = 1, \quad \omega(\pi/2)^2 = 1, \quad \omega(\pi)^2 = -3, \quad \omega(3\pi/2)^2 = 1.$$

The checked minimum-data helper records the negative minimum -3, with uncentered label 2 and momentum π, plus centered label -2 and momentum $-\pi$. Enabling its nonnegative finite-grid gate makes the helper raise an error. Thus the Hessian residual can vanish while the finite periodic symbol is unstable.

28.7 Doubler Counterexample

The Hessian condition is necessary but not sufficient. The symbol

$$\omega_\epsilon(k)^2 = m^2 + \frac{1}{\epsilon^2}\sum_a \sin^2(\epsilon k_a)$$

has the correct quadratic Hessian at $k = 0$ when $c = 1$. The checked example sets $m = 0$, so it is massless and has zero modes; by CONVENTIONS.md (j), this is outside the current positive-dispersion generator theorem. Its role is coefficient-level rejection: on even periodic grids it has multiple minima at Brillouin-zone points with $\sin(\epsilon k_a) = 0$. Therefore the compiler also needs a *single-patch* or *sector-selection* condition before a continuum relativistic scalar target is attached.

The current tests record this in $d = 1, 2, 3$ on the $L_a = 4$ periodic grids: the Hessian residual still vanishes at speed 1, while `symbol_minimum_data` finds 2^d zero minima with each centered label coordinate in $\{0, -2\}$. This is a finite-grid witness of extra patches, not a continuum species-count theorem.

28.8 Compiler Rule

For the scalar Gaussian tier, a Lorentz-target compile attempt must emit at least:

mass term	$\omega(0)^2$,
speed matrix	$-\frac{\epsilon^2}{2}\sum_r r_a r_b V_r$,
boost residual	$R_a(k)$,
rotation residual	$k_b\partial_a\omega^2 - k_a\partial_b\omega^2$,
boost-boost residual	$R_b X_a - R_a X_b$,
patch data	finite-grid minimum value, labels, momenta, and count.

Only the mass and speed entries are finite Taylor data. The rotation and boost-boost entries are now sampled smooth-symbol residuals in Julia, with the boost-boost entry represented as first-order X_c-coefficient data rather than as a finite-grid operator commutator. The residual and patch entries are what prevent an anisotropic or doubler model from being mislabeled as a single Klein-Gordon field.

29 Numerical Verification Suite for Gaussian Boson Examples

29.1 Status, Sources, and Dependencies

Status: checked Julia shard. This shard records the numerical invariants now implemented for the scalar Gaussian tier. The checks are not continuum proofs; they are finite algebraic witnesses that the coefficient compiler is computing the intended symbols and residuals. The massless doubler check is a coefficient-level rejection test, not a Fock-generator check. The finite periodic checks below do not assert an exact differential $[X_a, P_b] = i\delta_{ab}$ commutator.

The current numerical surface is deliberately narrow: it checks scalar coefficient structure, finite-periodic symbol values, finite stiffness spectra, assigned Fourier eigenvectors, periodic minimum data, and the implemented coefficient residuals/Hessians, including sampled rotation-Hamiltonian and boost-boost residual data on smooth-symbol momenta. It also checks centered periodic momentum labels for branch-aware low-energy windows, a loud finite-grid nonnegativity gate when requested, and test-local central finite-difference boost-time derivative oracles. It does not test finite boost generators, finite rotation generators, coordinate-generator commutators, or full generator closure.

Dependencies: CA-24–CA-27, INDEX.md, and CONVENTIONS.md (j).

29.2 Why These Tests

The numerical layer is not allowed to assert that a plot "looks relativistic". Each check must compare two independently constructed objects:

$$\text{real-space coefficient data} \quad \longleftrightarrow \quad \text{momentum-space symbol data.}$$

For Gaussian systems this is unusually powerful because the finite periodic stiffness matrix is exactly diagonalized by Fourier modes.

The boost-time derivative oracle follows the same rule. The production boost-time helper is compared to $\frac{1}{2}\nabla_k \omega(k)^2$ obtained by central finite differences of the separate scalar-symbol evaluator. Those tests use generic symmetric finite-range coefficient sets in $d = 1, 2, 3$, with off-axis terms in the higher-dimensional cases and sample momenta away from high-symmetry points.

The rotation and boost-boost residual tests are also smooth-symbol checks. They sample coefficient formulas at named momenta and compare residual sizes or tensor entries; they do not introduce $X_a = i\partial_{k_a}$ as a finite periodic matrix.

29.3 Numerical Tolerance Policy

CONVENTIONS.md (j) fixes the tolerance names used by this shard. With the `GAUSSIAN_` prefix and `_ATOL/_RTOL` suffix understood, the default pairs are

`SYMBOL_IMAG`	$(10^{-10}, 10^{-10})$,
`SYMBOL_VALUE`	$(10^{-10}, 10^{-10})$,
`EIGENVALUE`	$(10^{-10}, 10^{-10})$,
`MINIMUM_COUNT`	$(10^{-10}, 10^{-10})$,
`SMALL_SPACING_RESIDUAL`	$(5 \cdot 10^{-4}, 10^{-10})$.

The symbol-imaginary validator compares the complex symbol with its real part after the dimension and spacing preconditions are checked. The small-spacing pair is a named finite-sample tolerance, not a continuum error bound. The finite-difference derivative oracle is test-local and uses $(2\cdot 10^{-8}, 2\cdot 10^{-8})$; it is not part of the production tolerance surface.

29.4 Structural Coefficient Validation

The scalar Gaussian helpers now fail closed by default on coefficient data outside CONVENTIONS.md (j). The validator requires nonempty data, integer offsets, a common dimension $d \in \{1, 2, 3\}$, real coefficients, and aggregate paired equality $V_r = V_{-r}$ after duplicate offsets are summed. Negative tests cover a missing partner, unequal paired aggregates, complex coefficients, dimension mismatch, and non-integer offsets.

29.5 Periodic Stiffness Matrix

For finite periodic sizes $L_1, \ldots, L_d$, define the stiffness matrix

$$V_{xy} = \sum_r V_r \, \mathbf{1}_{y=x+r \bmod L}.$$

For the Fourier vector

$$\varphi_k(x) = |\Lambda|^{-1/2} e^{i\epsilon k \cdot x},$$

one has the exact identity

$$(V\varphi_k)(x) = \left(\sum_r V_r e^{i\epsilon k \cdot r} \right) \varphi_k(x) = \omega_\epsilon(k)^2 \varphi_k(x).$$

Therefore each dual label n gives a tested eigenvector with its assigned symbol value, and the sorted eigenvalues of the finite stiffness matrix must equal the sorted symbol values on the periodic Brillouin grid. These finite Fourier invariants are checked in Julia for $d = 1, 2, 3$, including mixed odd/even sizes with every $L_a > 2R$ and an off-axis symmetric pair $(1, 1), (-1, -1)$. They are not finite periodic first-moment or momentum-coordinate commutator tests; those require a branch, sawtooth, Fourier-interpolation, or other coordinate convention before they can become numerical claims.

The eigenvector suite also includes a one-sided shift sentinel to pin the implementation direction $y = x + r$. That sentinel is not a scalar Gaussian Hamiltonian claim. The strict default rejects it, and the test reaches this low-level matrix path only through an explicit non-validating keyword.

The eigenvalue oracle deliberately keeps the uncentered grid helper, because the tested symbol spectra are branch-insensitive. The separate centered grid helper returns the centered integer dual label and its physical representative

$$k_a = \frac{2\pi n_a}{\epsilon L_a}.$$

For even $L = 2r$, the branch is $n_a \in \{-r, \ldots, 0, \ldots, r-1\}$, matching the centered dual lattice in the OAR source. Thus, at $L = 4$ and $\epsilon = 1$, the uncentered representative $3\pi/2$ is recorded on the centered branch as $n = -1$ and $k = -\pi/2$. This is only a momentum-label helper; it is not a periodic position-coordinate or generator construction.

29.6 Implemented Example Systems

Nearest-neighbour Klein-Gordon. The tests check the coefficient symbol against

$$m^2 + \frac{2}{\epsilon^2} \sum_a (1 - \cos(\epsilon k_a)),$$

the boost-time symbol against $\sin(\epsilon k_a)/\epsilon$, the residual vector against $\sin(\epsilon k_a)/\epsilon - k_a$, and the low-energy Hessian residual against zero. The sampled rotation and boost-boost residual helpers are small on fixed low-energy momenta as ϵ is made small. The boost-time helper is also checked on non-Klein-Gordon finite-range symbols by the independent finite-difference oracle above.

Anisotropic cones. The tests build coefficients with speeds (c_1, c_2, c_3). They verify that the Hessian is

$$2\,\mathrm{diag}(c_1^2, c_2^2, c_3^2)$$

and that the Lorentz residual is nonzero when the target speed is 1. A separate sampled residual test checks that anisotropy also gives nonzero rotation-Hamiltonian and boost-boost residual data at a generic smooth momentum.

Hessian-passing unstable scalar. The tests build the one-dimensional coefficient witness

$$V_0 = 0, \qquad V_{\pm 1} = 1, \qquad V_{\pm 2} = -\frac{1}{2}.$$

Its speed-one Hessian residual vanishes, but finite periodic sampling at $L = 4$ has minimum -3. The minimum-data helper records one minimum at location 3, uncentered label 2, centered label -2, uncentered momentum π, and centered momentum $-\pi$. Requesting a nonnegative finite-grid symbol raises an error, so this example cannot pass as a stable scalar symbol.

Doubler symbol. The tests build

$$\omega_\epsilon(k)^2 = m^2 + \epsilon^{-2} \sum_a \sin^2(\epsilon k_a)$$

with $m = 0$. In $d = 1, 2, 3$, the speed-one Hessian residual vanishes on the chosen patch, but the $L_a = 4$ periodic grid has 2^d zero minima with centered label coordinates in $\{0, -2\}$. This catches the case where the Hessian at one minimum is correct but the compiler would need a multi-species, sector-selection, or zero-mode policy output. It is deliberately outside the positive-dispersion generator statements. The J6 symbol-residual test also samples a doubler momentum where the rotation residual is globally nonzero and an extra zero where R_a is nonzero relative to the selected $k = 0$ branch.

29.7 Current Julia Surface

The checked helper layer is:

`kg_nearest_neighbor_coefficients`	standard scalar KG coefficients,
`anisotropic_kg_nearest_neighbor_coefficients`	anisotropic coefficient witness,
`doubler_quadratic_coefficients`	coefficient-level doubler witness,
`validate_scalar_coefficients`	structural scalar-coefficient validator,
`scalar_quadratic_symbol`	$\sum_r V_r e^{i\epsilon k\cdot r}$,
`boost_time_symbol_from_coefficients`	$\frac{1}{2}\nabla\omega^2$,
`boost_time_residual_from_coefficients`	$\frac{1}{2}\nabla\omega^2 - c^2 k$,
`rotation_hamiltonian_residual_from_coefficients`	$k_b\partial_a\omega^2 - k_a\partial_b\omega^2$,
`boost_boost_residual_coefficients_from_coefficients`	coefficients of $R_b X_a - R_a X_b$,
`nearest_neighbor_integrated_energy_current_symbol`	1D integrated A4 current symbol,
`low_energy_hessian_from_coefficients`	$\partial_a\partial_b\omega^2(0)$,
`lorentz_hessian_residual`	$\frac{1}{2}\mathrm{Hess}(\omega^2)(0) - c^2 I$,
`periodic_stiffness_matrix`	finite periodic real-space matrix,
`periodic_momentum_grid`	uncentered spectrum grid,
`centered_periodic_momentum_grid`	centered labels and momenta,
`periodic_fourier_vector`	Fourier vector for a dual label,
`periodic_symbol_values`	finite Brillouin-grid eigenvalue oracle,
`symbol_minimum_data`	finite-grid minimum value, labels, momenta, and count,
`count_periodic_symbol_minima`	finite-grid minima counter.

This list is a coverage boundary, not a generator-algebra interface: no helper currently constructs a finite coordinate generator, a finite boost, a finite rotation, or their commutator table.

29.8 Next Numerical Milestones

Proposed next suite additions, none claimed as achieved here, are:

1. systematic sampled residual scans on shrinking low-energy windows,
2. comparison of open-boundary first moments with periodic symbol tests.

Any finite periodic coordinate-generator version of the first-moment comparison requires a branch or Fourier-interpolation convention before it can be promoted beyond a local experiment.

30 Gaussian Boson Real-Space Energy Density

30.1 Status, Sources, and Dependencies

Status: convention, not a theorem. This shard fixes how the scalar Gaussian Hamiltonian is split into site-attached real-space energy densities. It preserves the total Hamiltonian used in the sourced nearest-neighbour scalar model and in CA-24–CA-28, but it does not claim a continuum tensor object, a conservation law, or covariance.

Dependencies: CA-23–CA-28 and CONVENTIONS.md (j), (l).

30.2 Cell Volume and Notation

Let the hypercubic lattice spacing be ϵ, so one spatial cell has volume ϵ^d. In this shard h_x is the physical energy density attached to the cell at x:

$$H_\epsilon = \epsilon^d \sum_x h_x.$$

The integrated site energy is

$$e_x := \epsilon^d h_x, \qquad H_\epsilon = \sum_x e_x.$$

This distinction matters because the generic first-moment notation of CONVENTIONS.md (h) writes $H = \sum_x h_x$; for the Gaussian block, the object to insert there is e_x, not the density h_x.

For fields Φ_x, Π_x and a real symmetric translation-invariant kernel $V_r = V_{-r}$, use

$$H_\epsilon = \epsilon^d \left(\frac{1}{2} \sum_x \Pi_x^2 + \frac{1}{2} \sum_{x,r} \Phi_x V_r \Phi_{x+r} \right).$$

For the nearest-neighbour Klein-Gordon source, the renormalized coefficients are

$$V_0 = m^2 + \frac{2d}{\epsilon^2}, \qquad V_{\pm e_a} = -\frac{1}{\epsilon^2},$$

which give the dispersion recorded in CA-24.

30.3 Equal-Endpoint Bond Split

The chosen local density is

$$h_x = \frac{1}{2}\Pi_x^2 + \frac{1}{2} V_0 \Phi_x^2 + \frac{1}{4} \sum_{r\neq 0} V_r \big(\Phi_x \Phi_{x+r} + \Phi_{x-r}\Phi_x\big).$$

Each nonzero displacement bond is assigned half to each endpoint cell. Summing over x gives

$$\sum_x h_x = \frac{1}{2} \sum_x \Pi_x^2 + \frac{1}{2} \sum_{x,r} \Phi_x V_r \Phi_{x+r},$$

so this convention reproduces the total scalar Hamiltonian above after the cell-volume factor is restored.

The q, p notation in CA-24–CA-28 absorbs the cell volume into the coefficient bookkeeping. In that notation the same displayed formula defines e_x, and the physical density is $h_x = e_x/\epsilon^d$.

30.4 Boundary Policy

The default exact identities are infinite-lattice identities with summable tails, or finite periodic total-energy identities. On a finite open region, the density is obtained from the finite Hamiltonian actually named: include only bonds present in that Hamiltonian, and share each included bond equally between its included endpoints. No exterior ghost sites, omitted half-bonds, or boundary counterterms are implicit.

For periodic finite lattices, wrap-around bonds are included in the total energy split. This does not create a periodic position convention. Any periodic first moment still needs the branch, sawtooth, or interpolation policy required by CONVENTIONS.md (j).

30.5 Vacuum-Energy Policy

The density above is bare. No normal-ordering or vacuum subtraction is implicit. Once a state ω is named, the expectation-subtracted version may be recorded as

$$:h_x:_\omega := h_x - \omega(h_x)\mathbf{1}.$$

This changes H_ϵ by a scalar and therefore drops out of commutators, but it matters for absolute energies, expectation values, and Fourier modes of quadratic densities. The local subtraction source is the free-fermion Koo-Saleur setting, where the scaling-limit representation relies on subtracting vacuum or ground-state expectation values; this shard only records the analogous policy needed before such quantities are used in the scalar Gaussian block.

30.6 Improvement Policy

The default convention has no divergence improvement:

$$h'_x = h_x$$

unless a later shard explicitly names an edge field b, a discrete divergence $(\nabla_\epsilon \cdot b)_x$, and a boundary convention. Such a rewrite

$$h'_x = h_x + (\nabla_\epsilon \cdot b)_x$$

may leave the total energy unchanged on a periodic lattice or for decaying fields, but it is not harmless for position-weighted generators. A discrete summation-by-parts calculation shifts first moments by an edge-field term plus boundary contributions. For example, in one dimension with

$$(\nabla_\epsilon b)_n = \frac{b_{n+1/2} - b_{n-1/2}}{\epsilon}, \qquad x_n = \epsilon n,$$

one has

$$\epsilon \sum_n x_n (\nabla_\epsilon b)_n = -\epsilon \sum_n b_{n+1/2} + \text{boundary terms}.$$

Therefore improvements are separate data, not gauge choices to be silently modded out.

30.7 Compiler Consequence

A scalar Gaussian real-space generator proposal must now carry the tuple

$$\big(\epsilon^d,\ h_x,\ e_x = \epsilon^d h_x,\ \text{boundary policy},\ \text{subtraction policy},\ \text{improvement policy}\big).$$

The total symbol $\omega(k)^2$ remains sufficient for CA-24–CA-28, but it is not sufficient once a construction uses position weights or local quadratic modes. Those later constructions must cite this density convention or perform an explicit convention sweep.

31 One-Dimensional Gaussian Boson Stress-Energy Candidates

31.1 Status, Sources, and Dependencies

Status: proposal with checked T_{01} seed. This shard fixes a candidate dictionary for the finite open one-dimensional scalar Gaussian chain. Only the energy-current entry has a checked local witness. No convergence theorem, no continuum stress tensor, and no Virasoro representation is claimed.

Dependencies: CA-12, CA-23–CA-29, CONVENTIONS.md (j), (k), (l), and the A4 finite open-chain current tests.

31.2 Continuum Motivation and Boundary

The continuum notation $T_{\mu\nu}$ is used here as a target shape, not as an imported lattice theorem. Weinberg supplies the QFT motivation that spacetime translations lead to conserved currents grouped as an energy-momentum tensor and that translation generators are spatial integrals of time components. The same registered OCR also records two caveats that matter for the lattice dictionary: the canonical raised tensor need not be symmetric, while the Belinfante tensor is a modified conserved tensor with the same integrated translation generators and symmetry. The OCR around formulas is noisy, so this shard uses those anchors only for motivation and caveats, not for exact component signs. The self-adjoint commutator signs for any later Poincare comparison are the project convention in CONVENTIONS.md (k), not raw OCR import.

Schottenloher's Axiom 5 is narrower but sharper for the CFT endpoint: in a two-dimensional conformal field theory there are four fields $T_{\mu\nu},\ \mu,\nu \in \{0,1\}$, with symmetry and conservation. The later Luescher–Mack theorem gives tracelessness and two Virasoro mode families under the full Axioms 1–5. We do not use those CFT conclusions to define the lattice T_{10} or T_{11}. They are acceptance targets for a future continuum proof.

31.3 Finite Open Chain Setup

Work first with a finite open chain of N cells and spacing ϵ. The CA-29 density convention distinguishes the physical density h_j from the integrated cell energy

$$e_j = \epsilon h_j, \qquad H_\epsilon = \sum_{j=1}^{N} e_j.$$

Finite commutators use e_j. This avoids inserting an accidental factor of ϵ into a discrete continuity identity.

For the nearest-neighbour scalar Gaussian chain, the A4 helper implements the CA-29 equal-endpoint split of the finite Hamiltonian: kinetic and onsite terms belong to the site, and each present bond $q_j q_{j+1}$ is split equally between its two endpoints. Open-chain means no exterior ghost cells, no missing half-bonds, and no boundary counterterms are implicit.

31.4 Candidate Dictionary

T_{00}^{lat}. The lattice energy-density entry is the CA-29 density:

$$T_{00}^{\mathrm{lat}}(j) := h_j.$$

For exact finite open-chain algebra, the cell-integrated version is

$$E_j := e_j = \epsilon h_j.$$

Thus T_{00}^{lat} is the physical density label, while E_j is the object that appears in the tested quadratic commutators.

T_{01}^{lat}. The energy-current or energy-flux seed lives on open-chain edges:

$$J_{j+1/2} := i[e_j, e_{j+1}], \qquad 1 \leq j < N.$$

Set endpoint currents to zero,

$$J_{1/2} = J_{N+1/2} = 0.$$

The checked finite continuity identity is

$$i[H_\epsilon, e_j] = J_{j-1/2} - J_{j+1/2}, \qquad 1 \leq j \leq N.$$

This is the only stress-energy component in this shard with a Julia witness. It is the lattice candidate corresponding to T_{01} in the cell-integrated continuity law

$$\frac{d}{dt}\int_{\mathrm{cell}\ j} T_{00}\, dx = T_{01}(x_{j-1/2}) - T_{01}(x_{j+1/2}).$$

No continuum normalization or convergence of $J_{j+1/2}$ is claimed.

T_{10}^{lat}. Do not identify T_{10}^{lat} with T_{01}^{lat} at this stage. Define it instead as the future integrated momentum-density comparison target:

$$M_j \quad \text{with} \quad P^{\mathrm{lat}} = \sum_j M_j.$$

The required next witness is that the chosen M_j, or at least its total P^{lat}, matches the Gaussian one-particle flat-symbol momentum target after the appropriate $\mathrm{d}\Gamma(k)$ comparison is made. That is the S3 task, not an assumption in this shard.

T_{11}^{lat}. Define T_{11}^{lat} operationally as the future current of the chosen momentum density. Once M_j is fixed, the candidate stress component must come with edge terms $S_{j+1/2}$ and a checked equation

$$i[H_\epsilon, M_j] = S_{j-1/2} - S_{j+1/2}$$

with an explicit boundary convention. Until that witness exists, there is no lattice T_{11} formula. In particular, T_{11}^{lat} is not defined as $-T_{00}^{\mathrm{lat}}$, not derived from tracelessness, and not inferred from CFT holomorphic factorization.

31.5 Boundary, Subtraction, and Improvement

This candidate dictionary is open-boundary first. Endpoint currents are part of the statement, not small errors:

$$i[H_\epsilon, e_1] = -J_{3/2}, \qquad i[H_\epsilon, e_N] = J_{N-1/2}.$$

Periodic total-energy density splits are allowed by CA-29, but periodic first moments and periodic coordinate-dependent generator tests remain blocked until CONVENTIONS.md (j) fixes a coordinate branch, sawtooth, Fourier interpolation, or equivalent policy.

The default density is bare. A state-dependent subtraction $:h_j:_\omega = h_j - \omega(h_j)\mathbf{1}$ may later be named; it changes the Hamiltonian by a scalar and does not change commutators, but it is part of any claim about absolute energy or density modes. The default improvement is zero. Any later replacement $h'_j = h_j + (\nabla_\epsilon b)_j$ must name the edge field, discrete divergence, and boundary convention before first moments or currents are recomputed.

31.6 Compiler Payload and Acceptance

A compiler output that cites this shard must carry:

density convention	$h_j,\ e_j = \epsilon h_j,$ CA-29 split,
boundary policy	finite open chain, $J_{1/2} = J_{N+1/2} = 0$,
subtraction policy	bare unless a state ω is named,
improvement policy	$b = 0$ unless a discrete divergence is named,
checked T_{01} witness	$i[H_\epsilon, e_j] = J_{j-1/2} - J_{j+1/2}$.

The acceptance boundary is equally explicit. The checked witnesses are the finite open-chain continuity residual in `src/GaussianBosonCurrents.jl` and `test/runtests.jl`, plus the CA-31 translation-invariant current-symbol equality for the nearest-neighbour Klein-Gordon orientation. Still pending are the finite open-chain wave-packet comparison with the flat $d\Gamma(k)$ momentum target; the T_{11} witness consisting of a chosen momentum density M_j and checked current $S_{j+1/2}$; and the continuum witness: scaling maps, domains, state sequence, and covariance convergence. Without those witnesses, the dictionary remains a proposal with checked T_{01} evidence, not a stress-tensor theorem.

32 Gaussian Current-Symbol Equivalence in One Dimension

32.1 Status, Sources, and Dependencies

Status: checked translation-invariant symbol bridge. This shard adds the first bridge between the finite real-space current convention and the momentum-symbol layer. It is deliberately one-dimensional and nearest-neighbour. The checked statement is the equality of the translation-invariant integrated-current symbol with the CA-26 boost-time symbol for the scalar Klein-Gordon coefficients. No finite open-chain momentum operator, no finite periodic first moment, no $d\Gamma(k)$ theorem, and no continuum convergence statement is claimed.

Dependencies: CA-24, CA-26, CA-29–CA-30, and CONVENTIONS.md (j), (k), (l).

32.2 Real-Space Current Orientation

The finite open-chain A4 witness fixes the local current on an open bond by

$$J_{j+1/2} = i[e_j, e_{j+1}]$$

with endpoint currents zero. For a nearest-neighbour scalar chain with bond coefficient b, the checked closed form is

$$J_{j+1/2} = \frac{b}{2}\big(q_{j+1}p_j - q_j p_{j+1}\big).$$

This sign is not adjustable in this shard: it is the CONVENTIONS.md (l) orientation already used in the open-chain continuity test.

To compare with the boost-time symbol, pass from the local edge current to the first-moment integrated current density. In uniform spacing ϵ, the nearest-neighbour first-moment coefficient contributes the spacing factor ϵ. The translation-invariant integrated current therefore has symbol

$$\mathcal{J}_\epsilon(k) = -\epsilon b \sin(\epsilon k).$$

This is a Fourier-symbol identity for the infinite or translation-invariant nearest-neighbour expression. It does not introduce a periodic coordinate branch and does not construct an open finite-chain momentum operator.

32.3 Klein-Gordon Equality

For the one-dimensional nearest-neighbour lattice Klein-Gordon coefficients,

$$b = -\epsilon^{-2}, \qquad \omega_{\epsilon,m}(k)^2 = m^2 + 2\epsilon^{-2}(1 - \cos(\epsilon k)).$$

The integrated-current symbol becomes

$$\mathcal{J}_\epsilon(k) = \frac{\sin(\epsilon k)}{\epsilon}.$$

The CA-26 boost-time symbol is

$$B_\epsilon(k) = \frac{1}{2}\partial_k \omega_{\epsilon,m}(k)^2 = \frac{\sin(\epsilon k)}{\epsilon}.$$

Thus, at this checked symbol level,

$$\mathcal{J}_\epsilon(k) = B_\epsilon(k).$$

The mass drops out because the current and the derivative of the onsite mass term are both insensitive to m^2.

32.4 Julia Witness

The helper

`nearest_neighbor_integrated_energy_current_symbol`

implements

$$-\epsilon b \sin(\epsilon k)$$

for one-dimensional data only. The Klein-Gordon wrapper inserts $b = -\epsilon^{-2}$. The tests compare that wrapper against

`kg_lattice_boost_time_symbol`

for several momenta and spacings. They also include a sign sentinel: for positive k and the Klein-Gordon negative bond, the current symbol is positive and agrees with the boost-time target; reversing the bond sign gives a negative symbol and fails the target comparison. A small-spacing check compares the current symbol with the centered continuum momentum representative k near zero using the existing Gaussian small-spacing tolerance.

32.5 Open Obligations

This shard closes only the first symbol-level bridge. The following remain pending:

1. a finite open-chain wave-packet comparison with boundary residuals,
2. an identification with the one-particle $\mathrm{d}\Gamma(k)$ momentum target,
3. a periodic first-moment construction, which still needs a branch policy,
4. a scaling or continuum convergence theorem.

Any compiler output citing this shard must therefore say "current-symbol equality" rather than "finite-chain momentum" or "continuum stress tensor".

33 Higher-Dimensional Gaussian Cell-Current Proposal

33.1 Status, Sources, and Dependencies

Status: proposal and compiler-input specification. This shard states the support data and local equations a higher-dimensional Gaussian stress-energy candidate must carry in $d = 2, 3$. It is not a theorem, does not define a global orientation convention, and makes no convergence claim.

Dependencies: CA-13, CA-23, CA-29–CA-31, and CONVENTIONS.md (j), (k), (l).

33.2 Compiler Input: Oriented Support Data

For each finite or infinite hypercubic support in $d = 2$ or $d = 3$, the compiler input must include an oriented incidence structure:

$$\mathcal{C}, \qquad \mathcal{F}, \qquad \sigma(c, f) \in \{-1, 0, +1\}.$$

Here $\mathcal{C}$ is the set of energy cells, attached to sites $x_c \in \epsilon\mathbb{Z}^d$. The set $\mathcal{F}$ contains oriented codimension-one faces. In $d = 2$, these faces are oriented edges between neighbouring square cells. In $d = 3$, they are oriented plaquettes between neighbouring cube cells. If an interior face f is oriented from its minus cell $c_-(f)$ to its plus cell $c_+(f)$, set

$$\sigma(c_-(f), f) = +1, \qquad \sigma(c_+(f), f) = -1,$$

and set $\sigma(c, f) = 0$ for all other cells. This sign convention is local input for this proposal, not a repository-wide orientation convention.

Boundary faces are separate data. For an open box, $\mathcal{F}_\partial$ contains oriented exterior faces with only one adjacent cell; their incidence sign is recorded as $\sigma_\partial(c, f) = +1$ for outward flux. Setting a boundary flux to zero is then a named boundary condition, not an omitted term. For a periodic support, exterior faces are absent, but any first moment or coordinate-weighted test still needs the branch or Fourier policy required by CONVENTIONS.md (j).

33.3 Energy Density and Energy Flux

The energy component starts from CA-29:

$$T^{\text{lat}}_{00}(c) := h_c, \qquad e_c := \epsilon^d h_c, \qquad H_\epsilon = \sum_{c \in \mathcal{C}} e_c.$$

All finite commutator identities use the integrated cell energy e_c.

An energy-current candidate is face data

$$J_0(f) \quad (f \in \mathcal{F}),$$

with optional boundary data $J_0^\partial(f)$ on $f \in \mathcal{F}_\partial$. If f has normal direction a, then $J_0(f)$ is the proposed lattice T_{0a} flux through that oriented face. The required energy-continuity witness is the cell equation

$$i[H_\epsilon, e_c] + \sum_{f \in \mathcal{F}} \sigma(c, f) J_0(f) + \sum_{f \in \mathcal{F}_\partial} \sigma_\partial(c, f) J_0^\partial(f) = 0.$$

For an open no-leak boundary this specializes to $J_0^\partial(f) = 0$. For a driven or truncated-region diagnostic, the same displayed equation records the boundary residual explicitly. This shard does not propose a universal formula for $J_0(f)$; a later numerical suite must derive or solve for it from the chosen Gaussian density and then test the equation above.

33.4 Momentum Densities and Stress Currents

The momentum-density components are additional witnesses:

$$M_a(c) \quad (a = 1, \dots, d), \qquad P_a^{\text{lat}} = \sum_c M_a(c).$$

They are not inferred from the energy flux $J_0(f)$. The reason is already visible in CA-30: T_{10} was kept as a momentum-density target, not as a consequence of T_{01}. In higher dimensions this separation is mandatory because rotations require M_a before angular momentum even has a candidate density.

For each chosen momentum density $M_a(c)$, the stress components are its currents. If f has normal direction b, write

$$S_a(f) \equiv T_{ab}^{\text{lat}}(f).$$

The required momentum-continuity witness is

$$i[H_\epsilon, M_a(c)] + \sum_{f \in \mathcal{F}} \sigma(c, f) S_a(f) + \sum_{f \in \mathcal{F}_\partial} \sigma_\partial(c, f) S_a^\partial(f) = 0$$

for every a and every cell c. Until M_a and S_a are supplied with this witness, there is no lattice T_{a0} or T_{ab} claim.

33.5 Boost and Symbol Diagnostics

The boost first moment can be formed from the integrated energy cells:

$$K_a^{\text{lat}} := \sum_c x_c^a e_c.$$

Using the energy-continuity equation, $i[H_\epsilon, K_a^{\text{lat}}]$ reduces by discrete summation by parts to the integrated a-normal energy flux plus boundary terms. The higher-dimensional analogue of CA-31 is then a test, not an assumption:

$$\text{integrated } T_{0a} \text{ symbol} \quad \stackrel{?}{=} \quad \frac{1}{2}\partial_a \omega(k)^2.$$

Only after this equality is checked in the relevant local or symbol regime may it be compared with the flat one-particle momentum target k_a and the residuals of CA-26–CA-27. Passing the symbol equality would still not prove a continuum stress tensor.

33.6 Rotation and Antisymmetric-Stress Residual

For $a < b$, a rotation candidate requires the momentum-density witnesses:

$$J_{ab}^{\text{lat}} := \sum_c \big(x_c^a M_b(c) - x_c^b M_a(c)\big).$$

This formula is unavailable if only T_{00} and T_{0a} have been supplied. Applying the momentum-continuity equation gives a discrete summation-by-parts diagnostic of the form

$$i[H_\epsilon, J_{ab}^{\text{lat}}] = \epsilon \sum_{f \perp a} S_b(f) - \epsilon \sum_{f \perp b} S_a(f) + \text{boundary terms},$$

with the displayed sign tied to the incidence convention above. In continuum language this is the antisymmetric-stress obstruction: rotation conservation targets $T_{ab} = T_{ba}$, after boundary terms and any spin/improvement data have been named. Weinberg's registered anchors motivate that a symmetric Belinfante tensor can carry the same integrated translations, but the canonical tensor need not be symmetric. Therefore stress symmetry is a target or witness, not an assumption baked into the lattice proposal.

33.7 Trace, Improvement, and Boundary Alternatives

Trace diagnostics are also target-dependent. A compiler may report the diagonal stress combination built from h_c and the $S_a(f)$, but it must name the continuum target and metric/sign convention before calling it a trace residual. No massless-CFT tracelessness test, massive Klein-Gordon trace test, or improvement formula is imported here.

The CA-29 improvement policy remains active:

$$h'_c = h_c + (\nabla_\epsilon \cdot b)_c.$$

Any such change must name the edge or face field b, the discrete divergence, and the boundary convention. It must also recompute J_0, M_a, S_a, and the first-moment boundary terms; improvements are not silent gauge moves for boosts or rotations.

Two boundary regimes are admissible compiler inputs:

open box	use explicit $\mathcal{F}_\partial$ terms and test endpoint/face residuals,
periodic box	use no boundary faces, but fix a coordinate branch, sawtooth, or Fourier-interpolation policy before any first-moment test.

The existing Gaussian numerical layer is periodic-symbol first. The next stress-energy numerical suite, S5 in the action plan, should therefore start with open-boundary local continuity tests or with a separately sourced periodic coordinate construction, and should include anisotropic/doubler failure witnesses at the current-density level.

34 First Gaussian Stress-Energy Numerical Suite

34.1 Status, Sources, and Dependencies

Status: checked local-current numerical suite. This shard records the first stress-energy-specific numerical tests. The checked surface is narrow: finite open-chain T_{01} continuity for the CA-29/CA-30 density-current choice, and a one-dimensional nearest-neighbour current-symbol slope residual. It is not a finite boost construction, not a $d\Gamma(k)$ theorem, not a periodic first-moment test, and not a continuum stress-tensor claim.

Dependencies: CA-29–CA-32 and CONVENTIONS.md (j), (k), (l).

34.2 Finite Open-Chain Continuity

The first suite strengthens the A4 local-current witness instead of replacing it. For each tested open chain, the Julia layer constructs the CA-29 equal-endpoint site energies e_j, the total Hamiltonian

$$H = \sum_{j=1}^{N} e_j,$$

and the edge currents

$$J_{j+1/2} = i[e_j, e_{j+1}], \qquad 1 \le j < N.$$

The residual helper checks

$$i[H, e_j] - (J_{j-1/2} - J_{j+1/2}) = 0$$

with endpoint currents $J_{1/2} = J_{N+1/2} = 0$.

The test cases now cover several chain sizes and onsite/bond choices:

$$(N, a, b) \in \{(2, 0.75, -0.5), (3, 2.0, 1.25), (5, 7.25, -1.5), (7, 0.1, -0.875)\}.$$

For each case the suite asserts:

1. $\sum_j e_j = H$,
2. H_{qq} is the expected tridiagonal matrix with diagonal a and bond b,
3. $H_{pp} = I$,
4. $J_{j+1/2}$ equals the closed form $\frac{b}{2}(q_{j+1}p_j - q_j p_{j+1})$,
5. $J_{j+1/2}$ is independent of the onsite coefficient,
6. all continuity residual matrices vanish,
7. $i[H, e_1] = -J_{3/2}, \quad i[H, e_j] = J_{j-1/2} - J_{j+1/2}, \quad i[H, e_N] = J_{N-1/2}$.

The explicit endpoint tests are the important stress-energy addition: boundary signs are part of the local T_{01} claim, not discarded finite-size errors. The suite also checks that a one-site chain cannot produce a bond current and that an out-of-range open bond index raises an error.

34.3 Current-Symbol Slope Residual

The second suite is the smallest symbol-level current-density failure witness currently justified by CA-31. For the one-dimensional nearest-neighbour symbol

$$\mathcal{J}_\epsilon(k) = -\epsilon b \sin(\epsilon k),$$

the small-k slope is
$$\mathcal{J}'_\epsilon(0) = -\epsilon^2 b.$$
The new helper returns the speed residual
$$\rho_J(b, \epsilon, c) = -\epsilon^2 b - c^2.$$
It validates positive spacing, positive target speed, and real bond data before returning a number.

For the speed-c nearest-neighbour Klein-Gordon bond
$$b = -c^2\epsilon^{-2},$$
the residual is zero. The tests assert this for several spacings and speeds. They also assert failure witnesses:
$$b = -\epsilon^{-2} \quad \text{fails when } c \neq 1,$$
wrong bond magnitude gives the expected nonzero residual, and reversing the bond sign gives
$$\rho_J(c^2\epsilon^{-2}, \epsilon, c) = -2c^2.$$
These are current-density symbol failures for the CA-31 orientation. They are not Hessian-only failures and they do not use a periodic coordinate branch.

34.4 What This Falsifies

The suite falsifies three classes of local-current mistakes:

1. incorrect open-boundary signs in $i[H, e_j] = J_{j-1/2} - J_{j+1/2}$,
2. wrong orientation in $J_{j+1/2} = i[e_j, e_{j+1}]$,
3. nearest-neighbour current symbols whose low-energy slope does not match the selected c^2.

The onsite-independence checks also catch an accidental insertion of the mass or onsite term into the nearest-neighbour current.

34.5 Open Obligations

The suite deliberately leaves the following undone:

1. finite open-chain wave packets and comparison with $\mathrm{d}\Gamma(k)$,
2. periodic first moments or periodic coordinate generators,
3. higher-dimensional current-density formulas and incidence-sign tests,
4. next-nearest-neighbour or doubler current-density witnesses,
5. momentum-density T_{10} and stress-current T_{11} continuity tests,
6. scaling maps, state choices, domains, and continuum convergence.

The doubler evidence remains the CA-28 coefficient/minimum-data rejection. It should not be described as a current-density failure until a current formula for that non-nearest-neighbour symbol is derived and tested.

35 Qubit Nearest-Neighbour Symmetry Quest

35.1 Status, Sources, and Dependencies

Status: programme shard with checked seed. This shard scopes the next lattice-symmetry quest for qubit systems. CA-35 supplies the first checked finite algebraic obstruction. CA-36 and CA-37 state proposal-level diagnostics for $2+1$-dimensional Poincare tests and $1+1$-dimensional Witt/Virasoro tests.

Dependencies: CA-10–CA-17, CONVENTIONS.md (h), (k), (m), and the review orchestration in `reviews/2026-05-31_qubit_nn_symmetry/`.

35.2 Input Tier

The first input class is a translation-invariant qubit nearest-neighbour density. In the convention of CONVENTIONS.md (m),

$$h = \sum_{\alpha,\beta=0}^{3} h_{\alpha\beta}\, \sigma^\alpha \otimes \sigma^\beta, \qquad h_{\alpha\beta} \in \mathbb{R},$$

with $\sigma^0 = I$ and $\sigma^1, \sigma^2, \sigma^3 = X, Y, Z$. On a one-dimensional open chain the density h_j acts on sites $(j, j+1)$, so

$$H = \sum_j h_j.$$

In two spatial dimensions the same coefficient format may be used separately on horizontal and vertical edges, but the edge orientation, midpoint, boundary, and bulk-projection conventions are not fixed by this shard.

35.3 Question

The question is not whether a finite qubit lattice has exact Poincare or Virasoro symmetry. CA-15 already forbids that shortcut. The question is whether the finite algebra forced by the candidate generator ansatz gives cheap rejection tests:

$$\begin{gathered}\text{Pauli coefficient data} \longrightarrow \text{finite commutator residuals} \\ \longrightarrow \text{rule out impossible symmetry candidates early.}\end{gathered}$$

The first such test is the adjacent-density current

$$J(h) := i[h_{12}, h_{23}],$$

because CA-12 shows that the first energy moment

$$K = \sum_j x_j h_j$$

obeys $i[H, K]$ equal to the sum of adjacent commutators in the translation-invariant bulk. If this current vanishes, the first-moment boost ansatz produces zero spatial momentum.

35.4 Investigation Plan

One-dimensional Poincare filters. Compute $J(h)$ directly from $h_{\alpha\beta}$. Reject the density as a nontrivial Poincare candidate for the CA-12 first-moment route if the bulk current vanishes while the energy density is non-scalar. Then test stronger finite residuals such as $i[P, K] - H$, but only after boundary, scaling, and normalization policies are named.

Two-dimensional Poincare filters. Use horizontal and vertical edge densities h^x, h^y. Define ramp moments from edge midpoints and compute $P_x = i[H, K_x]$, $P_y = i[H, K_y]$ on open patches. Proposed finite residuals include current nontriviality, $i[H, P_a]$, $i[P_x, P_y]$, isotropy under $x \leftrightarrow y$, and later rotation closure after a local momentum-density split is fixed.

Witt/Virasoro filters. On a periodic one-dimensional chain, form Fourier modes of the density and of the adjacent current. Koo–Saleur supplies the sourced pattern: Hamiltonian-density modes plus commutator corrections. The first rejection test is again current collapse; central terms and finite-size non-closure are not early failure tests.

35.5 Immediate Rule-Outs

The first checked obstruction is deliberately modest. Under the symmetric nearest-neighbour split of a genuine on-site qubit Hamiltonian, $J(h) = 0$. Thus a fully local single-site Hamiltonian cannot provide nonzero bulk momentum through the CA-12 first-moment boost ansatz. The same obstruction also catches commuting classical nearest-neighbour densities, such as pure diagonal ZZ terms without noncommuting fields.

Nonzero $J(h)$ is not sufficient. It is only the first algebraic gate. A density that passes still needs the CA-15 evidence payload: local support policy, state sequence, renormalisation maps, convergence topology, low-energy sector or core, and observable covariance.

36 One-Dimensional Qubit Boost-Current Obstructions

36.1 Status, Sources, and Dependencies

Status: checked local derivation. This shard turns the CA-12 nearest-neighbour identity into explicit polynomial conditions on the two-qubit Pauli coefficient matrix. The obstruction is checked in Julia.

Dependencies: CA-12, CA-34, CONVENTIONS.md (h), (k), (m).

36.2 Adjacent Current in Pauli Coefficients

Let

$$h_j = \sum_{\alpha,\beta=0}^{3} h_{\alpha\beta}\,\sigma_j^\alpha \sigma_{j+1}^\beta.$$

The only noncommuting overlap between h_j and h_{j+1} is at site $j+1$. With

$$[\sigma^b, \sigma^c] = 2i \sum_{e=1}^{3} \epsilon_{bce}\sigma^e \qquad (b, c \in \{1, 2, 3\}),$$

one obtains

$$i[h_j, h_{j+1}] = -2 \sum_{\alpha,\delta=0}^{3} \sum_{e=1}^{3} \left(\sum_{b,c=1}^{3} h_{\alpha b} h_{c\delta} \epsilon_{bce} \right) \sigma_j^\alpha \sigma_{j+1}^e \sigma_{j+2}^\delta.$$

Equivalently, define right-leg and left-leg vectors

$$R_\alpha = (h_{\alpha 1}, h_{\alpha 2}, h_{\alpha 3}), \qquad L_\delta = (h_{1\delta}, h_{2\delta}, h_{3\delta}).$$

Then the current coefficients are the cross products

$$i[h_j, h_{j+1}] = -2 \sum_{\alpha,\delta=0}^{3} \sum_{e=1}^{3} (R_\alpha \times L_\delta)_e \sigma_j^\alpha \sigma_{j+1}^e \sigma_{j+2}^\delta.$$

Thus the simple current-collapse condition is

$$R_\alpha \times L_\delta = 0 \qquad \text{for every } \alpha, \delta.$$

When this holds, every adjacent-current contribution to the CA-12 bulk momentum candidate vanishes.

36.3 Fully On-Site Terms

For a genuine translation-invariant on-site density

$$A = \sum_{\alpha=0}^{3} a_\alpha \sigma^\alpha,$$

CONVENTIONS.md (m) embeds it into the two-site bond tier by the symmetric split

$$h_{\text{onsite}} = \frac{1}{2}(A \otimes I + I \otimes A).$$

Then the right-leg and left-leg vectors are all parallel in the only places where they overlap, so

$$i[(h_{\text{onsite}})_{12}, (h_{\text{onsite}})_{23}] = 0.$$

The Julia test checks this directly for a generic real linear combination of I, X, Y, Z.

Consequently the first-moment boost ansatz gives

$$i[H, K] = 0$$

in the translation-invariant bulk. If the target is a nontrivial Poincare representation with $i[H, K] = P$ and $i[P, K] = H$, this first-moment route cannot realise it unless the energy itself is trivial after the relevant compression or subtraction.

36.4 Other Immediate Rejections

The same current-collapse test catches commuting nearest-neighbour density families. A pure diagonal classical interaction such as ZZ has

$$[h_j, h_{j+1}] = 0$$

because the overlapping Pauli factor is the same diagonal matrix. Such a model may still be a well-defined lattice Hamiltonian, but it fails the first nontrivial bulk-symmetry diagnostic: the boost-current candidate supplies no spatial momentum.

By contrast, the tested transverse-Ising-style density with a ZZ bond and a symmetric X-field split has nonzero adjacent current. This does not prove Poincare invariance; it only passes the first obstruction.

36.5 The Next Residual

The next necessary Poincare relation is translation conservation:

$$i[H, P] = 0.$$

For the local candidate $P = \sum_j C_j$, $C_j = i[h_j, h_{j+1}]$, the minimal bulk density after the Jacobi cancellation of outer-overlap terms is

$$A_j = i[h_j + h_{j+1}, C_j].$$

The full infinite-chain equation is not necessarily $A_j = 0$, but

$$A_j = u_j - u_{j+1}$$

for a finite density u. CA-39 writes this and the next boost relation $i[P, K] = v^2 H$ as explicit Pauli-coefficient equations.

This residual is useful for detecting density-split artifacts. An asymmetric encoding of an on-site Hamiltonian,

$$h = U \otimes I + I \otimes V,$$

can create the fake current

$$i[h_{12}, h_{23}] = I \otimes i[V, U] \otimes I.$$

But the checked sentinel with $U = X$, $V = Z$ has nonzero

$$i[h_{12} + h_{23}, i[h_{12}, h_{23}]],$$

so it fails translation conservation even though the adjacent current is nonzero. The CA-39 boost equation then tests the remaining relation after the allowed density-divergence and scalar-energy subtraction have been fixed.

36.6 Boundary

The obstruction is conditional on the density convention. An asymmetric encoding of an on-site Hamiltonian as $A \otimes I + I \otimes B$ can create spurious adjacent commutators when A and B fail to commute. Such currents are density-split artifacts unless a separate convention justifies the split, and the conservation residual above is the first backstop against them. Passing the obstruction is also far weaker than CA-15 acceptance: boundary residuals, scaling maps, low-energy sectors, and observable covariance remain unproved.

37 Two-Dimensional Qubit Plaquette Symmetry Diagnostics

37.1 Status, Sources, and Dependencies

Status: proposal diagnostics. This shard gives a finite algebraic test plan for $2+1$-dimensional qubit lattices. No two-dimensional current or rotation convention is fixed yet, so every condition below is a proposed compiler filter.

Dependencies: CA-13, CA-15, CA-16, CA-32, CONVENTIONS.md (h), (k), (m).

37.2 Edge-Density Input

On an open square patch, use oriented horizontal and vertical edges:

$$h^a_{r,r+\hat{a}} = \sum_{\alpha,\beta=0}^{3} c^a_{\alpha\beta}\,\sigma^\alpha_r \sigma^\beta_{r+\hat{a}}, \qquad a \in \{x, y\}.$$

Let m_e be the midpoint of edge e, and write

$$H = \sum_e h_e, \qquad K_a = \sum_e m^a_e h_e.$$

The ramp-current candidate is

$$P_a := i[H, K_a] = \sum_{\{e,f\}} (m^a_f - m^a_e) i[h_e, h_f],$$

where only overlapping edge pairs contribute. This is the direct higher-dimensional analogue of CA-13's scalar-ramp formula.

37.3 Patch Diagnostics

The smallest useful checks are local:

1. three-site lines for straight x and y currents,
2. 2×2 plaquettes for corner commutators and C_4 covariance,
3. 3×3 open patches for P_x, P_y and $i[H, P_a]$,
4. 4×4 or larger patches for bulk-projected $i[P_x, P_y]$.

Boundary-supported residuals should be projected away or recorded separately. The diagnostic object is the interior Pauli-string residual, not raw equality on the whole open patch.

37.4 Proposed Rejection Filters

Current nontriviality. Both P_x and P_y should be nonzero in the bulk for a nontrivial $2+1$-dimensional Poincare candidate.

Translation conservation. The bulk residual $i[H, P_a]$ should vanish or be controlled in the intended finite algebra. A nonzero interior Pauli-string residual is an early rejection witness for exact finite bulk closure.

Translation commutativity. The bulk part of $i[P_x, P_y]$ should vanish or be a named boundary/irrelevant term before the model is promoted.

Boost-translation bracket. After normalisation, the residual

$$i[P_b, K_a] - \delta_{ab} H$$

is the direct $2+1$-dimensional Poincare diagnostic. This is strong and must be treated as a finite filter, not a theorem.

Isotropy. For scalar qubits with no internal spatial action, a microscopic 90° rotation should relate horizontal and vertical edge data after endpoint reversal. Obvious decoupled stacks, such as $c^y = 0$, fail this filter.

37.5 Rotation Is Not Free

A rotation candidate needs a local momentum-density split

$$P_a = \sum_v p_a(v)$$

before one can write

$$J_{xy} = \sum_v (x_v p_y(v) - y_v p_x(v)).$$

CA-13 already marks this as extra data. A plausible first convention is to assign each edge-pair current to the shared vertex, but that convention must be written and tested before rotation closure is claimed. Only then should one check

$$i[H, J_{xy}], \qquad i[P_x, J_{xy}] - P_y, \qquad i[P_y, J_{xy}] + P_x.$$

37.6 Rule-Out Classes

The proposal immediately targets four classes:

1. fully on-site Hamiltonians: $P_x = P_y = 0$,
2. commuting classical edge densities: all ramp currents vanish,
3. decoupled stacks: one momentum component vanishes,
4. anisotropic edge data: fail the scalar-qubit C_4 filter.

None of these rejections is a continuum theorem. They are finite algebraic filters for the specific candidate-generator route.

38 Qubit Nearest-Neighbour Witt/Virasoro Diagnostics

38.1 Status, Sources, and Dependencies

Status: proposal with local derivation targets. This shard is a finite-scale diagnostic proposal, not a continuum Virasoro claim.

Dependencies: CA-12, CA-14–CA-16, CA-34–CA-35, CONVENTIONS.md (m). CFT normalisations in CONVENTIONS.md (d) remain unfixed, so no central-charge extraction convention is available here.

38.2 Fourier Density and Current Modes

Let a periodic qubit chain of length N have

$$H = \sum_j h_j, \qquad h_j = \sum_{\alpha,\beta=0}^{3} h_{\alpha\beta}\sigma_j^\alpha \sigma_{j+1}^\beta.$$

For $q_n = 2\pi n/N$, define

$$H_n = \sum_j e^{iq_n j} h_j, \qquad J_n = \sum_j e^{iq_n j} j_j, \qquad j_j = i[h_j, h_{j+1}].$$

The nearest-neighbour commutator identity is

$$i[H_m, H_n] = (e^{iq_n} - e^{iq_m}) J_{m+n}.$$

This is the finite Pauli-chain analogue of the Koo–Saleur pattern in which Hamiltonian-density modes and commutator corrections approximate chiral stress-tensor modes.

38.3 First Failure Witness

CA-35 gives the coefficient condition

$$R_\alpha \times L_\delta = 0 \qquad \text{for all } \alpha, \delta,$$

which implies $j_j = 0$ and hence $J_n = 0$ for every Fourier mode. In that case the density modes cannot acquire the Koo–Saleur commutator correction needed to split left-moving and right-moving candidates. Fully on-site Hamiltonians under the symmetric split fail here, as do commuting classical densities.

38.4 Chiral-Rank Diagnostic

A later finite diagnostic may define tentative chiral candidates of the form

$$\ell_n = a_N{:}H_n{:} + \lambda_n J_n + \mu_N \delta_{n0} I, \qquad \bar{\ell}_{-n} = a_N{:}H_n{:} - \lambda_n J_n + \mu_N \delta_{n0} I.$$

Before testing brackets, one should check that H_n and J_n are linearly independent on the chosen low-energy sector P_N. A rank-collapse test such as

$$\det \mathrm{Gram}_{P_N}(H_n, J_n) \neq 0$$

is an early finite witness that the two proposed chiral directions have not collapsed to one.

38.5 Compressed Witt Residuals

The raw full Hilbert-space commutator is the wrong first test. Koo–Saleur already warns that finite-size quantities do not close as Virasoro generators and that commutators interact subtly with the scaling limit. The diagnostic should therefore choose a low-energy projection P_N, an intermediate projection Q_N, and test residuals such as

$$P_N\big(\ell_m Q_N \ell_n - \ell_n Q_N \ell_m - (m-n)\ell_{m+n}\big)P_N$$

for $m + n \neq 0$, together with the barred and cross-chiral versions.

The $m + n = 0$ brackets should not be the first rejection test: central terms require a Virasoro normalization, vacuum or ground-state subtraction, and a scaling state sequence. These are not fixed for arbitrary qubit chains.

38.6 Boundary

This shard gives finite algebraic diagnostics only. Passing them does not produce a Virasoro representation. A continuum claim still needs the full CA-15 payload plus Koo–Saleur-style scaling representation data: state sequence, subtraction policy, mode normalization, low-energy sector, operator topology, and convergence theorem.

39 Qubit Local-Algebra Equation Framework

39.1 Status, Sources, and Dependencies

Status: checked local framework. This shard supplies the algebraic language used by CA-39–CA-44. It does not assert that a solution has continuum Poincare or Virasoro symmetry.

Dependencies: CA-11, CA-12, CA-15, CA-34–CA-35, CONVENTIONS.md (m), (n).

39.2 Pauli Word Algebra

Use zero-based Pauli labels $a \in \{0, 1, 2, 3\}$, with $\sigma^0 = I$. For a finite interval, a Pauli word is

$$P_{\alpha_0 \dots \alpha_{\ell-1}} = \sigma^{\alpha_0} \otimes \cdots \otimes \sigma^{\alpha_{\ell-1}}.$$

The local product is encoded by constants M^c_{ab}:

$$\sigma^a \sigma^b = \sum_{c=0}^{3} M^c_{ab} \sigma^c,$$

where

$$M^c_{0b} = \delta_{bc}, \qquad M^c_{a0} = \delta_{ac},$$

and, for $a, b \geq 1$,

$$M^0_{ab} = \delta_{ab}, \qquad M^c_{ab} = i\epsilon_{abc} \quad (c \geq 1).$$

Thus for two ℓ-site coefficient tensors A, B,

$$(AB)_\gamma = \sum_{\alpha,\beta} A_\alpha B_\beta \prod_{r=0}^{\ell-1} M^{\gamma_r}_{\alpha_r \beta_r}.$$

The commutator coefficient is therefore the finite expression

$$(i[A, B])_\gamma = i \sum_{\alpha,\beta} A_\alpha B_\beta \left(\prod_r M^{\gamma_r}_{\alpha_r \beta_r} - \prod_r M^{\gamma_r}_{\beta_r \alpha_r} \right).$$

When two local densities have different supports, first embed both tensors into their common finite window by padding with Pauli label 0, then apply this formula.

39.3 Bulk Equality Modulo Divergence

Let τ be the unit shift of the infinite chain. A finite density R represents a zero bulk formal sum if it is a coboundary:

$$R = u - \tau(u).$$

If u has n Pauli indices, the coefficient form fixed in CONVENTIONS.md (n) is

$$(Du)_{a_0 \dots a_n} = u_{a_0 \dots a_{n-1}} \delta_{a_n 0} - \delta_{a_0 0} u_{a_1 \dots a_n}. \tag{38.1}$$

Consequently a local residual $R(h)$ passes the bulk equation precisely when there exists a finite tensor u such that

$$R_{a_0 \dots a_n}(h) - (Du)_{a_0 \dots a_n} = 0 \qquad \text{for every Pauli word } a_0 \dots a_n. \tag{38.2}$$

The scalar coefficient of u lies in the kernel of D, so it is a gauge variable. A coefficientwise condition $R(h) = 0$ is allowed as a stronger filter, but it is not the default infinite-chain equation.

39.4 What "Necessary" Means Here

Every equation in the next shards is necessary only for the named generator route: the fixed Pauli density, the fixed first-moment boost ansatz, and the fixed coboundary quotient. Solving these equations is not sufficient for continuum symmetry. Failing one of them is still useful: it excludes the Hamiltonian from this route before any low-energy sector, scaling map, or state-space analysis is attempted.

40 One-Dimensional Qubit Poincare Necessary Equations

40.1 Status, Sources, and Dependencies

Status: local derivation, partly checked. The current, coboundary, and residual tensors are implemented in Julia. The equations are necessary for the fixed CA-12 first-moment route, not sufficient for continuum Poincare symmetry.

Dependencies: CA-11, CA-12, CA-35, CA-38, CONVENTIONS.md (k), (m), (n).

40.2 Input and Current

Let

$$h_j = \sum_{\alpha,\beta=0}^{3} h_{\alpha\beta}\sigma_j^\alpha\sigma_{j+1}^\beta, \qquad h_{\alpha\beta} \in \mathbb{R}.$$

The first-moment route fixes

$$H = \sum_j h_j, \qquad K = \sum_j jh_j, \qquad P = i[H, K] = \sum_j p_j, \qquad p_j = i[h_j, h_{j+1}].$$

The current coefficients are

$$p_{aed} = -2\sum_{b,c=1}^{3} h_{ab}h_{cd}\epsilon_{bce}, \qquad e \in \{1, 2, 3\}, \tag{39.1}$$

with $p_{a0d} = 0$. Equivalently the right-leg vector (h_{a1}, h_{a2}, h_{a3}) must cross the left-leg vector (h_{1d}, h_{2d}, h_{3d}). If

$$p_{aed} = 0 \quad \text{for all } a, e, d, \tag{39.2}$$

then the candidate bulk momentum is $P = 0$. For a nontrivial speed $v^2 \neq 0$, the later equation $i[P, K] = v^2 H$ can then hold only if the Hamiltonian density is scalar-trivial in the coboundary quotient. The symmetric fully on-site split in CONVENTIONS.md (m) is checked to have $p = 0$; hence fully local non-scalar Hamiltonians are ruled out for this route.

40.3 Translation Conservation

Let

$$A_j = i[h_j + h_{j+1}, p_j].$$

The outer-overlap terms in $i[H, P]$ cancel in the formal sum by the Jacobi identity, so translation conservation is

$$A = Du \tag{39.3}$$

for some two-site tensor u_{rs}. In Pauli coefficients,

$$\begin{aligned} A_{rst} = i &\sum_{\alpha,\beta,a,b} h_{\alpha\beta}p_{abt}\left(M^r_{\alpha a}M^s_{\beta b} - M^r_{a\alpha}M^s_{b\beta}\right) \\ &+ i\sum_{\alpha,\beta,b,c} h_{\alpha\beta}p_{rbc}\left(M^s_{\alpha b}M^t_{\beta c} - M^s_{b\alpha}M^t_{c\beta}\right). \end{aligned}$$

Thus the explicit degree-three equations are

$$A_{rst} - u_{rs}\delta_{t0} + \delta_{r0}u_{st} = 0 \qquad (r, s, t = 0, \ldots, 3). \tag{39.4}$$

40.4 Boost-Translation Relation

Assume (39.3) has been witnessed by a chosen u. The remaining bulk representative of $i[P, K]$ is

$$B_j - u_j, \qquad B_j = i[p_j, h_{j+1}] + 2i[p_j, h_{j+2}].$$

The four-site coefficients of B are

$$\begin{aligned} B_{rstu} = i\delta_{u0} &\sum_{b,c,\alpha,\beta} p_{rbc} h_{\alpha\beta} \left(M^s_{b\alpha} M^t_{c\beta} - M^s_{\alpha b} M^t_{\beta c} \right) \\ &+ 2i \sum_{c,\alpha} p_{rsc} h_{\alpha u} \left(M^t_{c\alpha} - M^t_{\alpha c} \right). \end{aligned}$$

Allow an explicit scalar energy-origin subtraction

$$\bar{h}_{rs} = h_{rs} - e\,\delta_{r0}\delta_{s0}.$$

The necessary speed-v boost equation $i[P, K] = v^2 H$ is that there exists a three-site tensor w such that

$$\begin{aligned} 0 = B_{rstu} &- u_{rs}\delta_{t0}\delta_{u0} - v^2 \bar{h}_{rs}\delta_{t0}\delta_{u0} \\ &- w_{rst}\delta_{u0} + \delta_{r0} w_{stu} \qquad (r, s, t, u = 0, \ldots, 3). \end{aligned} \tag{39.5}$$

Equations (39.1), (39.4), and (39.5), together with real variables u, w, e, v^2, are the advertised explicit finite polynomial conditions on $h_{\alpha\beta}$ for the one-dimensional first-moment Poincare route.

40.5 Boundary

The scalar h_{00} cannot be detected by commutators; it must be handled by the explicit e variable or by a state-dependent normal ordering. The coboundary witnesses are also not unique. These are convention choices, not loopholes: changing the density by a divergence changes $K = \sum_j j h_j$, so it changes the equations.

41 Computer Algebra for One-Dimensional Qubit Residuals

41.1 Status, Sources, and Dependencies

Status: checked. This shard documents the finite algebra engine behind CA-35 and CA-39.
Dependencies: CA-35, CA-38, CA-39, CONVENTIONS.md (m), (n).

41.2 Operator and Coefficient Roundtrip

The file `src/QubitPauliLattice.jl` implements

$$\sum_{\alpha_0,\ldots,\alpha_{n-1}=0}^{3} c_{\alpha_0\ldots\alpha_{n-1}}\sigma^{\alpha_0}\otimes\cdots\otimes\sigma^{\alpha_{n-1}}$$

for $n = 1, 2, 3$ and for general n. The inverse map uses

$$c_\alpha = 2^{-n}\operatorname{Tr}(P_\alpha A).$$

The tests assert coefficient roundtrips for two-site densities and for the five-site residual representatives. The code rejects arrays whose axes are not all length 4, and rejects non-self-adjoint matrix inputs.

41.3 Checked Residual Helpers

The current helper returns

$$p = i[h_{12}, h_{23}]$$

and the direct coefficient helper returns exactly (39.1). The conservation helper

$$A = i[h_{12} + h_{23}, p]$$

is the minimal three-site density for $i[H, P]$ after the Jacobi cancellation of outer-overlap terms.
The file `src/QubitPauliResiduals.jl` adds

$$(Du)_{a_0\ldots a_n} = u_{a_0\ldots a_{n-1}}\delta_{a_n 0} - \delta_{a_0 0}u_{a_1\ldots a_n}$$

and

$$B = i[p_{123}, h_{23}] + 2i[p_{123}, h_{34}].$$

The test suite checks that the coboundary coefficients reconstruct the operator identity

$$Du = u_{12} - u_{23}$$

on a three-site window, including the fact that the scalar $u_{00}I$ is in the kernel.

41.4 Sentinels

The checked examples are deliberately small:

- A generic real one-site density embedded by the symmetric split has $p = 0$, $A = 0$, and $B = 0$, while the embedded Hamiltonian density itself is nonzero. This is the concrete fully-local exclusion witness for CA-39.

- A pure classical ZZ density also has $p = 0$.
- A transverse-Ising-style density has nonzero p and nonzero boost residual density B. This only means it survives the first obstruction.
- An asymmetric fake on-site split creates a nonzero current, but the conservation sentinel is larger than the current and catches the density-split artifact.

41.5 Current Gap

The code does not yet solve the polynomial system (39.1), (39.4), (39.5), and does not build moment matrices. Its role is narrower: it makes every displayed coefficient tensor executable and gives mutation-friendly invariants before an SDP exporter is added.

42 Two-Dimensional Qubit Poincare Equation Schema

42.1 Status, Sources, and Dependencies

Status: proposal-level local schema. This is a finite algebraic target for a future checker. It is not yet a canonical stress tensor or a continuum $2+1$-dimensional Poincare theorem.

Dependencies: CA-13, CA-36, CA-38, CONVENTIONS.md (o).

42.2 Edge Densities and Vertex Currents

On $\mathbb{Z}^2$, write positively oriented edge densities

$$h_r^a = \sum_{\alpha,\beta=0}^{3} c^a_{\alpha\beta}\,\sigma_r^\alpha \sigma_{r+\hat{a}}^\beta, \qquad a \in \{x, y\}.$$

At a vertex v, order incident edges as W, S, E, N, with midpoint offsets

$$m_W = (-1/2, 0), \quad m_S = (0, -1/2), \quad m_E = (1/2, 0), \quad m_N = (0, 1/2).$$

Use shared-leg coefficient tensors

$$q^W_{\lambda\mu} = c^x_{\lambda\mu}, \quad q^E_{\lambda\mu} = c^x_{\mu\lambda}, \quad q^S_{\lambda\mu} = c^y_{\lambda\mu}, \quad q^N_{\lambda\mu} = c^y_{\mu\lambda}.$$

For ordered incident pairs $A < B$ in $W < S < E < N$, the local commutator coefficient on outer leg A, center v, outer leg B is

$$J^{AB}_{\lambda\kappa\rho} = -2 \sum_{\mu,\nu=1}^{3} q^A_{\lambda\mu} q^B_{\rho\nu} \epsilon_{\mu\nu\kappa}. \tag{41.1}$$

The ramp-generated momentum densities are

$$p_a(v) = \sum_{A<B} (m^a_B - m^a_A) J^{AB}(v). \tag{41.2}$$

For $AB = WS, WE, WN, SE, SN, EN$, the x-weights are

$$1/2,\ 1,\ 1/2,\ 1/2,\ 0,\ -1/2,$$

and the y-weights are

$$-1/2,\ 0,\ 1/2,\ 1/2,\ 1,\ 1/2.$$

42.3 Divergence-Relaxed Residuals

Every residual below is a finite Pauli tensor R on a named patch. The strong filter is $R = 0$. The default bulk filter is the existence of finite-support tensors U_x, U_y with

$$R = (1 - \tau_{-\hat{x}})U_x + (1 - \tau_{-\hat{y}})U_y. \tag{41.3}$$

Coefficientwise, this is an explicit linear equation in the auxiliary divergence tensors and a polynomial equation in c^x, c^y, using the Pauli product constants of CA-38.

42.4 Necessary Bulk Equations

Momentum conservation is

$$C_a := \sum_{e:\operatorname{supp} h_e \cap \operatorname{supp} p_a(0) \neq \varnothing} i[h_e, p_a(0)] = \nabla \cdot U^{(C_a)}, \qquad a = x, y. \tag{41.4}$$

The commuting-translation equation is

$$T_{xy} := \sum_{r:\operatorname{supp} p_x(0) \cap \operatorname{supp} p_y(r) \neq \varnothing} i[p_x(0), p_y(r)] = \nabla \cdot U^{(T)}. \tag{41.5}$$

Let $e_0 = h_0^x + h_0^y$ denote the chosen local energy cell. With m_e^a the a-coordinate of the edge midpoint,

$$B_{ab} := \sum_{e:\operatorname{supp} h_e \cap \operatorname{supp} p_b(0) \neq \varnothing} m_e^a \, i[p_b(0), h_e] - \delta_{ab} v_a^2 e_0 = \nabla \cdot U^{(B_{ab})}. \tag{41.6}$$

This is the $i[P_b, K_a] = \delta_{ab} v_a^2 H$ equation after the coordinate weighted conservation term is removed by (41.4).

If one defines the rotation candidate

$$J = \sum_v (v_x p_y(v) - v_y p_x(v)),$$

then the translation-rotation residual is

$$R_c = \sum_r (r_x \, i[p_c(0), p_y(r)] - r_y \, i[p_c(0), p_x(r)]) - (\delta_{cx} p_y(0) - \delta_{cy} p_x(0)) = \nabla \cdot U^{(R_c)}. \tag{41.7}$$

This last equation is quartic in c^x, c^y before the quadratic subtraction. It depends on the chosen momentum-density split, so it is less canonical than the boost equations.

42.5 Implementation Boundary

CA-41 supplies the finite equations a checker should implement. The next implementation step is to enumerate the patch supports in (41.4)–(41.7), embed every edge tensor into the common patch, and emit Pauli coefficients plus the divergence auxiliaries. Until that exists, the $2+1$-dimensional claims remain proposal-level.

43 Qubit Witt/Virasoro Necessary Equations

43.1 Status, Sources, and Dependencies

Status: exact finite algebra plus proposal residuals. The mode identities are finite local algebra. Their use as Virasoro diagnostics is a necessary filter only.

Dependencies: CA-14, CA-35, CA-37–CA-39, CONVENTIONS.md (d), (m), (n).

43.2 Exact Density-Mode Algebra

On a periodic chain of length N, set $z_n = \exp(2\pi i n/N)$ and

$$H_n = \sum_j z_n^j h_j, \qquad C_j = i[h_j, h_{j+1}], \qquad J_n = \sum_j z_n^j C_j.$$

Then $H_n^* = H_{-n}$, $J_n^* = J_{-n}$, and the nearest-neighbour identity is

$$[H_m, H_n] = -i(z_n - z_m)J_{m+n}. \tag{42.1}$$

This is the exact lattice antecedent of the Koo-Saleur $H(f), P(f)$ commutator. In Pauli coefficients C_j is the tensor p from (39.1).

Define higher-current densities

$$D_{r,j} = i[h_{j+r}, C_j] \quad (r = 0, 1, 2), \qquad F_j^{(r)} = i[C_j, C_{j+r}] \quad (r = 1, 2).$$

Using the Jacobi cancellation of the $r = -1$ term,

$$i[H_m, J_n] = \sum_j z_{m+n}^j \left(D_{0,j} + z_m D_{1,j} + (z_m^2 - z_n)D_{2,j}\right), \tag{42.2}$$

and

$$i[J_m, J_n] = \sum_j z_{m+n}^j \left((z_n - z_m)F_j^{(1)} + (z_n^2 - z_m^2)F_j^{(2)}\right). \tag{42.3}$$

The D-coefficients are degree three polynomials in $h_{\alpha\beta}$; the F-coefficients are degree four.

43.3 Chiral Residuals

A Koo-Saleur-style chiral ansatz has the form

$$\ell_n = A_n{:}H_n{:}_\omega + B_n{:}J_n{:}_\omega + \mu\delta_{n0}I,$$
$$\bar\ell_n = A_n{:}H_n{:}_\omega - B_n{:}J_n{:}_\omega + \mu\delta_{n0}I,$$

with $A_{-n} = \overline{A_n}$ and $B_{-n} = \overline{B_n}$. The necessary Virasoro residuals for a chosen finite mode set are

$$R_{mn}^{++} = [\ell_m, \ell_n] - (m-n)\ell_{m+n} - \frac{c}{12}m(m^2-1)\delta_{m+n,0}I, \tag{42.4}$$

$$R_{mn}^{--} = [\bar\ell_m, \bar\ell_n] - (m-n)\bar\ell_{m+n} - \frac{c}{12}m(m^2-1)\delta_{m+n,0}I, \tag{42.5}$$

and

$$R_{mn}^{+-} = [\ell_m, \bar\ell_n]. \tag{42.6}$$

For example,

$$\begin{aligned}[\ell_m, \ell_n] = {}& -iA_mA_n(z_n - z_m)J_{m+n} - iA_mB_n\mathcal{D}_{mn} \\ & + iB_mA_n\mathcal{D}_{nm} - iB_mB_n\mathcal{F}_{mn},\end{aligned}$$

where $\mathcal{D}_{mn} = i[H_m, J_n]$ and $\mathcal{F}_{mn} = i[J_m, J_n]$ are given by (42.2)–(42.3).

43.4 Necessary Filter

At finite size, the Koo-Saleur source explicitly warns that the lattice quantities need not close. Therefore the safe condition is not "finite Virasoro symmetry". The safe condition is: for the chosen ansatz and mode set, every Pauli coefficient of R^{++}, R^{--}, R^{+-} outside the named span

$$\mathrm{span}\{I, H_{m+n}, J_{m+n}\}$$

must either vanish, be a registered finite-size correction, or be annihilated in the chosen low-energy/state restriction. CA-43–CA-44 turn the last option into moment constraints.

44 Vacuum Moment Constraints for Qubit Residuals

44.1 Status, Sources, and Dependencies

Status: proposal-level SDP interface backed by sourced state/GNS architecture. The moment equations are finite algebra. The existence of a continuum vacuum or positive-energy representation is not claimed.

Dependencies: CA-15, CA-38–CA-42, CONVENTIONS.md (n).

44.2 Finite Moment Data

Let P_s be a finite Pauli word on $\mathbb{Z}$, padded by identities outside its support. A candidate translation-invariant global state ω has moments

$$y_s = \omega(P_s), \qquad y_\emptyset = 1, \qquad y_{\tau^n s} = y_s. \tag{43.1}$$

For a finite word set $\mathcal{W}$, positivity is the moment-matrix constraint

$$M_{\mathcal{W}}(y)_{u,v} = \omega(P_u^* P_v) = \sum_s \mu_{u,v}^s y_s \succeq 0, \tag{43.2}$$

where the constants $\mu_{u,v}^s$ are computed from the Pauli product table of CA-38. Hermiticity gives

$$M_{\mathcal{W}}(y)_{v,u} = \overline{M_{\mathcal{W}}(y)_{u,v}}.$$

The SDP may be represented over real variables by splitting real and imaginary parts.

44.3 Zero Expectations

The infinite sums H, P, K are not finite observables. The finite data are their local densities and residuals. After choosing the scalar subtraction $\bar{h} = h - eI$, a vacuum-normalized test may impose

$$\omega(\bar{h}_j) = 0, \qquad \omega(p_j) = 0, \tag{43.3}$$

and, for the residuals of CA-39,

$$\omega(A_j - Du_j) = 0, \qquad \omega(B_j - u_j - v^2\bar{h}_j - Dw_j) = 0. \tag{43.4}$$

These are only the level-zero moment consequences. They are usually too weak to exclude much.

44.4 GNS Relation Constraints

The stronger finite restriction says that a residual vanishes on the vacuum sector as a relation in the GNS representation. For each chosen residual density $R(h)$ and probe words X, Y,

$$\omega(X^* R(h) Y) = 0. \tag{43.5}$$

For fixed h, fixed v^2, e, u, w, and fixed residual coefficients, (43.5) is linear in the moments y_s. The same rule applies to the finite Witt/Virasoro residual densities $r_{mn}^{++}, r_{mn}^{--}, r_{mn}^{+-}$ from CA-42:

$$\omega(X^* r_{mn}^{\pm\pm}(h) Y) = 0, \qquad \omega(X^* r_{mn}^{+-}(h) Y) = 0. \tag{43.6}$$

These equations are the promised finite data extracted from the demand that a global state ω make the generators and relations vanish in expectation on a finite probe space.

44.5 Highest-Weight Caveat

If a chiral vacuum sector is named, one may also impose finite analogues of

$$\ell_n \Omega = 0, \quad \bar{\ell}_n \Omega = 0 \qquad (n > 0),$$

for example

$$\omega(\ell_{-n}\ell_n) = 0. \tag{43.7}$$

But central charge, normal ordering, and finite-volume mode normalisation are still convention-sensitive. Until CONVENTIONS.md (d) is fixed for this block, such constraints are optional diagnostics rather than canonical Virasoro vacuum equations.

45 SDP Exclusion Hierarchy for Qubit Hamiltonians

45.1 Status, Sources, and Dependencies

Status: rigorous proposal. The hierarchy is a finite-state consequence of the sourced state/GNS framework and the local residual derivations. No solver is implemented yet.

Dependencies: CA-38–CA-43, CONVENTIONS.md (n).

45.2 Level Data

Fix a Hamiltonian coefficient matrix $h_{\alpha\beta}$. Also fix, or scan externally, the scalar subtraction e, speed v^2, and any coboundary witnesses u, w used in CA-39. This is important: if those witnesses are unknown variables multiplied by unknown moments, the problem is no longer an SDP without an additional lifting relaxation.

At level N, choose finite sets

$$\mathcal{W}_N = \{\text{Pauli words supported in } [-N, N]\},$$

$$\mathcal{A}_N = \{\text{probe words whose residual products fit the window}\},$$

and residual densities

$$\mathcal{R}_N(h) = \{A - Du,\ B - u - v^2\bar{h} - Dw\}$$

plus any selected finite Witt/Virasoro residuals from CA-42.

45.3 SDP Constraints

The variables are finitely many moments y_s. The constraints are

$$y_\emptyset = 1, \qquad y_{\tau s} = y_s, \tag{44.1}$$

$$M_{\mathcal{W}_N}(y) \succeq 0, \tag{44.2}$$

and optional stationarity constraints for local probes:

$$\omega(\delta_H(X)) = 0, \qquad \omega(\delta_P(X)) = 0, \qquad X \in \mathcal{A}_N. \tag{44.3}$$

The generator-relation constraints are

$$\omega(X^* R Y) = 0, \qquad R \in \mathcal{R}_N(h), \quad X, Y \in \mathcal{W}_N, \tag{44.4}$$

whenever the Pauli product X^*RY fits inside the moment support. Each entry in (44.1)–(44.4) is affine in y for fixed residual coefficients, so this is a semidefinite feasibility problem.

45.4 Exclusion Statement

If a translation-invariant global state ω exists whose GNS representation carries the chosen local residual relations, then every finite restriction of ω satisfies every level N. Therefore:

$$\text{SDP level } N \text{ infeasible} \quad \Longrightarrow \quad h_{\alpha\beta} \text{ is excluded for this generator route.} \tag{44.5}$$

The hierarchy is monotone: increasing N adds moment variables and relation tests, and cannot turn a previously infeasible level into evidence of symmetry.

45.5 Non-Implications

Feasibility at all checked levels does not prove that a global state exists, does not prove uniqueness or clustering, does not prove positive energy, and does not prove a continuum Poincare or Virasoro representation. It only means that the tested finite restrictions have not yet found a contradiction.

When $h_{\alpha\beta}$ is also a variable, the residual coefficients are polynomial in h. Then the feasibility problem becomes a polynomial moment problem over both Hamiltonian and state variables. That is a separate relaxation; the first implementation should instead take h as input and return either an SDP infeasibility certificate or "not excluded at this level".

46 Roadmap for Qubit Symmetry Exclusion

46.1 Status, Sources, and Dependencies

Status: implementation plan. This closes the current qubit necessary-equation expansion and names the next build steps.

Dependencies: CA-34–CA-44, CONVENTIONS.md (m)–(o).

46.2 What We Have Now

The one-dimensional first-moment route now has explicit necessary equations on the Pauli coefficient matrix $h_{\alpha\beta}$:

$$p_{aed} = -2 \sum_{b,c=1}^{3} h_{ab} h_{cd} \epsilon_{bce},$$

$$A(h) = Du, \qquad B(h) - u - v^2(h - eI) = Dw.$$

These equations already rule out fully local symmetric on-site Hamiltonians and commuting classical densities for any nontrivial Poincare route of this form, because their generated momentum current is zero.

The two-dimensional shard gives a square-lattice edge schema, not yet code. The Witt/Virasoro shard gives exact finite mode commutators and residuals, but keeps Koo-Saleur's finite-nonclosure warning active. The vacuum/SDP shards turn "there exists a global state ω" into finite positive moment data and linear relation constraints.

46.3 Implementation Steps

1. Add symbolic coefficient builders for $A(h)$, $B(h)$, and coboundary equations, returning sparse polynomial tensors rather than dense matrices.

2. Add a solver-facing algebraic filter: for a fixed h, check current collapse; optionally solve the linear coboundary witnesses u, w after p, A, B are computed.

3. Build a Pauli-word moment enumerator for intervals $[-N, N]$, including product constants, translation-invariance canonicalisation, and Hermitian realification.

4. Export the fixed-h feasibility problem (44.1)–(44.4) through a local SDP interface. Record every run under `runs/` with the command line, h, level N, parameters e, v^2, u, w, and solver status.

5. Add mutation tests: perturb a known zero-current density to make the current nonzero; perturb the coboundary sign to make the coboundary reconstruction test fail; perturb a residual coefficient and require a small SDP level to change.

46.4 Acceptance for an Exclusion Claim

A future statement "h is excluded at level N" must point to:

- the exact Pauli coefficient matrix $h_{\alpha\beta}$ and density convention;
- the algebraic residual tensors and any witnesses e, v^2, u, w;

- the finite word sets $\mathcal{W}_N, \mathcal{A}_N$;
- the SDP solver transcript or certificate in a run bundle;
- a worklog entry explaining why the infeasible constraints are necessary for the named route.

46.5 Open Gaps

The central charge convention remains unfixed. The two-dimensional edge schema has no implementation. The SDP hierarchy is fixed-h; scanning continuous families of $h_{\alpha\beta}$ requires a separate polynomial relaxation. None of the filters replaces CA-15's scaling-limit obligations. They are designed to cheaply reject impossible candidates before the expensive continuum work begins.

47 Qubit SDP Implementation Contract

47.1 Status, Sources, and Dependencies

Status: checked implementation contract. This shard records the Julia layer that implements the fixed-h hierarchy of CA-43–CA-45.

Dependencies: CA-38–CA-45, CONVENTIONS.md (n).

47.2 Package Split

The implementation is deliberately layered.

- `QubitPauliWords.jl`: exact positioned Pauli words, exact multiplication phases, translation canonicalisation, and coefficient-term extraction.
- `QubitPoincareWitnesses.jl`: linear algebraic witnesses for $A = Du$ and $B - u - \lambda h + \mu I = Dw$, used before invoking an SDP.
- `QubitMomentSDP.jl`: finite moment bases, moment matrices, relation constraints $\omega(X^*RY) = 0$, JuMP model construction, and Mosek-backed solve helpers.
- `QubitHamiltonianScreening.jl`: deterministic sentinel Hamiltonians and current-collapse filters.

The fixed-h boundary from CA-44 is enforced in the API: an SDP instance takes residual terms as fixed coefficients. If h, u, w, and λ are all variables, the problem is no longer this SDP; it becomes a separate polynomial/lifted hierarchy.

47.3 Verdicts

The solver helper returns one of three finite-level statuses:

`:excluded`, `:not_excluded_at_level`, `:solver_unknown`.

Only the first is a mathematical rejection, and only for the named level and named residual constraints. The second is not evidence for symmetry; it means that this finite relaxation did not find a contradiction.

An exclusion claim must cite the coefficient matrix $h_{\alpha\beta}$, the residuals, the moment/probe word sets, the solver status, and a run bundle or test that reproduces the result.

48 Pauli Word Moment Basis for Infinite Qubit Chains

48.1 Status, Sources, and Dependencies

Status: checked. The word algebra is implemented without dense matrices and checked against exact Pauli identities.

Dependencies: CA-38, CA-43, CA-46.

48.2 Positioned Words

A word is stored as sorted non-identity labels on integer sites. Identities are not stored, but gaps are preserved:

$$X_0Z_2 \quad \leftrightarrow \quad \texttt{sites=(0,2), labels=(1,3).}$$

The empty word is the identity. Multiplication is sitewise with

$$XY = iZ, \quad YZ = iX, \quad ZX = iY,$$

and the reversed products carrying $-i$. The implementation returns

$$P_uP_v = \phi(u,v)P_{u\cdot v}, \qquad \phi(u,v) \in \{1,-1,i,-i\}.$$

The tests assert $XY = iZ$, $YX = -iZ$, $X^2 = I$, and commutation of disjoint words.

48.3 Translation Quotient

The SDP row/column words are not quotiented by translation. They are actual local probes in a finite window. Translation invariance enters only when a moment variable is named:

$$y_s = \omega(P_s) = \omega(P_{\tau^n s}).$$

The canonical representative shifts the first non-identity site to 0. Thus $X_{-3}Z_{-1}$ and X_0Z_2 share the same moment variable.

Because each phase-free Pauli word is self-adjoint, each canonical moment variable is real. Complex numbers enter only as multiplication phases.

49 Moment Matrices and Positivity Constraints

49.1 Status, Sources, and Dependencies

Status: checked. Moment matrices are constructed by `build_qubit_sdp_model`.

Dependencies: CA-43–CA-47.

49.2 Affine Moment Matrix

For a finite actual probe set $\mathcal{W}$, define

$$M_{u,v} = \omega(P_u^* P_v) = \phi(u,v) y_{\mathrm{can}(u \cdot v)}.$$

The identity moment is the constant 1. All other canonical words appearing in products or relation constraints receive real JuMP variables.

Write $M = A + iB$, with A, B real affine matrices. Positivity of the complex Hermitian moment matrix is imposed by the real symmetric cone

$$\begin{pmatrix} A & -B \\ B & A \end{pmatrix} \succeq 0. \tag{48.1}$$

For the one-site probe set $\{I, X, Y, Z\}$, the implementation creates three real moment variables and an 8×8 realified PSD block. This exact dimension is checked in the Julia suite.

49.3 Tiny Feasibility Tests

The zero-residual one-site moment problem solves with Mosek as

`:not_excluded_at_level`.

The artificial relation $\omega(I) = 0$ solves as `:excluded`. These are smoke tests for the solver path, not physical Hamiltonian claims.

50 Compiling Poincare Residuals into Moment Equations

50.1 Status, Sources, and Dependencies

Status: checked local implementation. Dense matrix residual oracles from CA-40 are converted into coefficient terms and then into moment equations.

Dependencies: CA-39–CA-40, CA-43–CA-48, CONVENTIONS.md (n).

50.2 Algebraic Witnesses

For fixed h, the first algebraic gate solves

$$A(h) = Du$$

as a real least-squares linear system after fixing the scalar gauge $u_{00} = 0$. Passing this gate supplies a concrete u.

The next gate solves the linearized boost equation

$$B(h) - u - \lambda h + \mu I = Dw, \tag{49.1}$$

again with the scalar gauge $w_{000} = 0$. Here $\lambda = v^2$ and $\mu = v^2 e$. A positive λ is the nontrivial-speed branch; a solution with $\lambda = 0$ is only the trivial current-collapsed branch.

50.3 Moment Relations

Once residual terms $R = \sum_r c_r P_r$ are fixed, each pair of probe words gives

$$\omega(X^* RY) = \sum_r c_r \, \phi(X, r, Y) \, y_{\mathrm{can}(XrY)} = 0. \tag{49.2}$$

The implementation splits (49.2) into real and imaginary affine constraints. This is the finite SDP condition promised by CA-43.

50.4 Important Boundary

Witness solving is an algebraic necessary filter. The SDP layer does not optimize over h, u, w, or λ. It checks whether a fixed residual system is compatible with a finite positive moment restriction.

51 SDP Levels and the Mosek Backend

51.1 Status, Sources, and Dependencies

Status: checked implementation. The solver path is exercised by tests and by a recorded smoke run.

Dependencies: CA-44–CA-49.

51.2 Level Data

An implemented level consists of:

$$\mathcal{W}_{\mathrm{psd}}, \qquad \mathcal{W}_{\mathrm{rel}}, \qquad \mathcal{R}.$$

The first set is the row/column basis for the PSD moment matrix. The second set supplies X, Y in $\omega(X^*RY) = 0$. The third is a list of fixed residual coefficient terms.

The convenience constructor uses full Pauli-word windows:

$$\mathcal{W} = \{P_s : \operatorname{supp} s \subseteq [a, a+L-1]\}.$$

Optional weight truncation is an implementation diagnostic, not the canonical hierarchy.

51.3 Solver Status

The feasibility problem has no objective. The status map is:

MOI termination status	hierarchy verdict
`OPTIMAL`	`:not_excluded_at_level`
`INFEASIBLE`, `INFEASIBLE_OR_UNBOUNDED`	`:excluded`
anything else	`:solver_unknown`.

This map is intentionally conservative.

51.4 Smoke Run

The run bundle

`runs/2026-05-31-qubit-sdp-smoke/`

records three tiny Mosek solves: zero residual, identity-zero relation, and a forced $ZZ = 0$ relation. The first is not excluded at the level; the latter two are excluded. The ZZ case is artificial: it forces $\omega(ZZ \cdot ZZ) = \omega(I) = 0$.

52 Sentinel Hamiltonians for the SDP Hierarchy

52.1 Status, Sources, and Dependencies

Status: checked sentinels. These examples test the implementation and do not assert physical criticality.

Dependencies: CA-35, CA-39–CA-50.

52.2 Matrices

The package records five 4×4 Pauli coefficient matrices:

- a symmetric fully onsite split of $0.25I + 0.7X - 0.3Y + 0.5Z$;
- a commuting classical ZZ density;
- a transverse-Ising-style density $-ZZ - 0.35(XI + IX)$;
- a generic currentful density;
- the fake asymmetric onsite split $XI + IZ$.

The deterministic current filter gives

$$\text{onsite}, ZZ : \texttt{:current\_collapsed},$$

and

$$\text{transverse-Ising, generic, fake} : \texttt{:currentful}.$$

52.3 Gate Outcomes

The onsite and ZZ examples are excluded before SDP for the nontrivial first-moment route because $p = 0$. Their boost-witness solve has only the trivial $\lambda = 0$ branch.

The generic and fake examples have nonzero current but fail $A = Du$. This is the second algebraic gate catching more than current collapse.

The transverse-Ising-style example has nonzero current and passes $A = Du$, but the implemented linear boost witness does not solve (49.1) with the tested data. It is therefore a useful near-candidate sentinel: it reaches a deeper gate before failing.

53 Candidate Hamiltonian Selection Roadmap

53.1 Status, Sources, and Dependencies

Status: roadmap. This shard connects the implementation to the north star: identifying Hamiltonians worth expensive continuum analysis.

Dependencies: CA-15, CA-42–CA-51.

53.2 Screening Order

For a fixed nearest-neighbour qubit density $h_{\alpha\beta}$, use:

1. current collapse $p = 0$;
2. conservation witness $A = Du$;
3. boost witness $B - u - \lambda h + \mu I = Dw$ with $\lambda > 0$;
4. finite moment SDP feasibility for the chosen residuals;
5. only then, low-energy/scaling and Koo-Saleur/Virasoro tests.

The point is to reject impossible candidates cheaply. A Hamiltonian surviving all finite checks is not accepted as symmetric; it is merely promoted to the next, more expensive analysis.

53.3 Next Implementation Steps

The immediate next step is a run format for candidate scans: input $h_{\alpha\beta}$, current status, witness data, level data, solver status, and a headline verdict. Continuous families require external parameter scans or a later polynomial hierarchy. Named physical model families need registered sources before the report can claim criticality, universality class, or continuum CFT content.

54 Qubit Candidate Scan Contract

54.1 Status, Sources, and Dependencies

Status: checked implementation. The scan layer packages the CA-46–CA-52 fixed-h gates into reproducible candidate runs.

Dependencies: CA-39, CA-46–CA-52, CONVENTIONS.md (m)–(q).

54.2 Contract

Each scan point is a real 4×4 Pauli coefficient matrix $h_{\alpha\beta}$. The gate order is:

$$p = i[h_j, h_{j+1}], \qquad A = Du, \qquad B - u - \lambda h + \mu I = Dw,$$

followed, only when requested, by finite moment-SDP checks.

The serialized row records the family, parameters, coefficient matrix, current norm, witness residual norms, speed data, terminal gate, and scoped verdict. The scope field records the fixed first-moment route.

54.3 Verdict Boundary

`excluded_current_collapsed` and `excluded_no_conservation_witness` are early algebraic failures for the fixed current route. `excluded_no_boost_witness` means the exact local boost-witness equation failed; it does not reject other lattice stress-tensor constructions such as Koo–Saleur/free-fermion routes. This distinction is essential for sourced critical transverse Ising.

55 Pauli Coefficients for Scan Families

55.1 Status, Sources, and Dependencies

Status: checked implementation convention. The family constructors are implemented in the Julia backend.

Dependencies: CA-34–CA-53, CONVENTIONS.md (m), (q).

55.2 Implemented Maps

The scan uses (I, X, Y, Z) row/column order. A one-site field $f_a \sigma_j^a$ is represented on bonds by

$$h_{a0} = h_{0a} = f_a/2.$$

Thus the self-dual transverse-Ising sample is

$$h = -ZZ - \frac{1}{2}(XI + IX),$$

which is the bond-density version of the sourced transverse-Ising form.

The XXZ constructor uses

$$h = J(XX + YY + \Delta ZZ),$$

and Heisenberg is the $\Delta = 1$ specialization. The transverse-XY constructor uses $J_x XX + J_y YY + h_z Z$ with the same symmetric field split.

55.3 Synthetic Families

XYZ, compass, DM-style, field-deformed Heisenberg, and deterministic dense generic matrices are stress tests. They exercise the gates but carry no criticality or universality-class claim in this lab book.

56 Locally Sourced Spin-Chain Inputs

56.1 Status, Sources, and Dependencies

Status: source audit. This shard records which family names the first scan may treat as physical rather than synthetic.

Dependencies: CA-14, CA-52–CA-54.

56.2 Allowed Physical Labels

The defect-gauging source gives an effective qubit Hamiltonian proportional to $\sum_i(I+Z_iZ_{i+1}+X_i)$, identifies it as transverse Ising, and states that it gives a critical theory. The Ising anyon-chain source independently identifies the localized chain with critical TFIM and the Ising CFT.

The lattice-fermion source supports transverse XY as a Jordan–Wigner image and also records TFIM/Koo–Saleur convergence on the even observable subalgebra. The XXZ source gives a Pauli-chain Hamiltonian and the gapless/free-boson continuum anchors for $-1 < \Delta \leq 1$, with Heisenberg as a special point.

56.3 Excluded From Physical Labeling

The first scan does not make physical claims for XYZ, DM, compass, clock, Potts, or Ashkin–Teller. Potts is qutrit in the local source. The sourced Ising zigzag/Ashkin–Teller relation uses a three-site term, outside the present nearest-neighbour two-qubit input class.

57 Synthetic and Stress-Test Hamiltonian Grids

57.1 Status, Sources, and Dependencies

Status: checked implementation input. These grids are generated by the candidate-scan sample constructor.

Dependencies: CA-53–CA-55, CONVENTIONS.md (q).

57.2 Grids

The synthetic block includes XYZ triples on $\{-1,0,1\}^3 \setminus \{0\}$, XY anisotropy, compass XX/ZZ, DM-style antisymmetric $XY - YX$, Heisenberg with one field component, and 24 deterministic dense generic matrices. The generic matrices are generated by fixed sine/cosine formulas in the code rather than by an ambient random number generator.

57.3 Purpose

These samples answer engineering questions: Does the conservation gate reject generic dense inputs? Do structured bilinear families reach the boost gate? Do current-collapsed one-axis cases remain caught? They do not answer whether XYZ or DM chains are critical.

58 Current, Conservation, and Boost Witness Gates

58.1 Status, Sources, and Dependencies

Status: checked scan gates. The implementation calls the CA-40 and CA-49 helpers pointwise.
Dependencies: CA-39–CA-40, CA-49, CA-53–CA-56.

58.2 Gates

The current gate checks whether $p = i[h_j, h_{j+1}]$ collapses. The conservation gate solves $A = Du$. The boost gate solves

$$B - u - \lambda h + \mu I = Dw$$

with $w_{000} = 0$ and requires nonzero positive speed for the nontrivial operator route.

58.3 Interpretation

The first full scan has no survivor of the exact boost gate. The self-dual TFIM point, which is locally sourced as critical, passes current and conservation but fails the exact boost witness. Therefore this gate is best read as a test for an exact UV first-moment construction, not as a general criterion for CFT emergence.

59 Optional SDP Levels for Scan Survivors

59.1 Status, Sources, and Dependencies

Status: implemented option, unused by the first broad run. The scan API can call `solve_qubit_sdp` after a nontrivial boost witness.

Dependencies: CA-44, CA-50, CA-53–CA-57.

59.2 Level Data

An SDP row must record the PSD window, relation window, moment count, relation constraint count, optimizer, and solver status. The status vocabulary remains

`:excluded`, `:not_excluded_at_level`, `:solver_unknown`.

59.3 First-Run Boundary

The 99-point run has no nontrivial exact boost witness, so it does not invoke Mosek per point. This keeps the run honest: adding SDP rows before fixing a residual witness would mix the fixed-h SDP with a polynomial variable-witness problem.

60 Reproducible Scan Bundle Format

60.1 Status, Sources, and Dependencies

Status: checked run bundle. The bundle is

`runs/2026-05-31-qubit-candidate-scan/`.

Dependencies: CA-53–CA-58 and AGENTS.md Law 3.

60.2 Files

`inputs.toml` records the Pauli basis, indexing rule, family inventory, source-kind inventory, and sample count. `summary.toml` records verdict counts and notable sourced points. `results.toml` records one row per sample, including coefficients, parameters, terminal gate, residual norms, and scope.

60.3 Acceptance

The run command prints the same headline counts that the TOML stores: 99 samples, 9 current-collapsed, 27 conservation failures, and 63 exact boost-witness failures.

61 First Candidate Scan Results

61.1 Status, Sources, and Dependencies

Status: checked run result. This shard reports only the run bundle named in CA-59.

Dependencies: CA-53–CA-59.

61.2 Headline Table

The 99 samples split as:

verdict	count
`excluded_current_collapsed`	9
`excluded_no_conservation_witness`	27
`excluded_no_boost_witness`	63

No point reaches `not_excluded_algebraic`.

61.3 Notable Points

The self-dual TFIM sample

$$-ZZ - \frac{1}{2}(XI + IX)$$

is currentful and has a conservation witness, but its exact boost residual norm is nonzero in the run. The Heisenberg point and the XX point behave the same way: they reach the boost gate and fail it.

Because TFIM and XXZ have local source anchors as critical or gapless models, the conclusion is about the route: the exact CA-39 boost witness is too strict as a universal candidate filter.

62 From Scan Failures to Scaling Work

62.1 Status, Sources, and Dependencies

Status: roadmap from checked failures. This is the handoff after CA-60.

Dependencies: CA-14, CA-15, CA-42, CA-52–CA-60.

62.2 Queue

Do not retry the same exact boost-witness gate expecting TFIM to pass. The next candidate path should use the sourced Koo–Saleur/free-fermion/OAR evidence chain for TFIM and transverse XY, and the Koo–Saleur/XXZ literature for XXZ. Those routes use low-energy or scaling-limit stress tensors, not an exact finite-support UV boost density satisfying CA-39.

62.3 SDP Upgrade

The state-level hierarchy should next learn how to quotient or relax the unknown Dw term without turning the problem into an uncontrolled polynomial relaxation. Until then, the implemented SDP layer is still useful for fixed residual systems and artificial contradictions, but the broad scan has shown where the exact-local route stops.